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碩士論文

MDS-UPDRS 手指敲擊測驗評估，比較 ChatGPT-4 與兩位神經科
醫生的評分

MDS-UPDRS Finger Tapping Test Evaluation Comparing
ChatGPT-4 and Two Neurologists

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Chinese abstract

研究介紹: 巴金森氏症 (PD) 是一種進行性神經退化性疾病，根據 2021 年全球疾病負擔研究，全球有超過 1100 萬人被診斷患有此病，對全球失能調整生命年造成了顯著影響。巴金森氏症的一個關鍵診斷標誌是運動遲緩，可以通過運動障礙協會統一巴金森氏症評定量表 (MDS-UPDRS) 手指敲擊 (FT) 測試進行量化。運動遲緩的特徵是啟動和執行動作的緩慢，在手指敲擊測試中，透過速度、振幅和節奏來評估。目前，這些評分是由臨床神經科醫師決定的，不僅耗時、成本高，而且可能存在主觀性。通用的大型語言模型 (LLMs) 每季度在各種分析和邏輯測試中創下新紀錄，有時甚至超越人類表現，因此測試大型語言模型在手指敲擊測試評分方面的表現具有重要意義。

研究目標: 本研究旨在測試一個通用大型語言模型——ChatGPT-4 (版本: gpt-4-turbo-2024-04-09) ——在有無範例指示的情況下對手指敲擊評分的表現，並與臨床神經科醫師的表現進行比較。

研究方法: 本研究招募了 16 名巴金森氏症的受試者，並使用智慧型手機 (720p/240 FPS) 錄製了 53 個手指敲擊測試的影片。這些影片由兩名神經科醫師獨立評估，以建立真實評分標準並評估人類評分者間信度。同時，ChatGPT-4 基於使用 CoTracker 從單個智慧型手機提取食指和拇指上標記的時間序列軌跡以及 MDS-UPDRS 手指敲擊指示，對每個影片進行了 10 次獨立的嚴重程度評估。研究執行了四項實驗來測試模型的表現：第一項實驗 (Exp. A) 使用動作的第 2.9 至 9.4 秒 (平均 4.0 秒) 起開始的 10 秒片段資料進行分析。第二項實驗 (Exp. B) 則在前者基礎上加入了利用 Python 程式碼根據極值辨識敲擊特徵的過程。第三項實驗 (Exp. C) 聚焦在前 10 次敲擊動作，以評估早期動作特徵。第四項實驗 (Exp. D) 則選取了第三項實驗中表現最差的四個例子，結合神經科醫師的評分，作為 few-shot (少樣本) 學習的範例，以探討是否能提升模型表現。

研究結果: 在實驗 D 中，ChatGPT-4 的評分者內信度達到組內相關係數 ICC (2,1) 為 0.45，優於神經科醫師間的評分者間信度 ICC (2,1) 為 0.37，顯示其在評分一致性上的潛力。不同實驗設計對模型表現的影響，分析如下：實驗 A 的 Wasserstein 距離為 1.22，ICC (2,1) 為 0.05，準確率為 18.2 %，幾乎與隨機猜測無異；實驗 B 納入以 Python 程式進行的特徵偵測，Wasserstein 距離降至 0.79，ICC (2,1) 提升至 0.16，準確率為 22.7 %，表現顯著提升；實驗 C 聚焦於前 10 次敲擊動作，強調早期動作特徵，Wasserstein 距離為 0.58，ICC (2,1) 為 0.21，準確率為 50.0 %，準確度顯著提升；實驗 D 採用了包含四個範例的少樣本學習方法，在剩餘測試案例上達到 Wasserstein 距離為 0.63，ICC (2,1) 為 0.45，準確率為 61.1 %，顯示少樣本學習策略在信度上帶來了顯著提升。在實驗 D 中，ChatGPT-4 與神經科醫師之間的評分者間信度 ($ICC(3,k)=0.85$) 高於過去幾項研究中神經科醫師彼此之間的評分者間信度。不過，在解讀此比較結果時，應考量不同資料集中受試者特性與評分者背景的差異。

研究結論: 在實驗 D 中加入具體範例後，顯著提升了 ChatGPT-4 的評分者內信度與準確率。這些結果顯示，在設計良好的輸入引導下，ChatGPT-4 能夠產出可與臨床神經科醫師相當的評分結果。

關鍵字: 巴金森氏症，手指敲擊測試，嚴重性評估，大型語言模型，ChatGPT-4，沃瑟斯坦距離，評分者內信度，評分者間信度

Abstract

Introduction: Parkinson’s disease (PD) is a progressive neurodegenerative disorder that significantly impacts global disability-adjusted life years, with over 11 million people diagnosed worldwide according to the Global Burden of Disease Study 2021. A key diagnostic marker of PD, bradykinesia, can be quantified using the Movement Disorder Society-Sponsored Revision of the Unified Parkinson’s Disease Rating Scale (MDS-UPDRS) Finger tapping (FT) test. Bradykinesia is characterized by slowness in initiating and executing movements, which in the FT is assessed by evaluating the speed, amplitude, and rhythm. Currently, these scores are determined by clinical neurologists, which is costly, time-consuming, and can suffer from subjectivity. General-purpose large language models (LLMs) are setting new records each quarter in various analytical and logical tasks, and in some cases even outperform human performance. Therefore, it is of interest to test how well LLMs can perform in scoring finger-tapping tests.

Objective: This study aims to test how well a general-purpose LLM—ChatGPT-4 (version: gpt-4-turbo-2024-04-09)—can perform the scoring of FT based on instructions with and without examples, and compare its performance to that of clinical neurologists.

Methods: The study enrolled 16 subjects with PD and recorded 53 videos of finger-tapping tests using smartphones (720p/240 FPS). These videos were independently evaluated by two neurologists to establish ground truth scores and evaluate human inter-rater reliability. Concurrently, ChatGPT-4 performed 10 independent severity evaluations of each video based on time-series trajectory of markers placed on the index finger and thumb extracted from a single smartphone using CoTracker and the MDS-UPDRS FT instruction. Four experiments were conducted to test the model’s performance: the first used a 10-second segment data of the recording taken 2.9-9.4 (mean 4.0) seconds after the start (Exp. A), the second added

Python code for identifying tapping features based on extrema (Exp. B), the third focused on the initial 10 taps to assess early movement characteristics (Exp. C), and the fourth included four examples of poorest performance from Exp. C together with the neurologists' score to see if few-shot learning improves model performance (Exp. D).

Results: ChatGPT-4 achieved an intra-rater reliability of 0.45 using the Intraclass Correlation Coefficient (ICC(2,1)) in Exp D, surpassing the inter-rater reliability (ICC(2,1)) of 0.37 by the neurologists, highlighting its potential for consistent evaluations. The influence of different experimental designs on performance was analyzed as follows: Exp. A with Wasserstein distance (WD) 1.22, ICC(2,1) 0.05, and accuracy 18.2 % is indistinguishable from random guessing. Exp. B with WD 0.79, ICC(2,1) 0.16, and accuracy 22.7 % incorporated Python-based feature detection, leading to significant improvements. Exp. C with WD 0.58, ICC(2,1) 0.21, and accuracy 50.0 % focused on the initial 10 taps, emphasizing early movements, and achieved significantly better accuracy. Exp. D, employed few-shot learning with four examples and achieved WD 0.63, ICC(2,1) 0.45, and accuracy 61.1 % on the remaining test cases, demonstrating substantial improvement in reliability through few-shot learning. The inter-rater reliability (ICC(3,k)) of 0.85 for ChatGPT-4 versus neurologists in Exp. D is higher than values reported between neurologists in past few studies. However, differences in subject characteristics and rater backgrounds across datasets should be taken into account when interpreting this comparison.

Conclusion: The inclusion of specific examples in Experiment D substantially improved the intra-rater reliability and accuracy of ChatGPT-4. These results demonstrate that, when guided by well-designed inputs, ChatGPT-4 can produce scoring outcomes comparable to those of clinical neurologists.

Keywords: Parkinson's disease, finger-tapping test, severity assessment, Large Language Models, ChatGPT-4, Wasserstein Distance, intra-rater reliability, inter-rater reliability

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Chapter 1

Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder and represents an increasingly significant global health concern. In 2021, approximately 11.8 million individuals were living with PD worldwide, representing a 274% increase in prevalence since 1990. That same year, PD accounted for an estimated 7.5 million disability-adjusted life years (DALYs), a 162% increase over the same period. Among individuals aged 60-79 years, PD ranked fifth in neurological DALYs and rose to third among those aged 80 years and older, underscoring its rapid global growth (Steinmetz et al., 2024).

PD is characterized by a wide spectrum of motor and non-motor symptoms, which include bradykinesia, tremor, rigidity, and sleep disturbances (Schapira et al., 2017). To standardize the clinical evaluation of symptom severity, the Movement Disorder Society-Sponsored Revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS) was developed (Goetz et al., 2008). It comprises 65 items structured to facilitate precise assessments, building upon the original UPDRS established by Fahn et al. (1987).

The MDS Clinical Diagnostic Criteria (Postuma et al., 2015) specify that diagnosing PD requires bradykinesia in combination with either rest tremor or rigidity. Bradykinesia manifests as slowness in initiating and executing movement, typically evident as progressive impairment during repetitive motor tasks like finger or foot tapping. It is a cardinal feature of PD and a required component for clinical diagnosis (Berardelli et al., 2013; Abdo et al., 2010).

A widely used method for assessing bradykinesia is the finger tapping (FT) test, listed as item 3.4 in the MDS-UPDRS. This simple yet effective test emphasizes high-frequency,

repetitive, and precise movements, making it particularly sensitive to subtle motor deficits. Ling et al. (2012) indicates that FT movements are more severely impaired in PD patients compared to other movements like hand opening and closing or pronation and supination. A notable decline in rhythm and amplitude over time during these repetitive tasks highlights the impact of bradykinesia on daily activities, leading to significant functional limitations.

To address these challenges, recent advances in artificial intelligence (AI) offer promising alternatives to traditional scoring methods. Although clinical expertise remains essential, it introduces inherent variability and subjectivity. Moreover, the rising prevalence of PD underscores the urgent need for scalable, objective, and standardized assessment tools. In this context, large language models (LLMs)—which exhibit advanced reasoning and analytical capabilities—present an opportunity to enhance the consistency and interpretability of clinical evaluations (Zhao et al., 2023).

Given this context, this study aims to evaluate whether ChatGPT-4, a popular LLM, can evaluate the severity of finger tapping performance—a key clinical indicator of bradykinesia—according to MDS-UPDRS criteria. Specifically, we examine the ability of GPT-4 to determine the severity scores comparable to those provided by neurologists.

Previous works to objectively assess FT severity have predominantly employed machine learning (ML) and traditional deep learning (DL) techniques, which are well-suited for handling complex tasks that often demand human-like cognitive processing (Sarker, 2022), such as distinguishing subtle variations in motor function. These models, while effective in classification tasks, typically provide scores without explanatory context. This can limit their utility in clinical settings where understanding the basis of diagnostic decisions is crucial.

In contrast, LLMs offer significant enhancements in this domain (Minaee et al., 2024). Trained on extensive databases containing diverse forms of textual data, including websites, online books, research papers, Wikipedia, and Github repository, LLMs are adept at discerning complex patterns and features across various areas and datasets. Their ability to understand natural language queries and produce coherent text responses makes LLMs a powerful tool for applications requiring detailed narrative generation.

In the medical domain, several LLMs have demonstrated strong performance in discriminative tasks; among them, GPT-4 consistently outperforms models such as Flan-PaLM, Med-PaLM 2, GPT-3.5 (Nori et al., 2023), and even exceeds human expert performance on bench-

marks like the United States Medical Licensing Examination (USMLE) and PubMedQA (Zhou et al., 2023). Beyond medicine, GPT-4 has also demonstrated superior capabilities across various domains compared to models like LLaMA (Touvron et al., 2023) and PaLM (Chowdhery et al., 2023; OpenAI, 2023).

Leveraging these capabilities, we evaluated ChatGPT-4’s effectiveness in assessing FT severity in PD. We collected video data from subjects performing FT tests and provided the FT trajectory data, MDS-UPDRS criteria, and data collection methodology as inputs for interaction with ChatGPT-4. To further guide the model’s evaluation process, we supplemented the instruction with a Python script and representative features corresponding to each severity level. The model subsequently generated step-by-step evaluations and severity scores based on the provided inputs.

To provide broader context on existing finger tapping data sources, we also identified several publicly available datasets through online search. Appendix G summarizes these datasets, including subject characteristics and extracted features. These datasets were not used in the current study but are included to illustrate the diversity of available resources in this domain.

We conducted four experiments using distinct prompt designs to examine ChatGPT-4’s performance in evaluating finger tapping severity. Specifically, we aimed to identify the features considered by the model, understand its reasoning process, and compare its outputs to those provided by neurologists. This study explores ChatGPT-4’s behavior and potential relevance in the context of motor symptom interpretation.

1.1 Automatic Assessment of Finger Tapping Severity

Recent scientific approaches to assessing the severity of Parkinson’s disease (PD) frequently involve the collection of sensor-based and video-based data to capture motor symptoms. A summary of studies applying machine learning and deep learning models to classify FT severity in PD is presented in Table 1.1, with corresponding search strings and the PRISMA flowchart provided in Appendix F. Features such as the number of taps, and the speed of finger opening and closing, vary with the severity of the condition (Butt et al., 2018).

Table 1.1: Classification of FT severity in PD. HC is healthy control subject. TF means the tapping features. N/A means the information is not available. L is left hand and R is right hand. DT: Decision Tree, SVR: Support Vector Regression, LightGBM: Light Gradient Boosting Machine, DNN: Deep Neural Network, SVM: Support Vector Machine, RF: Random Forest, LR: Logistic Regression, FCN: Fully Connected Network, CNN: Convolutional Neural Network.

Study	Subject	Sample	Input	Model	Performance
Liu et al. (2024)	93 PD 27 HC	565	TF	ATST-Net	accuracy: 0.84
Shin et al. (2024)	45 PD	207	TF	SVM	accuracy: 0.87, f1-score: 0.86
Hsu et al. (2024)	N/A	253	joint	LSTM + DNN	accuracy: 0.77 (L), 0.85 (R)
Singh et al. (2023)	20 PD 8 HC	48	accelerometer data	XceptionTime	accuracy: 0.71
Yu et al. (2023)	50 PD	75	TF	DT	accuracy: 0.8, acceptable accuracy: 0.93
Islam et al. (2023)	172 PD 78 HC	489	TF	SVR	accuracy: 0.52
Islam et al. (2023)	172 PD 78 HC	489	TF	LightGBM	MAE: 0.58, MSE: 0.54
Yang et al. (2022)	N/A	611	TF	DNN	f1-score: 0.88 (L), 0.84 (R)
Yang et al. (2022)	N/A	611	TF	SVM	f1-score: 0.80 (L), 0.73 (R)
Yang et al. (2022)	N/A	611	TF	RF	f1-score: 0.84 (L), 0.73 (R)
Yang et al. (2022)	N/A	611	TF	LR	f1-score: 0.83 (L), 0.74 (R)
Li et al. (2022)	120	252	TF	FCN	accuracy: 0.8
Li et al. (2021)	157 PD	744	joint+TF	CNN + ST-Attention	accuracy: 0.72, acceptable accuracy: 0.98
Liu et al. (2019)	60 PD	120	TF	SVM	accuracy: 0.9
Martinez-Manzanera et al. (2015)	25 PD 10 HC	70	TF	SVM	accuracy: 0.9

Liu et al. (2024) proposed ATST-Net (Attention Time Series Two-stream Networks), a deep learning method to assess FT severity. The authors extracted feature of amplitude and velocity from recorded videos. The network uses a Two-stream architecture that processes

both amplitude and speed features in parallel, with an enhanced attention mechanism to capture the long-term trends in continuous movements. To address data imbalance issues, they implemented Focal Loss for reweighting.

Shin et al. (2024) used the Leap Motion optical motion-capture device to collect hand movement data, including finger tapping. They calculated 31 movement patterns and extracted 69 statistical features from each pattern, generating 2139 total features. Their novelty lies in implementing a two-stage feature selection process: first using Random Forest to rank features by Gini impurity and select the top 15%, then applying Sequential Forward Floating Selection with Support Vector Machine (SVM) to identify optimal feature combinations. This approach achieved impressive classification accuracies on three hand movements (FT, hand opening and closing, and forearm pronation and supination).

Hsu et al. (2024) used OpenPose to extract joint coordinates and introduced “Multitemporal Feature Average Pooling” for temporal feature extraction. Their approach combined three inputs: LSTM Counting Model outputs (which measure movement frequency in 10-second windows using bidirectional LSTM layers), basic coordinate information, and features from their proposed pooling method. This pooling method extracts activation values from each of the three bidirectional LSTM layers across multiple time segments, then computes their average to create a unified temporal feature representation that captures movement patterns at different levels of abstraction simultaneously. Shapley additive explanation analysis demonstrated that this pooling approach significantly enhanced performance by contributing key temporal features.

Singh et al. (2023) used wearable accelerometers and feed the processed motion data into XceptionTime architecture to classify movement data into low or high severity levels. Their key innovation lies in two techniques: first, enhancing the training dataset by including motion data from healthy older adults; second, using unsupervised pre-training to initialize the model, enabling it to learn representations by predicting missing parts of motion signals. This approach addresses the challenge of limited labeled PD patient data in deep learning applications. Their method achieved a 9.4% improvement in accuracy for finger tapping compared to baseline models. The results demonstrate that incorporating motion data from healthy older adults can implicitly improve representations of PD motion patterns, while unsupervised pre-training enhances classification performance without requiring additional

annotated PD data.

Yu et al. (2023) employed the Mediapipe pose estimation technique to capture the joint coordinates of the thumb and index finger. They extracted 15 features, including 5 features of tapping amplitude and 5 features of tapping velocity. These features were then input into a decision tree to enhance the interpretability of the decision-making process. Additionally, the paper provided definitions for hesitation and halts in the context of their research.

Islam et al. (2023) collected FT video data from 250 participants, comprising 172 individuals with PD and 78 healthy controls. They evaluated the finger-tapping severity scores assigned by three neurologists based on the MDS-UPDRS FT. Models used interpretable features such as speed, amplitude, frequency, period, and acceleration extracted from hand keypoints. Among several models tested, the LightGBM regressor achieved the best performance with a mean absolute error (MAE) of 0.58, mean squared error (MSE) of 0.54.

Yang et al. (2022) utilized the MMPose pose estimation technique to capture the joint coordinates of the thumb and index finger from 5-second FT video data. The model inputs are three extracted features: tapping rate, variation in tapping amplitude, and the tapping frozen times. These features were classified using several models, including SVM, Random Forest (RF), Logistic Regression (LR), and a three-layer Deep Neural Network (DNN), each with 8 hidden nodes—to evaluate motor symptom severity according to the MDS-UPDRS scale. Notably, the DNN model outperformed the others. With the feature extraction method, the performance improved compared to inputting the joint coordinates directly.

Li et al. (2022) used the Mediapipe pose estimation technique to capture the joint coordinates of the thumb and index finger from six-second FT video data, requiring at least 10 taps. The distance between the thumb and index finger was input into a 4-layer Fully Connected Network (FCN) with 5-fold cross-validation. The performance is better compared to several studies using skeleton data.

Li et al. (2021) employed OpenPose for pose estimation in 5-second FT videos, capturing full hand's joint coordinates. They inputted these coordinates and the features of distances between joints into a CNN based spatial-temporal attention mechanism (ST-Attention), utilizing 4-fold cross-validation. The study implemented balanced mini-batch sampling to address class imbalance and improve classification. Additionally, it defined acceptable accuracy as predictions within one point of the agreement score. Implementing mini-batch sampling for

unbalanced data increases accuracy and improves recognition of categories with smaller sample sizes.

Liu et al. (2019) proposed a pose estimation approach based on deep learning to capture joint coordinates, inputting amplitude and tap interval features into an SVM with the radial basis function kernel (RBF-SVM) and employing 5-fold validation to assess performance. The study found that the predicted scores highly correlated with amplitude, interval, and interval variation, and showed a weak correlation with amplitude variation, as determined by Pearson's correlation analysis.

Martinez-Manzanera et al. (2015) affixed a 9 degrees of freedom orientation sensor to the index finger to capture FT data. They input 7 features from 21 movement signal features into an SVM, employing leave-one-out cross-validation (LOOCV) for evaluation. SVM with the radial basis function kernel (RBF-SVM) enables the repeated input of specific features to achieve a better decision boundary and enhance classification performance.

1.2 ChatGPT-4 Compared with Other LLMs

OpenAI (2023) demonstrates that ChatGPT-4 achieves state-of-the-art performance across multiple benchmarks, surpassing both its predecessor, ChatGPT-3.5, and other advanced LLMs in few-shot evaluations. On the MMLU (Massive Multitask Language Understanding) benchmark, which evaluates knowledge across 57 subjects, 5-shot ChatGPT-4 achieves 86.4%, outperforming 5-shot ChatGPT-3.5 (70.0%) and 5-shot U-PaLM (70.7%) (Tay et al., 2022). In code generation, 0-shot ChatGPT-4 generates more accurate and functional code snippets on the HumanEval benchmark, achieving 67.0% compared to 0-shot ChatGPT-3.5 (48.1%) and 0-shot PaLM (26.2%) (Chowdhery et al., 2023). In reading comprehension, 3-shot ChatGPT-4 exhibits superior performance on the DROP (Discrete Reasoning Over Paragraphs) benchmark with an F1 score of 80.9, compared to 3-shot ChatGPT-3.5 (64.1) and 1-shot PaLM (70.8). Furthermore, ChatGPT-4 demonstrates enhanced problem-solving abilities in mathematics. On the GSM-8K (Grade School Math 8K) benchmark, 5-shot-CoT ChatGPT-4 achieves 92.0%, significantly surpassing 5-shot ChatGPT-3.5 (57.1%) and even the 8-shot Minerva model (58.8%) (Singh and Singh, 2024).

Lim et al. (2023) conducted a research on the performance of ChatGPT-3.5, ChatGPT-4, and Google Bard in responding to 31 myopia-related questions. The responses were evaluated by three consultant-level pediatric ophthalmologists and categorized as “poor”, “border-line”, or “good”. The results indicated that ChatGPT-4 outperformed the others, achieving a “good” rating for 80.6% of its responses, whereas ChatGPT-3.5 and Google Bard received “good” ratings for 61.3% and 54.8% of their responses, respectively.

1.3 Review of Prompt Engineering for LLMs

A variety of prompting strategies have been developed to optimize the performance of LLMs, enabling them to tackle diverse tasks with improved accuracy and efficiency. Among the most prominent techniques are In-Context Learning, Chain-of-Thought prompting, prompt tuning, and Retrieval-Augmented Generation.

In-Context Learning (ICL) (Dong et al., 2022) consists of three approaches: zero-shot, which uses only a task description; one-shot, which includes a single example; and few-shot, which provides multiple examples. LLMs leverage these contexts to generate predictions without requiring gradient updates or fine-tuning. For instance, Brown et al. (2020) demonstrates that on the LAMBADA dataset, GPT-3’s accuracy increased from 76.2% in zero-shot to 86.4% in few-shot settings. This leap underscores the value of contextual examples in boosting predictive accuracy. In this study, we apply zero-shot ICL in Experiment A and few-shot ICL in Experiment D to evaluate their effectiveness in different scenarios.

Chain-of-Thought (CoT) prompting (Wei et al., 2022), often implemented as few-shot-CoT, enhances LLM performance by embedding step-by-step reasoning within the prompt. This approach mirrors human problem-solving, excelling in tasks that demand logical or computational rigor. PaLM 540B showed a clear improvement on GSM8K, a benchmark of math word problems, with accuracy rising from 18% using standard few-shot prompting to 57% when given eight CoT examples (Wei et al., 2022). This dramatic improvement highlights CoT’s ability to structure complex reasoning. Drawing on this, our Experiments B through D incorporate code in the prompt to capture the tapping features, adapting CoT’s principles to guide the model systematically.

A simpler variant, zero-shot-CoT (Kojima et al., 2022), elicits reasoning by appending

the phrase “Let’s think step by step” to the prompt. Using the InstructGPT model (text-davinci-002), zero-shot-CoT achieves 40.7% accuracy on the GSM8K benchmark. While this lags behind the 48.7% accuracy of 8-shot CoT, combining the two approaches yields a robust 51.5%, suggesting that hybrid strategies can further refine performance.

Prompt tuning (Lester et al., 2021) optimizes the input by fine-tuning a small set of prompt-specific parameters, typically a soft prompt vector added to the embedding layer, while keeping the LLM’s core parameters fixed. On the GLUE benchmark, this method achieved 84.8% accuracy with just 76.8k parameters, closely rivaling full fine-tuning’s 84.9% with 220M parameters. An advanced version, Decomposed Prompt Tuning (DEPT) (Shi and Lipani, 2023), further refines this by shrinking the soft prompt vector, boosting accuracy to 85.9% with the same parameter count. These results demonstrate prompt tuning’s efficiency in enhancing LLM performance with minimal computational overhead.

Retrieval-Augmented Generation (RAG) (Gao et al., 2023) addresses LLMs’ limitations—such as hallucinations or outdated knowledge—by integrating real-time external data into the generation process. For example, when queried about recent events, RAG retrieves current information from external sources, bypassing the model’s static training data. This adaptability makes RAG ideal for tasks requiring up-to-date or domain-specific insights. In this study, RAG motivate us to include the MDS-UPDRS FT criteria in the prompt of each experiment. We specifically focus on few-shot prompting in Experiment D, examining the effects of providing different scoring features.

The impact of prompt phrasing is evident in a study by Caruccio et al. (2024), which tested three distinct prompt styles—differing in structure and wording—on medical diagnostic tasks. Across models like ChatGPT-3.5, Google BARD, MedBot, and GPT-Neo, ChatGPT-3.5 achieved a peak accuracy of 66%, revealing how subtle prompt variations can significantly influence outcomes. This suggests that tailored prompts are effective to harnessing LLMs for specialized applications like healthcare.

Similarly, Nori et al. (2023) enhanced ChatGPT-4 with Medprompt, a strategy that uses an embedding model to select similar questions for chain-of-thought guidance and shuffles answer options to improve reliability. Outperforming benchmarks like Med-PaLM and Flan-PaLM, Medprompt illustrates how systematic prompt engineering can elevate LLM accuracy in medical contexts without extensive retraining.

1.4 Evaluations of Rater Reliability in PD Assessment

Clinical ratings often show significant variability among human raters (McHugh, 2012), requiring reliable methods to measure and ensure consistency. Rater reliability is essential for clinical assessment and content analysis, forming the foundation for valid data collection and interpretation (Lombard et al., 2002). This section reviews the literature on rater reliability in the FT test for Parkinson’s PD assessment. Various statistical measures have been employed to evaluate both inter-rater reliability (the level of agreement among different raters) and intra-rater reliability (temporal stability of a single rater’s evaluations).

Post et al. (2005) investigated the reliability of UPDRS motor examination ratings, including the FT test. The researchers compared assessments from two nurse practitioners, two neurology residents, and a movement disorder specialist, using a senior movement disorder specialist as the reference standard. The inter-rater reliability of FT was evaluated using weighted Cohen’s κ , with values ranging from 0.71 to 0.86. A subset of raters - one nurse, one neurology resident, and one senior movement disorder specialist - performed follow-up assessments of the same patients after 6-8 weeks to establish intra-rater reliability, with values ranging from 0.76 to 0.87. Notably, neurology residents achieved the highest reliability scores, surpassing even the movement disorder specialists, as shown in Table 1.2.

Heye et al. (2024) explored FT reliability in a video-based study distinguishing two PD motor states: the OFF state (no dopaminergic medication for over 12 hours) and the ON state (after medication). The study analyzed 64 videos from 31 PD patients and 96 videos from 49 healthy controls, including 20 paired ON/OFF assessments. Three raters participated: two who completed the MDS-UPDRS official online training program and one with over 10 years of experience rating the MDS-UPDRS. Inter-rater reliability, assessed using weighted Cohen’s κ , ranged from 0.42 to 0.51. This result is presented in Table 1.2.

Table 1.2 summarizes rater reliability studies employing weighted Cohen’s κ for FT. Post et al. (2005) reported an intra-rater reliability with a score of 0.8 and substantial inter-rater reliability with scores all exceeding 0.7. Notably, neither study specified whether linear or quadratic weighting was applied to the weighted Cohen’s κ calculation.

Kenny et al. (2024) assessed the inter-rater reliability of the finger-tapping test among 20 patients, conducted by 8 clinicians, consisting of two neurologists (one with a special interest in PD; one highly experienced in neurology), two geriatricians (who ran a specialized

Table 1.2: Review of rater reliability using weighted Cohen's κ in PD

Method	Study	Subject	Raters	Type	Reliability
wCK	Post et al. (2005)	50 PD	2 (nurses)	inter-rater	0.72-0.83
wCK	Post et al. (2005)	50 PD	2 (residents)	inter-rater	0.76-0.86
wCK	Post et al. (2005)	50 PD	1 (movement disorder specialists)	inter-rater	0.71-0.76
wCK	Post et al. (2005)	50 PD	1 (nurse)	intra-rater	0.76-0.78
wCK	Post et al. (2005)	50 PD	1 (resident)	intra-rater	0.83-0.87
wCK	Post et al. (2005)	50 PD	1 (movement disorder specialist)	intra-rater	0.78-0.81
wCK	Heye et al. (2024)	64 PD 49 HC	3 (others)	inter-rater	0.42-0.51

PD clinic), three senior geriatric registrars each with a minimum of three months of clinical experience in a PD clinic, and one PD specialist nurse. Initial intraclass correlation coefficient (ICC)(2,1) was 0.32, indicating poor reliability. Following clinical training by reviewing the four high-variance videos and discussing MDS-UPDRS criteria together, six of eight raters (excluding a neurologist and a senior geriatric registrar) evaluated an additional 10 patients after 3-5 weeks, with the ICC improving to about 0.55. This result is presented in Table 1.3. Williams et al. (2023) assessed the inter-rater reliability of 21 neurologists on the MDS-UPDRS FT task. The raters scored 30 videos randomly selected from a pool of 133, achieving moderate agreement with an ICC(2,1) of 0.53 and an ICC(3,1) of 0.58, as shown in Table 1.3.

Islam et al. (2023) collected FT video data from 250 participants, comprising 172 individuals with PD and 78 healthy controls. They evaluated the finger-tapping severity scores assigned by three neurologists. Among the 489 valid videos, full agreement among all three raters was achieved in 30.7% of cases, and at least two raters agreement in 93% of cases. The inter-rater reliability was high, with an ICC of 0.88 and a Krippendorff's alpha of 0.69.

Table 1.3: Review of rater reliability using Intraclass correlation coefficient in PD

Method	Study	Subject	Raters	Type	Reliability	Note
ICC(2,1)	Kenny et al. (2024)	20	8 (clinicians)	inter-rater	0.32	before training
ICC(2,1)	Kenny et al. (2024)	10	6 (clinicians)	inter-rater	0.55	after training
ICC(2,1)	Williams et al. (2023)	133	21 (neurologists)	inter-rater	0.53	
ICC(3,1)	Williams et al. (2023)	133	21 (neurologists)	inter-rater	0.58	
ICC	Islam et al. (2023)	250	3 (neurologists)	inter-rater	0.88	no type

The average pairwise difference between expert ratings was 0.53 points. Models used interpretable features such as speed, amplitude, frequency, period, and acceleration extracted from hand keypoints. Among several models tested, the LightGBM regressor achieved the best performance with a MAE of 0.58 and a Pearson correlation coefficient (PCC) of 0.66.

Table 1.3 summarizes studies assessing rater reliability of the FT test in PD using the ICC. Besides wCK and ICC, researchers have employed additional metrics to evaluate FT rater reliability in PD (Table 1.4). Furthermore, several automatic machine learning models have been developed to assess FT in PD patients, with reliability measured against human raters (Table 1.5). It should be noted that Islam et al. (2023) did not specify the ICC type, limiting precise interpretation of the metric’s assumptions.

Yang et al. (2022) collected 368 videos of left-hand and 243 videos of right-hand FT from patients. Three experienced doctors rated the severity of FT, with an initial average Cohen’s κ score of 0.13. After modifying the evaluation method from counting 10 taps to using a 5-second time interval and adjusting the scoring criteria to cover the entire section, the Cohen’s κ score improved from 0.13 to 0.41. The result is presented in Table 1.4.

Martínez-Martín et al. (1994) evaluated the inter-rater reliability of the finger-tapping test using Fleiss’ κ , obtaining a value of 0.47, which indicates moderate agreement among raters. The result is presented in Table 1.4. The evaluation involved 40 Parkinson’s disease patients

Table 1.4: Review of rater reliability for other metrics in PD. α is the Krippendorff’s alpha.

Method	Study	Subjects	Rater	Type	Reliability	Note
α	Islam et al. (2023)	250	3 (neurologists)	inter-rater	0.69	
Fleiss’ κ	Martínez-Martín et al. (1994)	40	6 (clinicians)	inter-rater	0.47	
Cohen’s κ	Yang et al. (2022)	611	3 (doctors)	inter-rater	0.13	
Cohen’s κ	Yang et al. (2022)	611	3 (doctors)	inter-rater	0.41	Changed criteria from 10 taps to 5 seconds.
Pearson’s r	Islam et al. (2023)	250	3 (neurologists)	inter-rater	0.72	

assessed by six raters, which included three neurologists, two resident physicians specializing in geriatrics, and one unknown.

Park et al. (2021) employed a SVM model to assess the severity of FT in 55 patients. The model’s inter-rater reliability was compared to that of an experienced movement disorder specialist, achieving a wCK of 0.79 and an ICC of 0.93. These results are presented in Table 1.5.

Table 1.5: Reliability review for models compared to human rating in PD. Park et al. (2021) and Islam et al. (2023) didn't report the type of ICC. wCK is weighted Cohen's κ .

Method	Study	Subject	Model	Type	Reliability	Note
ICC	Park et al. (2021)	110 videos	SVM	inter-rater	0.93	no type
ICC	Islam et al. (2023)	172 PD 78 HC	LightGBM	inter-rater	0.76	no type
wCK	Park et al. (2021)	110 videos	SVM	inter-rater	0.79	
Kendall's τ	Islam et al. (2023)	172 PD 78 HC	LightGBM	inter-rater	0.51	
Spearman's ρ	Islam et al. (2023)	172 PD 78 HC	LightGBM	inter-rater	0.64	
Pearson's r	Islam et al. (2023)	172 PD 78 HC	LightGBM	inter-rater	0.66	

Chapter 2

Theory and Methods

2.1 Study Design and Ethical Approval

We conducted a study to evaluate the performance of an automated ChatGPT-4 pipeline in scoring the MDS-UPDRS finger-tapping sub-test, using independent ratings from two neurologists as the reference standard. Data for four MDS-UPDRS hand-motor sub-tests were collected to comprehensively characterize upper-limb motor function in individuals with Parkinson’s disease; the present report analyzes the finger tapping sub-test only. All recordings were obtained at Kaohsiung Medical University Hospital (KMUH, Kaohsiung, Taiwan) between March 2020 and May 2021. The complete protocol (Ashyani et al., 2022)—encompassing the four hand-motor sub-tests (finger tapping, rest tremor, hand movements, postural tremor)—was approved by the National Cheng Kung University Hospital Institutional Review Board (IRB No. B-ER-108-362, December 2019).

The research design included multiple FT recording sessions per subject, each captured under a consistent environmental setup (see Sections 2.4-2.3). Subsequent sections detail the participant recruitment process (Section 2.2), the finger-tapping protocol (Section 2.4), and the analytical framework used to compare LLM-generated scores with those from neurologists.

2.2 Participants

2.2.1 Recruitment

Eighteen individuals with PD were prospectively screened by a movement disorder specialist and invited to participate at KMUH between March 2020 and May 2021. All participants gave written informed consent before participation.

2.2.2 Demographic Profile

The final analytic cohort comprised 16 participants (6 women and 10 men; all Taiwanese). Baseline age—defined as chronological age on the day of the first recording—was 69 years (SD = 10.14; median = 68.5; range = 48–83). Disease severity was mild to moderate, with a mean Hoehn and Yahr (H.Y.) stage of 2.1 (SD = 0.44; median = 2; range = 1–3), based on the modified five-point staging scale that classifies Parkinson’s disease progression according to motor symptoms (Goetz et al., 2004). Age-band and sex distributions appear in Table 2.1.

2.2.3 Session Definition and Sample Size

A *session* was defined as a single smartphone video recording in which one hand (dominant or non-dominant) completed the finger-tapping task on a given day. For example, a participant recorded on two separate days with both hands contributes four sessions (2 days $\times$ 2 hands). A total of 55 sessions were initially collected from the 16 enrolled participants.

2.2.4 Exclusion Criteria

Two participants were excluded due to visible facial features in their recordings, which violated our privacy protocol prohibiting facial capture. Accordingly, the videos were not submitted for neurologist scoring or included in further analysis. From the remaining 16 participants, two sessions were excluded due to persistent occlusion of the hand stickers, which rendered trajectory extraction unreliable (see Section 2.6). The final dataset thus comprised 53 analyzable sessions.

Table 2.1: Demographic Characteristics of Participants ($N=16$). SD denotes standard deviation.

Characteristic	Participants (<i>n</i>)	Percentage (%)
Sex		
Female	6	37.5
Male	10	62.5
Age (years)		
Mean ± SD	69.0 ± 10.14	
Median (range)	68.5 (48–83)	
Female Mean ± SD	64.5 ± 11.63	
Male Mean ± SD	71.7 ± 7.99	
Age Band (years)		
45-49	1	6.3
50-59	2	12.5
60-69	5	31.3
70-79	5	31.3
80-84	3	18.8

2.3 Apparatus and Recording Environment

Finger-tapping tests were recorded with a standardized apparatus detailed in our motor assessment protocol (Ashyani et al., 2022). A green filming box was placed on the table, with internal illumination maintained between 500 and 700 lux to ensure consistent video quality. This lighting range was based on the recommended lighting in examination rooms (Alzubaidi and Soori, 2012). Within the box, four chessboard patterns were affixed to enable three-dimensional calibration, and a tablet displayed visual instructions to guide participants through the task from start to finish.

Three Samsung Galaxy S7 edge smartphones (model SM-G935FD) were mounted on tripods around the filming box to capture hand movements from multiple angles—left, right, and back. Each smartphone recorded video at 240 frames per second and a resolution of 1280×720 , enabling the capture of micromovements imperceptible to the naked eye. Although the protocol included using two OAK-D cameras mounted on tripods to record hand movements, this study analyzed only the smartphone recordings, as the neurologists also based their severity scores on those videos. The complete setup is shown in Figure 2.1.

To support automated tracking, a red sticker was placed on the thumb and a blue sticker on the index finger. The strong chromatic contrast between the stickers, the skin, and the green

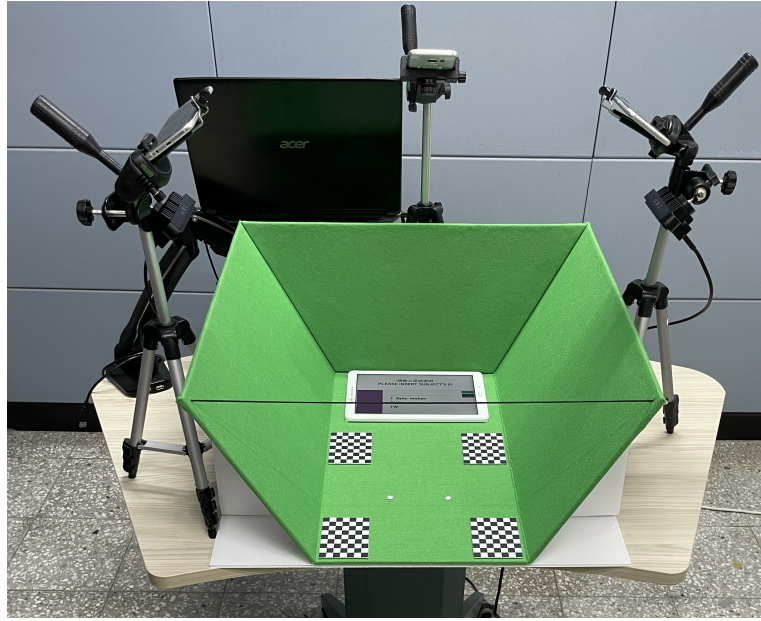


Figure 2.1: Experiment setup used for all tests. It includes a filming box, tripods with smart-phones, chessboards and a tablet.

background facilitated robust frame-by-frame marker extraction (Figure 2.2). Full device specifications and experiment setup are provided in our protocol (Ashyani et al., 2022).

2.4 Finger-Tapping Task Protocol

2.4.1 Task Description

Participants were instructed by on-site researchers to tap the tip of the index finger against the thumb as rapidly and widely as possible for 15 seconds per hand, while keeping the movement within a green filming box for smartphone capture. The MDS-UPDRS clinical training videos demonstrate 10 taps performed in front of the chest for each severity score from 0 to 4 (Goetz et al., 2010); however, our protocol extended the duration to 15 seconds to enable a consistent stop criterion that could be easily implemented in the tablet app. Preliminary testing confirmed that even the slowest participants capable of completing 10 taps could do so within this period. To support stable and natural movement, participants sat comfortably and positioned their hands in a relaxed posture within the filming box. The setup allowed flexibility in hand height and minimized the likelihood of capturing the participant's face during recordings from multiple smartphones.

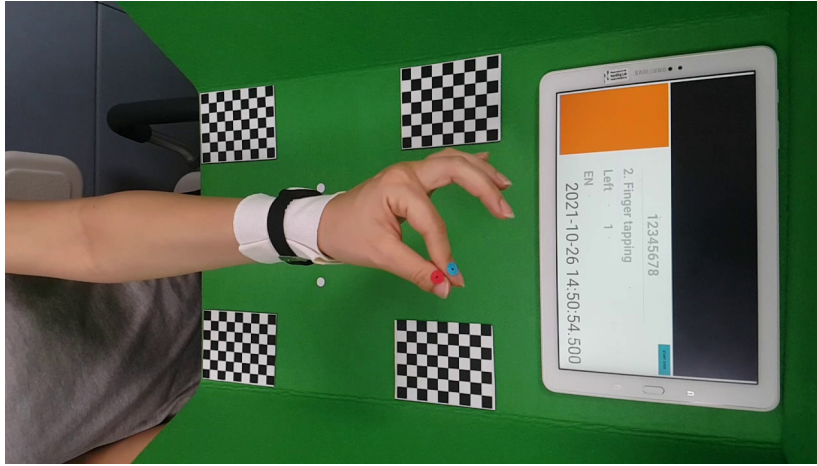
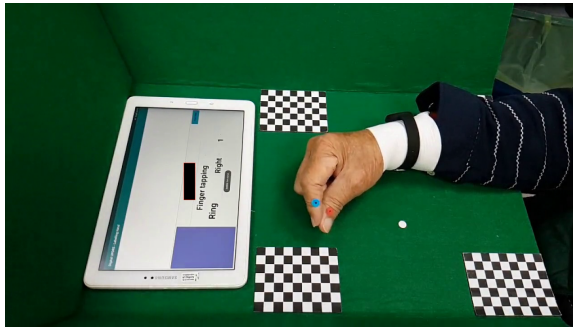
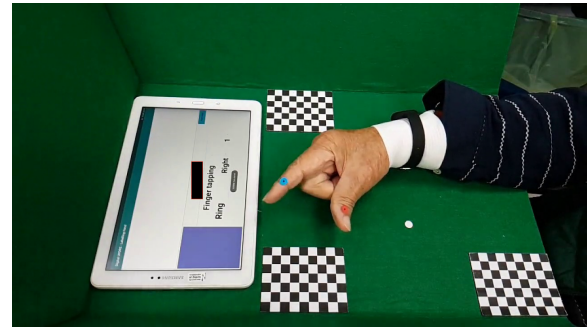


Figure 2.2: Smartphone view of hand movements during the test. The smartphone at the back captures this scene, showing the hand and stickers on the thumb and index finger.



(a) Thumb and index finger in contact (a tap).



(b) Thumb and index finger fully separated (peak aperture).

Figure 2.3: Demonstration of the finger-tapping task following MDS-UPDRS instructions. (a) A tap is defined by contact between the thumb and index finger. (b) Peak aperture corresponds to the maximal separation of the two fingers.

The finger-tapping task emphasized continuous motion to capture variations in speed and amplitude, enabling the detection of interruptions, movement slowing, and amplitude decrement. Representative frames illustrating a tap (thumb–index contact) and the subsequent peak aperture are shown in Figure 2.3.

2.4.2 Researcher Checklist

To ensure consistency and data quality, the researchers adhered to a standardized procedure based on MDS-UPDRS FT criteria throughout the entire data collection process, as detailed in our protocol (Ashyani et al., 2022). Researchers demonstrated the movement beforehand to ensure full task comprehension. Any trial with compromised data quality—such as marker

detachment, poor posture, or procedural errors—was repeated to preserve dataset integrity.

2.4.3 Examiner Checklist (MDS-UPDRS Finger-Tapping criteria)

When scoring the FT tests, both neurologists in our study adhered to the MDS-UPDRS criteria, which evaluate speed, amplitude, hesitations, halts, and amplitude decrement across the initial ten taps. To ensure that ChatGPT-4 interpreted and applied the correct and complete version of the MDS-UPDRS FT criteria, we included the full scoring guidelines directly within the prompt text (see Section 2.7). The official scoring criteria, reproduced from Goetz et al. (2008), is presented below:

Instructions to the examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

0: Normal: No problems.

1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps.

2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence.

3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap.

4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.

2.5 Neurologist Evaluations

Two movement disorder specialists were recruited to serve as independent raters: one from KMH and the other from NCKUH. For anonymity, they are referred to as Neurologist X and Neurologist Y throughout this thesis.

Table 2.2: Distribution of concordant MDS-UPDRS scores in the 22 concordant sessions.

Score	Count	Percent
0	3	13.6
1	13	59.1
2	5	22.7
3	1	4.5
4	0	0

Each neurologist independently reviewed all video recordings without discussion and was blinded to the other’s ratings. To facilitate the evaluation process, we organized all video recordings in a hierarchical folder structure by subject and test date. Each folder contained videos: three smartphone views (left, back, right) for each hand.

Both neurologists rated the recordings in accordance with the official MDS-UPDRS criteria, scoring based on the first 10 finger taps of each finger-tapping test. Neurologist X adhered to the standard MDS-UPDRS protocol. Neurologist Y, in addition, provided specific procedural details: evaluations were completed over 2–3 partial days, and scoring began at the initiation of continuous tapping motion. He used only visual information (audio muted) and initially selected one smartphone view per test; if that view lacked sufficient clarity, he switched to an alternative angle to ensure accurate assessment.

2.5.1 Concordant Sessions

Out of the 53 total recorded sessions, both neurologists independently assigned identical MDS-UPDRS finger-tapping scores in 22 sessions. These are defined as *concordant sessions*, with the corresponding values referred to as *concordant scores*. This terminology is consistent with prior literature from the Movement Disorder Society, where “concordant scores” refer to cases in which independent raters assign the same score without discussion (Goetz et al., 2010).

Table 2.2 summarizes the distribution of concordant scores across the 22 sessions. These sessions serve as a high-confidence subset when evaluating LLM performance against a stringent ground truth.

2.6 Video Pre-processing and Trajectory Extraction

2.6.1 Temporal Segmentation Strategy

The MDS-UPDRS instructions emphasise the first 10 taps. To ensure sufficient temporal context, we also evaluated fixed-duration segments of 10 seconds to determine which approach yields more reliable severity estimates. Each recording was analyzed twice: (1) a fixed-count segment containing the first 10 taps; and (2) a fixed-duration segment comprising 10 seconds of continuous finger-tapping motion.

The finger-tapping motion onset was defined as the frame at which continuous tapping began, consistent with Neurologist Y’s approach (see Section 2.5). From this point, we extracted both the first 10 taps and a 10-second segment.

To characterize the duration of fixed-count segments, Figure 2.4 shows the distribution of time intervals between the first and tenth tap across 53 sessions (mean: 5.1 s; median: 4.9 s). In parallel, the 10-second segment began at a time point ranging from 2.9 to 9.4 seconds after motion onset (mean: 4.0 s; median: 3.7 s). The segment start was defined by detecting the onset of a blinking signal on the tablet screen, which appeared after the on-site researcher confirmed correct task execution and pressed “start.” This blinking cue was used as a consistent and automatically detectable temporal marker across sessions. Figure 2.5 illustrates the distribution of these start times.

For each session, the trajectory of the thumb–index finger distance is plotted in Appendix J, annotated with the 10-second window and the completion of the first 10 taps. Both segments were independently processed by the ChatGPT-4 scoring pipeline (Section 2.7). Agreement and reliability metrics were then computed against neurologist scores to evaluate the performance of each segmentation strategy.

2.6.2 CoTracker feature tracking

We began with 55 video sessions and first identified, for each, the smartphone view that retained unobstructed visibility of both stickers throughout the required 10-s interval and the initial 10 taps. Two sessions failed this quality screen because sustained occlusion prevented reliable point tracking; these were excluded, leaving 53 sessions for downstream analysis.

Prior studies (Li et al., 2021, 2022; Yang et al., 2022) utilized pose estimation tools like

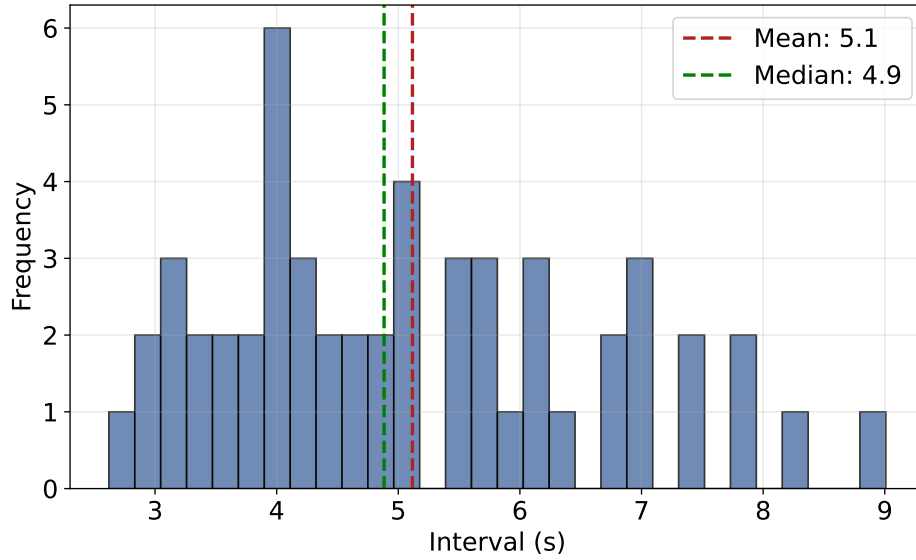


Figure 2.4: Distribution of the time interval between the first and tenth tap across 53 sessions. Red and green dashed lines indicate the mean (5.1 s) and median (4.9 s), respectively.

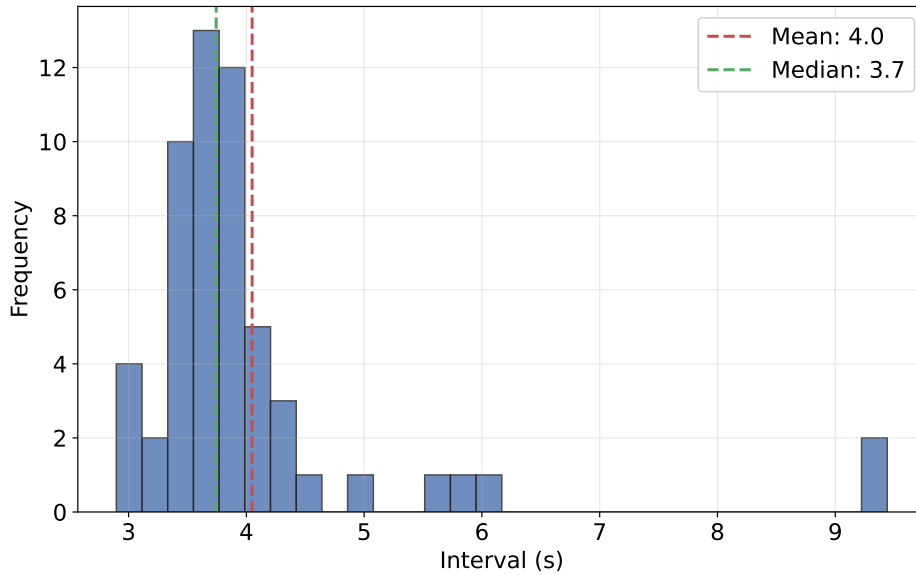


Figure 2.5: Distribution of elapsed time between tapping onset and the start of the analysed 10-s segment (53 sessions). Red and green dashed lines indicate the mean (4.0 s) and median (3.7 s), respectively.

OpenPose (Cao et al., 2017), Mediapipe (Lugaresi et al., 2019), and MMPose (Sengupta et al., 2020) to obtain coordinates of thumb and index finger joints from video data. In this study, to track the point on the stickers attached to the thumb and index finger from videos, we employed CoTracker, a state-of-the-art transformer-based feature tracking model developed by Meta AI in 2023 (Karaev et al., 2023). CoTracker is designed to track a large number of

2D points in video sequences. Unlike traditional methods that track points independently, CoTracker uses a joint tracking approach to account for dependencies among points. This results in improved accuracy, robustness, and the ability to track occluded points or points outside the smartphone’s view.

Shi and Nordling (2024) integrated CoTracker with Deep Feature Encoding (Chang and Nordling, 2021) to enhance tracking accuracy, achieving a minimal pixel error of 0.82, an improvement over using CoTracker alone, which had a 1.16 pixel error on a hand mole. We used CoTracker alone because our study began prior to the availability of the integrated method. The tracking example is shown in Figure 2.6.

For our processing, we used an NVIDIA GeForce RTX 3090 SUPRIM X 24G GPU. Due to GPU memory constraints, videos were cropped to 360×360 pixels, centered approximately at the midpoint between the two stickers in the initial frame. To ensure that both stickers remained within view throughout the video, we visually inspected each cropped video before tracking. If either sticker moved too close to the cropping boundary, we manually regenerated the video with adjusting the cropping center to maintain complete visibility of both stickers across all frames.

We labeled the initial positions of the thumb and index finger stickers in the first frame and provided these coordinates, along with the cropped video, to the CoTracker model, which predicted the sticker positions across all frames. This yielded the trajectories of the two stickers throughout the video.

Despite CoTracker’s robustness, two videos encountered tracking issues due to occlusions, affecting several frames. Ultimately, we obtained reliable sticker trajectories for 53 of the 55 sessions, which formed the dataset used for subsequent analysis and evaluation of finger-tapping severity based on the MDS-UPDRS item 3.4 criteria using ChatGPT-4.

2.7 ChatGPT-Based Scoring Pipeline

2.7.1 Model Description

In this study, we employ the gpt-4-turbo-2024-04-09 version of ChatGPT-4, the most recent iteration in the GPT series developed by OpenAI when starting the evaluation. This sophisticated language model is built on the Transformer architecture (Vaswani et al., 2017)

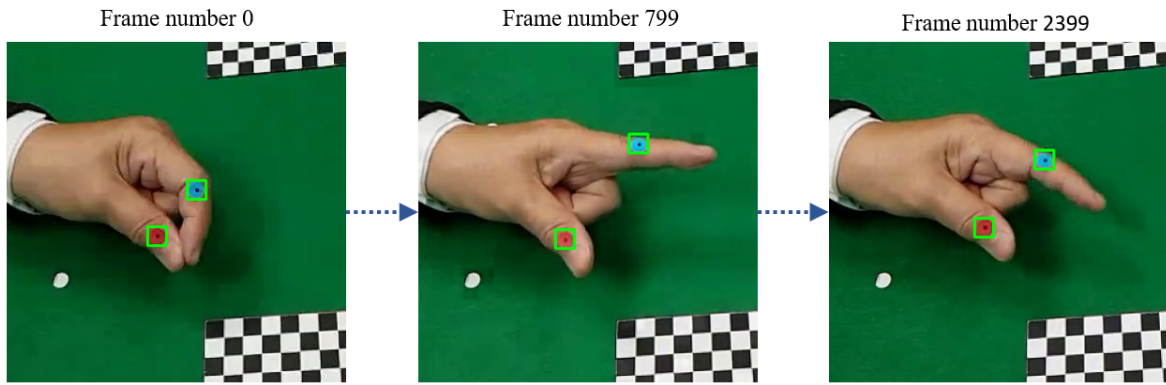


Figure 2.6: Sticker tracking through frames using the CoTracker method. This figure demonstrates the sticker’s trajectory across three frames, frame number 0, frame number 799, and frame number 2399.

and undergoes an initial training phase focused on next-token prediction. This architecture eliminates recurrent layers in favor of multiple layers of self-attention mechanisms and feed-forward networks, enabling parallel data processing during training. This shift not only significantly reduces the time required compared to traditional models reliant on sequential data processing but also enhances accuracy and model performance. The ability of the Transformer to efficiently handle and interpret sequential data enables GPT-4 to attend to all parts of the input regardless of their positions, enhancing performance in complex tasks like machine translation and other sequence-to-sequence problems.

GPT-4 was trained using a large-scale collection of publicly sourced and licensed datasets, enabling it to build strong general language capabilities (OpenAI, 2023). Subsequently, the model was subjected to Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017) to refine its outputs and align them with desired behaviors. This extensive pre-training empowers the model to adapt its responses according to the context, a capability vital for diverse natural language processing (NLP) tasks such as conversational AI, question answering, and technical explanations. As detailed in OpenAI (2023), GPT-4 not only performs better than its predecessor, GPT-3.5, but also outperforms other state-of-the-art systems across a variety of languages and benchmarks. For example, on simulated exams and traditional NLP benchmarks, GPT-4 exceeds performance metrics of previous models and most current systems, which are often tailored specifically to those benchmarks.

In the medical domain, GPT-4 has shown promise in analyzing and managing medical data, supporting clinical diagnoses, managing medical records, and identifying candidates for

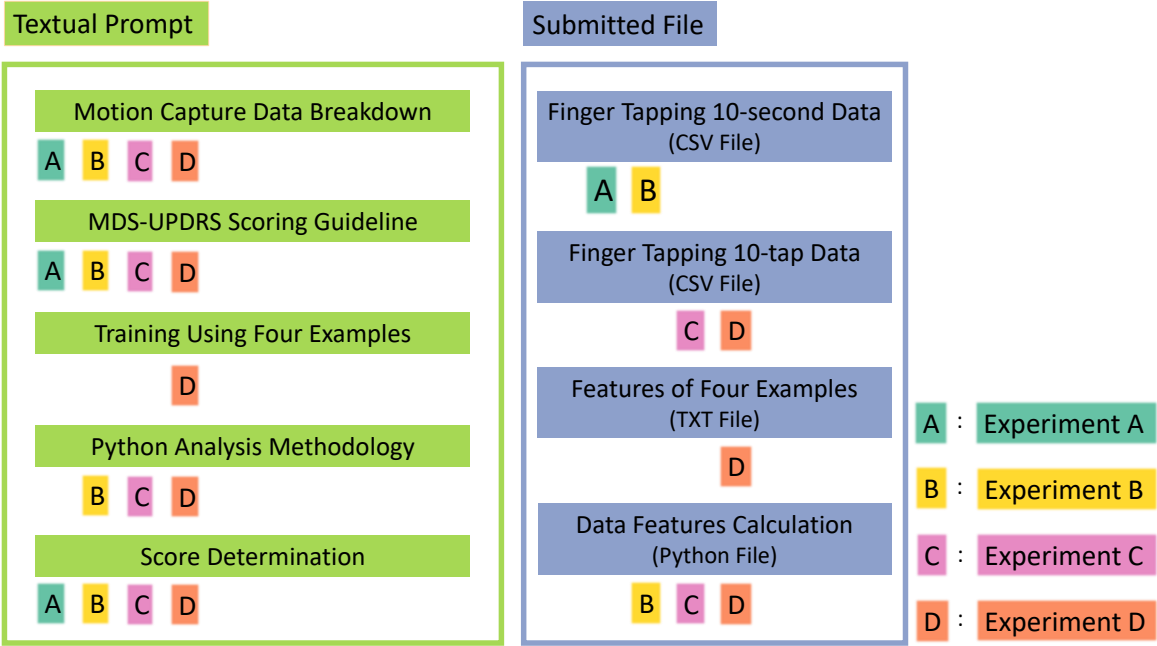


Figure 2.7: Visualization of prompt elements across different experiments. A, B, C, and D in the figure correspond to the prompts used in Experiment A, Experiment B, Experiment C, and Experiment D, respectively. In Prompt D, the paragraph “Python Analysis Methodology” includes not only instructions for getting tapping features of target data but also a directive to compare the features of the target data with the example score features provided in a TXT file.

clinical trials (Dave et al., 2023). Overall, GPT-4’s ability to integrate and interpret complex medical guidelines and patient data makes it interesting to study how it can assist in the objective and consistent scoring of Parkinson’s disease symptoms, potentially facilitating the refinement of diagnostic processes. In this study, we harnessed the advanced language understanding and generation capabilities of GPT-4 to analyze FT data based on the MDS-UPDRS criteria and score descriptions.

2.7.2 Prompt Engineering Strategy

We conducted four distinct experiments: Experiment A-D, each utilizing different prompts: Prompt A-D, respectively, to evaluate the performance of ChatGPT-4 across 53 sessions. The prompts are illustrated in Figure 2.7 and Table 2.3, 2.4, 2.5, 2.6, and 2.7. The CSV, TXT, and Python files with all prompts and answers from the interactions are in Figshare DOI: (10.6084/m9.figshare.25735248).

Different experiments involve different prompt elements, which are visualized in Figure

2.7, and describe as follows:

1. Motion Capture Data Breakdown: Explanation of the data format and data collection method, as shown in Table 2.3.
2. MDS-UPDRS Scoring Guideline: The scoring criteria and instructions for the MDS-UPDRS FT, including evaluation factors (speed, amplitude, hesitations, halts, and decrementing amplitude), as shown in Table 2.4.
3. Training Using Four Examples: Instructions for analyzing the example score features in a TXT file, as shown in Table 2.5. These features are crucial for identifying the characteristics associated with each scoring level. This element is only used in Prompt D.
4. Python Analysis Methodology: Instructions for ChatGPT-4 to analyze the features of the FT data in Python, as shown in Table 2.6. In Prompt D, it further instructs to compare the features of the target FT data with the example score features.
5. Score Determination: Instructions for ChatGPT-4 to analyze the FT data thoroughly and assign a score based on the MDS-UPDRS criteria, with the score printed as a single number after the detailed motivation, as shown in Table 2.7. In Prompt D, it further instructs to correlate the features of the target FT data with the example score features for scoring.
6. FT 10-second Data (CSV File): A CSV file containing the 2D trajectory of stickers attached to the thumb and index finger, with columns for frame number, thumb coordinates (x, y), and index finger coordinates (x, y). The data spanned 10 seconds at 240 frames per second.
7. FT 10-tap Data (CSV File): A CSV file containing the 2D trajectory of stickers attached to the thumb and index finger, with columns for frame number, thumb coordinates (x, y), and index finger coordinates (x, y) for 10 individual taps.
8. Data Features Calculation (Python File): A method to calculate the distance between the two stickers (thumb and index finger), as well as peak values and tap intervals.

9. Features of Four Examples (TXT File): Contains features of peak values and tap intervals from four reference data, each representing a different score level (0 to 3) for MDS-UPDRS FT.

Table 2.3: Motion Capture Data Breakdown. This element explains the data format and data collection method. The key difference between the prompts is that the descriptions in Prompts A and B correspond to 10 seconds of FT data, while the descriptions in Prompts C and D correspond to 10 taps of FT data. The bold text highlights the difference across the prompts.

Prompt	Motion Capture Data Breakdown
Prompt A	The provided data consists of five columns: frame, x_thumb, y_thumb, x_index, and y_index. The first column represents the frame number. The second and third columns represent the x and y coordinates, respectively, of the thumb sticker. The fourth and fifth columns represent the x and y coordinates, respectively, of the index finger sticker. This data comes from a subject performing the MDS-UPDRS finger tapping test, with stickers fixed to the thumb and index finger. The subject was instructed to tap the index finger on the thumb to perform continuous finger tapping, and a 10-second segment from this activity was selected for analysis. Cameras recording at a frame rate of 240 frames per second (fps) captured the movement, employing a tracking technique to record the x and y coordinates of the stickers on the thumb and index finger in each frame throughout the 10-second interval.

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Table 2.3 – *Continued from previous page*

Prompt	Motion Capture Data Breakdown
Prompt B	The provided CSV file contains data with five columns: frame, X_thumb, Y_thumb, X_index, and Y_index. The first column represents the frame number. The second and third columns represent the x and y coordinates of the thumb sticker, respectively. The fourth and fifth columns represent the x and y coordinates of the index finger sticker, respectively. This data comes from a subject performing the MDS-UPDRS finger tapping test, with stickers fixed to the thumb and index finger. The subject was instructed to tap the index finger on the thumb to perform continuous finger tapping. A 10-second segment from this activity was selected for analysis. Cameras recording at a frame rate of 240 frames per second (fps) captured the movement, employing a tracking technique to record the x and y coordinates of the stickers on the thumb and index finger in each frame throughout the 10-second interval.
Prompt C	The provided CSV file contains data with five columns: frame, X_thumb, Y_thumb, X_index, and Y_index. The first column represents the frame number. The second and third columns represent the x and y coordinates of the thumb sticker, respectively. The fourth and fifth columns represent the x and y coordinates of the index finger sticker, respectively. This data comes from a subject performing the MDS-UPDRS finger tapping test, with stickers fixed to the thumb and index finger. The subject was instructed to tap the index finger on the thumb 10 times to perform continuous finger tapping. Cameras recording at a frame rate of 240 frames per second (fps) captured the movement.

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Table 2.3 – *Continued from previous page*

Prompt	Motion Capture Data Breakdown
Prompt D	The provided CSV file contain target data with five columns: frame, X_thumb, Y_thumb, X_index, and Y_index. The first column represents the frame number. The second and third columns represent the x and y coordinates of the thumb sticker, respectively. The fourth and fifth columns represent the x and y coordinates of the index finger sticker, respectively. This target data comes from a subject performing the MDS-UPDRS finger tapping test, with stickers fixed to the thumb and index finger. The subject was instructed to tap the index finger on the thumb 10 times to perform continuous finger tapping. Cameras recording at a frame rate of 240 frames per second (fps) captured the movement.

Table 2.4: MDS-UPDRS Scoring Guideline. The scoring criteria in all prompts are the same, with the focus on assessing the patient’s performance according to MDS-UPDRS FT. The bold text highlights the difference across the prompts.

Prompt	MDS-UPDRS Scoring Guideline
Prompt A and Prompt B	The objective is to score the subject in accordance with MDS-UPDRS item 3.4: FINGER TAPPING based on the selected 10 seconds of continuous finger tapping. The instruction and scoring criteria of MDS-UPDRS item 3.4: FINGER TAPPING are as follows: “Instructions to the examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude. 0: Normal: No problems. 1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps. 2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence. 3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap. 4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.”

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Table 2.4 – Continued from previous page

Prompt	MDS-UPDRS Scoring Guideline
Prompt C	The objective is to score the subject in accordance with MDS-UPDRS item 3.4: FINGER TAPPING. The instruction and scoring criteria of MDS-UPDRS item 3.4: FINGER TAPPING are as follows: “Instructions to the examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude. 0: Normal: No problems. 1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps. 2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence. 3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap. 4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.”

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Table 2.4 – Continued from previous page

Prompt	MDS-UPDRS Scoring Guideline
Prompt D	The objective is to score the target subject in accordance with MDS-UPDRS item 3.4: FINGER TAPPING. The instruction and scoring criteria of MDS-UPDRS item 3.4: FINGER TAPPING are as follows: “Instructions to the examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude. 0: Normal: No problems. 1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps. 2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence. 3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap. 4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.”

Table 2.5: Training Using Four Examples. This element is used only in Prompt D to describe the submitted TXT file.

Prompt	Training Using Four Examples
Prompt A, Prompt B, and Prompt C	None
Prompt D	The provided TXT file contains features from four reference data, each representing a different score level (0 to 3) for MDS-UPDRS item 3.4: FINGER TAPPING. These reference data, scored by medical professionals as accurate representations. They are crucial for identifying the characteristics associated with each scoring level.

Table 2.6: Python Analysis Methodology. This element is not used in Prompt A. Prompts B and C describe the use of a Python file, while Prompt D provides instructions for comparing the features of the target data, generated by the Python file, with those in the TXT file. The bold text highlights the difference across the prompts.

Prompt	Python Analysis Methodology
Prompt A	None
Prompt B and Prompt C	The provided Python file includes a method for calculating the number of taps, intervals between taps, and the amplitudes. Use this method for the analysis.
Prompt D	The provided Python script includes a method for calculating the number of taps, intervals between taps, and the amplitudes. Execute the entire script as provided, without any simplifications or modifications. Print out the values of features of the target data and each reference data to align the target data's performance with the reference standards. Specifically concentrate on the occurrences of interruptions, variations in intervals that indicate slowing, and decrements in amplitude, to assess the correlation between these specific characteristics and the established scoring levels.

Table 2.7: Score Determination. The key difference in this element across prompts is that Prompt D emphasizes that the score of the target data should correlate with the example features. The bold text highlights the difference across the prompts.

Prompt	Score Determination
Prompt A	To determine the subject’s score, analyze the provided data thoroughly and provide a detailed motivation for the assigned score based on the MDS-UPDRS finger tapping instruction. Print the score using only one number on the last line of your answer, after the detailed motivation.
Prompt B and Prompt C	To determine the subject’s score, analyze the provided data thoroughly and give a detailed motivation for the assigned score based on the MDS-UPDRS finger tapping instruction. Print the score using only one number on the last line of your answer, after the detailed motivation.
Prompt D	Determine the score for the target data file based on this correlation and alignment. Provide a detailed motivation for the assigned score, and print the score using only one number on the last line of your answer, after the detailed motivation.

2.7.3 Experiment A (Exp. A)

Exp. A is the first experiment we conducted, designed to assess the performance of 10 evaluations across 53 sessions. Each evaluation consists of an interaction with ChatGPT-4 in a new chat using Prompt A. Prompt A includes the submitted 10-second stickers position data from the FT test, along with a textual description explaining the data format, data collection method, MDS-UPDRS FT criteria, and final instructions for ChatGPT-4 to analyze the data and assign a score based on the MDS-UPDRS FT criteria. The elements of Prompt A are illustrated in Figure 2.7, and the detailed description of each element is provided in Table 2.3, 2.4, 2.5, 2.6, and 2.7. This experiment was conducted between May 4, 2024, and August 11, 2024.

2.7.4 Experiment B (Exp. B)

In Exp. A, ChatGPT-4 encountered difficulties in accurately calculating the intervals between taps and peak amplitudes from the FT data. These features are critical for assessing FT severity, as defined in MDS-UPDRS FT. Specifically, it evaluates: interruptions (seen as irregular amplitudes and tap intervals), slowing (manifested as slow speed, or gradual decrease in tap-

ping speed), and amplitude decrements (a reduction in peak amplitudes).

To address these challenges, we refined Prompt B by incorporating the following improvements: (1) We submitted an additional Python file containing methods specifically designed to calculate the number of taps, the intervals between taps, and the peak amplitudes from the FT data. This script is essential for accurately analyzing FT data in line with the MDS-UPDRS FT. (2) In the prompt element “Python Analysis Methodology” (as referenced in Table 2.6), we included a detailed textual description of the Python methods used for data analysis. This clarification ensures that the methodology for calculating the required features is clearly understood and can be properly executed.

We also made the following additional modifications: (1) Prompt element “Motion Capture Data Breakdown” (shown in Table 2.3) was also updated. The textual description of the column names was revised from “frame, x_thumb, y_thumb, x_index, and y_index” to “frame, X_thumb, Y_thumb, X_index, and Y_index”. This change ensures consistency with the actual column names in the CSV file, thereby eliminating confusion when parsing the data. While ChatGPT-4 would likely resolve this discrepancy by refining the Python script on its own, this ensures it does not impact the analysis. (2) Minor adjustments were made to the phrasing in the textual descriptions of elements “Motion Capture Data Breakdown” and “Score Determination”. For example, the phrase “provide a detailed motivation” was changed to “give a detailed motivation.” These changes are purely semantic and we do not expect them to impact the analysis.

With these modifications, ChatGPT-4 can now run the submitted Python file and get the correct taps and amplitude values. This refined analysis provides measures for evaluating the criteria for MDS-UPDRS FT, enabling a more precise evaluation of interruptions, slowing, and amplitude decrements in the FT task. This experiment was conducted between May 15, 2024, and August 18, 2024.

2.7.5 Experiment C (Exp. C)

To align with the MDS-UPDRS FT instruction, which considers only 10 taps, in Exp. C, we focus on the first 10 taps of the FT movement seen in our videos as our data, which was then submitted to ChatGPT-4. As part of these changes, we modified the text in Prompt B to reflect the shift from 10-second data to 10-tap data, resulting in the updated version, Prompt

C. This experiment was conducted between August 1, 2024, and August 17, 2024.

2.7.6 Experiment D (Exp. D)

The scoring criteria for MDS-UPDRS FT lacks clear definitions for the following aspects: (1) Amplitude Decrement: What constitutes a decrease in amplitude during the tapping sequence? (2) Tapping Interval: How much slowing is “slight” or “mild” slowing? (3) Interruptions: How should interruptions be defined? Given these ambiguities, ChatGPT-4 assigns scores based on its own medical knowledge, categorizing them into the following levels: Normal, Slight, Mild, Moderate, and Severe.

To address on these ambiguities, we implemented Few-Shot Prompting in Exp. D to guide the model in recognizing the characteristics of various scores. ChatGPT-4 was provided with a single TXT file containing features of tap intervals and amplitude values extracted from example sessions representing different score groups.

The standard MDS-UPDRS training videos were not used as few-shot examples, as preliminary tests indicated that CoTracker failed to reliably track finger tapping points across all severity levels (see Appendix H). To ensure the consistency and quality of visual features, we instead used internally recorded sessions with successful tracking.

The distribution of the sessions with concordant score groups is as follows: three sessions scored 0, thirteen sessions scored 1, five sessions scored 2, one session scored 3, and no sessions scored 4 (only one session was scored as 4 by a single doctor). For our examples, we selected features from four sessions corresponding to scores 0-3: session 2 (score 0), session 17 (score 1), session 46 (score 2), and session 52 (score 3).

Criteria for selecting these specific example sessions are outlined below:

- **Step 1:** Exclude the 17 sessions influenced by instructor interference during the assessment.
- **Step 2:** Exclude the two sessions with one interruption marked by the Neurologist Y, since the rare presence of these interruptions may not be accurately interpreted by the model and result in the example being misleading.
- **Step 3:** Identify sessions where both doctors assigned the same score. Group these 22 sessions into five categories based on their scores: 0, 1, 2, and 3.

- **Step 4:** Within each score group, order the sessions based on the true positive rate of 10 ChatGPT-4 prediction scores from Exp. C, from lowest to highest.
- **Step 5:** For sessions with identical true positive rates, rank them by the deviation of ChatGPT-4 prediction scores from the actual scores, from largest to smallest.
- **Step 6:** Select the session with the lowest true positive rate as the example from each group. If multiple sessions meet this criterion, choose the one with the greatest deviation per **Step 5**.

In Exp. D, we use the same 10-tap data as in Exp. C. Key updates in Prompt D compared to Prompt C: (1) We submit the TXT file with different score features to ChatGPT-4. Features including number of taps, tap intervals, and amplitude values. (2) In prompt element “Motion Capture Data Breakdown” (as referenced in Table 2.3) and “MDS-UPDRS Scoring Guideline” (as referenced in Table 2.4), we clarify the concept of “target data” to ensure that ChatGPT-4 understands the data collection method used for the target data in the CSV file, distinguishing it from the example features provided in the TXT file. (3) A new prompt element “Training Using Four Examples” (as referenced in Table 2.5), has been introduced to clarify that the reference data features are accurate representations of the corresponding MDS-UPDRS scores. This step enhances the model’s understanding of the reference data’s reliability. (4) The prompt element “Python Analysis Methodology” (as referenced in Table 2.6) has been updated to include specific instructions for comparing the features of the target data (from the Python output) against the reference features in the TXT file. Particular attention is given to identifying occurrences of interruptions, variations in intervals indicating slowing, and amplitude decrements, which are key to assessing the correlation with established scoring levels. (5) In the prompt element “Score Determination” (as referenced in Table 2.7), we highlight to correlate and align the target data’s features with the reference features in the TXT file for determining a final MDS-UPDRS score.

To evaluate whether ChatGPT-4 can accurately understand the correlation between the features of the target data and the example features in each score group, we conducted a test. Specifically, we tested the four example sessions, each corresponding to a different score, by submitting each session’s data individually as a target file along with Prompt D. Each target file was evaluated 10 times to assess the consistency of ChatGPT-4’s predictions. The goal

was to determine whether ChatGPT-4 could correctly identify the score associated with each target file.

The results, as shown in Figure 3.2 (denoted in orange), demonstrate that ChatGPT-4 achieved perfect accuracy (10 out of 10 correct predictions) for sessions with scores 0, 1, and 3, while correctly predicting 8 out of 10 evaluations for sessions with score 2. These high prediction accuracy rates suggest that ChatGPT-4 is able to recognize the correlation between the features of the target data and the corresponding example features.

Actually, our original approach was to submit the reference data as four separate CSV files along with the target data to ChatGPT-4, with the intention of having it run a Python script to generate distance figures and analyze the features of both the target and reference data. However, this approach quickly encountered the usage limitations of ChatGPT-4, which resulted in interruptions after approximately 10 evaluations, requiring a wait time of three hours before further interaction could occur despite using the paid ChatGPT subscription. To address this issue, we revised the format of the reference data, changing it from the original 10-tap CSV format to a simplified TXT file containing only the features of the tapping. In the updated Python script, we instructed ChatGPT-4 to process only the target data, eliminating the need to generate distance figures. This modification helped reduce the occurrences of hitting the ChatGPT-4 usage limitation. This experiment was conducted from August 4, 2024, to October 16, 2024.

2.7.7 Screening of Output from ChatGPT-4

The experiments provided different types of information input to ChatGPT-4. Exp. A utilized MDS-UPDRS scoring criteria, while Exp. B through D included instructions for identifying tap and peak features using a Python script. Additionally, Exp. D incorporated reference features for correlation, provided in a TXT file.

For Exp. B–D, we screened ChatGPT-4’s outputs based on three criteria: correct analysis of target data, proper handling of reference features, and successful execution of the Python script. We excluded results when the output failed to meet any of these criteria or did not clearly list the features of the data and align them with the MDS-UPDRS criteria.

Across all experiments (A–D), we also excluded cases where ChatGPT-4 returned “NaN” scores, which occurred when it either could not analyze the data or had difficulty distinguish-

ing between two possible scores. The total number of exclusions remained below 15 cases per experiment (see Appendix E for example).

In Exp. D, four example session features were used as reference data for correlation analysis. To evaluate how well ChatGPT-4 generalizes to other sessions, we divided the experiments into two categories: (1) those including the example sessions (Exp. A to F), and (2) those excluding them (Exp. A-EE to F-EE). This design enables us to assess ChatGPT-4’s performance specifically on the remaining sessions in Exp. D—those not used as examples—and to compare it with its performance in the other experiments (A-EE to F-EE), which likewise exclude these reference sessions.

2.7.8 Inference Procedure

For each experiment, 10 independent evaluations were conducted per session, with each evaluation defined as a separate interaction with ChatGPT-4 that assigned a score ranging from 0 to 4. Given the stochastic nature of ChatGPT’s text generation, where responses are sampled from probabilistic distributions informed by its training data, multiple evaluations were performed to account for potential variability.

2.8 Baseline Scoring Conditions

2.8.1 Constant-Score Baseline (Exp. E)

Due to the class imbalance in the dataset seen based on the neurologist ratings: with more than half being rated as 1 in the concordant sessions, we designed Exp. E where all scores were always set to 1. The purpose of this approach is to compare the outcomes of ChatGPT-4 to this trivial estimator yielding the constant score of 1. This is a biased estimator. This comparison help us determine if ChatGPT-4 can outperform it or not.

2.8.2 Random-Score Baseline (Exp. F)

To assess whether ChatGPT-4 performs better than chance, we implemented a Monte Carlo random-score baseline (Exp. F). For each of the 53 sessions, we independently sampled ten scores from a discrete uniform distribution over the integers $\{0, 1, 2, 3, 4\}$, repeating this

process 1000 times to generate 1000 random rating matrices of size 53×10 . Each matrix was evaluated using the performance metrics defined in Section 2.10.

To quantify statistical uncertainty, we computed 95% confidence intervals (CIs) by extracting the 2.5th and 97.5th percentiles from the empirical distribution of metric values across the 1000 replicates. If ChatGPT-4’s performance does not significantly exceed this random baseline, it would suggest that the model does not reliably assess finger-tapping severity.

2.9 Statistical Analysis

2.9.1 Bootstrap Resampling

To estimate statistical uncertainty in Exp. A-E, we applied the bootstrap resampling method due to the limited sample size. This approach involved generating 1,000 bootstrap samples by sampling with replacement from the original scores, with each sample matching the original sample size. A 95% CI was then calculated using the 2.5th and 97.5th percentiles of the sorted bootstrap distribution.

In this study, we employed the bootstrap method over the Jackknife, as it enables the generation of a larger number of resamples and yields more stable confidence interval estimates under limited data conditions. By comparison, the Jackknife produces a fixed number of resamples equal to the sample size through systematic leave-one-out sampling.

We adopted two distinct bootstrap strategies to compute CIs for intra-rater reliability: (1) session-level resampling, in which entire sessions were sampled with replacement to account for uncertainty due to the limited number of subjects; and (2) evaluation-level resampling, in which the 10 repeated evaluations within each session were resampled to capture within-session variability.

2.9.2 Kolmogorov-Smirnov Test

We evaluated the distributions of Wasserstein distances (Section 2.10.4) between ChatGPT-4 evaluation scores and neurologist ratings across all experimental conditions. To assess whether any pair of experiments yields significantly different Wasserstein distance distri-

butions, we applied the two-sample Kolmogorov-Smirnov (KS) test. The KS test is non-parametric: it compares the empirical cumulative distribution functions (eCDFs) of two independent samples and reports the largest vertical distance between them, together with a two-sided p -value derived from the exact (small n) or asymptotic (large n) null distribution. Because it is sensitive to differences in location, dispersion, and overall shape, the KS test is well suited to the skewed, heavy-tailed Wasserstein-distance data analysed here. Throughout, we label two experiments as statistically distinguishable when $p < 0.05$.

2.10 Outcome Measures

2.10.1 Error Measures

Statistical measures were employed to evaluate the model's prediction accuracy across n predictions, comparing median predicted values (calculated from 10 evaluations per session, $\hat{y}_i$) against actual observations (y_i). The Mean Absolute Error (MAE) quantifies the average magnitude of prediction errors

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (2.1)$$

where $\hat{y}_i$ represents the median of 10 evaluations per session, and n denotes the session numbers. The Root Mean Square Error (RMSE) measures the standard deviation of prediction residuals

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (2.2)$$

RMSE's quadratic nature emphasizes larger deviations, making it valuable for assessing outlier impact.

The Standard Deviation (SD) quantifies each session prediction consistency by measuring error spread

$$\text{SD} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (e_i - \bar{e})^2}, \quad (2.3)$$

where e_i represents individual evaluation scores from the 10 evaluations per session and $\bar{e}$ denotes their mean. We use the unbiased estimator (denominator $n - 1$) as these predictions

represent a sample of possible model outputs, not the entire population of possible predictions. The final SD for each experiment is reported as the average of SDs across the 22 concordant sessions and 53 overall sessions.

2.10.2 Classification Accuracy

The model's classification performance was assessed using direct accuracy (Acc) metrics, comparing median predictions against true values

$$\text{Acc} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \times 100 \quad (2.4)$$

To accommodate minor rating variations, threshold-based acceptable accuracy (AAcc) includes predictions within one point of true values is also being used

$$\text{AAcc} = \frac{\text{Number of predictions with } |y_i - \hat{y}_i| \leq 1}{\text{Total number of predictions}} \times 100 \quad (2.5)$$

where $\hat{y}_i$ represents the median of 10 evaluations per session, as used in the error measures.

2.10.3 Reliability Metrics Selection for Ordinal Data

Clinical ratings can exhibit significant variability from one human rater to another (McHugh, 2012), necessitating robust methods for measuring and ensuring rating consistency. This variability makes the measurement of agreement fundamental for establishing rater reliability in clinical settings. Reliability serves as a critical component in content analysis and clinical assessment, providing the foundation for valid data collection and interpretation (Lombard et al., 2002). Reliability comprises three key types: inter-rater reliability (consistency between different raters evaluating the same subjects), intra-rater reliability (consistency of a single rater's evaluations of the same subjects over multiple sessions), and test-retest reliability (stability of measurements across repeated assessments under similar conditions). While all three types contribute to the overall understanding of measurement consistency, this study specifically focuses on rater reliability—encompassing both inter-rater and intra-rater reliability—to evaluate how ChatGPT-4's assessments compare with those made by neurologists.

Appropriate reliability metrics vary by data type, which can be classified into four cate-

gories: (i) nominal, (ii) ordinal, (iii) interval, and (iv) ratio data.

- i. Nominal data consists of unordered categories. Examples include gender (male, female), color (yellow, blue, green), and blood type (A, B, AB, O). Suitable reliability metrics for nominal data include Krippendorff's α , Cohen's κ , Scott's π , Cramer's V , and Fleiss' κ (Zapf et al., 2016; Parker et al., 2013; Krippendorff, 2011; Hayes and Krippendorff, 2007).
- ii. Ordinal data contains categories with meaningful rank ordering but potentially unequal intervals between categories. Examples include disease severity (mild, moderate, severe), student performance rankings (poor, fair, good, excellent), and Likert scales (strongly disagree to strongly agree) (Joshi et al., 2015). Suitable reliability metrics for ordinal data include Krippendorff's α , weighted Cohen's κ (both linear and quadratic), intraclass correlation coefficient, Spearman's rank correlation coefficient, and Kendall's τ_b (De Raadt et al., 2021; Vanbelle, 2016; Hayes and Krippendorff, 2007).
- iii. Interval data features numerical values with equal intervals between consecutive values but lacks an absolute zero. Value differences are interpretable, but comparing values as ratios is not valid. Examples include Celsius temperatures (0°C doesn't indicate absence of temperature), and calendar dates. Suitable reliability metrics for interval data include Krippendorff's α , intraclass correlation coefficient, Pearson's correlation coefficient, Spearman's rank correlation coefficient, and Kendall's τ (De Raadt et al., 2021; Hayes and Krippendorff, 2007; McGraw and Wong, 1996).
- iv. Ratio data has both equal intervals between consecutive values and an absolute zero point, making both differences and ratios between values meaningful. Examples include weight (0 kg means no weight), height, time duration, and age. Suitable reliability metrics for ratio data include Krippendorff's α , intraclass correlation coefficient, Pearson's correlation coefficient, Spearman's rank correlation coefficient, and Kendall's τ (Akoglu, 2018; Hayes and Krippendorff, 2007).

In the context of the MDS-UPDRS, FT severity is rated on an ordinal scale from 0 to 4. While various reliability metrics have been used with ordinal data, their appropriateness

varies. De Raadt et al. (2021) examined seven commonly used metrics: Cohen's κ , linear weighted Cohen's κ , quadratic weighted Cohen's κ , intraclass correlation coefficient, Pearson's correlation coefficient, Spearman's rank correlation coefficient, and Kendall's τ_b . However, critical evaluation reveals that not all these metrics are equally suitable for ordinal data. The distance between each category is important for ordinal data because it represents how much the two ratings differ. For example, when rating severity on a scale from 1 to 5, a disagreement between ratings of 1 and 2 should be considered less severe than a disagreement between ratings of 1 and 5, as the latter indicates a much larger discrepancy in the raters' assessments. Pearson's correlation coefficient requires normal distribution and is thus not optimal for ordinal data (Schober et al., 2018; Mukaka, 2012). Fleiss' κ (Fleiss, 1971) was previously used by (Martínez-Martín et al., 1994) to assess inter-rater reliability for PD FT severity ratings. However, Fleiss' κ , like Cohen's κ , treats all disagreements equally and does not account for the ordinal nature of severity ratings (Fleiss, 1971; Hayes and Krippendorff, 2007).

Based on these considerations, we selected six metrics appropriate for our ordinal FT severity ratings: (1) Linear weighted Cohen's κ (accounting for linear magnitude of disagreements). (2) Quadratic weighted Cohen's κ (accounting for quadratic magnitude of disagreements). (3) Krippendorff's α (offers versatility across data types and sample sizes). (4) Intraclass correlation coefficient (quantifies reliability by comparing within-subject and between-subject variances). (5) Spearman's rank correlation coefficient (operates on ranks rather than absolute values). (6) Kendall's τ_b (particularly suitable for tied ranks).

After selecting the appropriate reliability metrics for ordinal data, we provide a detailed examination of each metric in this section. We present their descriptions, mathematical definitions, and step-by-step calculation processes. To demonstrate how these metrics are computed and interpreted, we use the example dataset shown in Table 2.8, which represents ordinal ratings from two doctors assessing three patients.

Table 2.8: Example dataset for demonstrating reliability metrics calculations. The data shows ordinal ratings assigned by two doctors to three patients, serving as a consistent example throughout this section.

Patient	Doctor X	Doctor Y
A	1	1
B	1	2
C	3	3

2.10.4 Wasserstein Distance

The Wasserstein Distance (WD), also known as Earth Mover’s Distance (Villani and Villani, 2009), measures the minimal amount of work required to transform one distribution into another, where work is defined as the amount of distribution weight moved times the distance it is moved. In the context of discrete, one-dimensional data with integer values, the Wasserstein Distance provides a precise way to quantify differences in distributions over an ordinal scale. This is particularly useful for assessing agreement in settings where ratings or counts are compared across categorical distances.

For a discrete, one-dimensional case, the formal definition of the WD between two distributions P and Q is given by

$$\text{WD}(P, Q) = \frac{\sum_{i=1}^M \sum_{j=1}^N d_{ij} f_{ij}^*}{\sum_{i=1}^M \sum_{j=1}^N f_{ij}^*}. \quad (2.6)$$

Here, d_{ij} represents the distance between elements of two distributions P and Q , reflecting the cost of moving mass from the i -th element of P to the j -th element of Q . f_{ij}^* denotes the optimal flow, or the best possible way to redistribute mass from P to Q , minimizing the overall work done during the transformation.

The numerator sums the work done for all movements, while the denominator, the total flow, normalizes the total work, making the distance metric scale-invariant. This normalization ensures that the WD does not depend on the total mass of the distributions, allowing comparisons across different scales.

In this application, we calculate the Wasserstein Distance between predicted evaluations and actual scores for discrete integer scores from 0 to 4, based on ground truth distributions across multiple sessions. For each ground truth score j (where $j = 0, 1, 2, 3, 4$), we calculate

the WD between the predicted evaluations (P) and the ground truth distribution (Q) for that score. For instance, if 3 sessions are ground truth score 0, and each includes 10 evaluation scores from ChatGPT-4, then P would consist of 30 evaluation scores, while Q would consist of 30 instances of score 0. This results in a per-score Wasserstein Distance $WD_j(P, Q)$, given by

$$WD_j(P, Q) = \frac{\sum_{i=0}^4 d_{ij} f_{ij}^*}{\sum_{i=0}^4 f_{ij}^*}, \quad j = 0, 1, 2, 3, 4 \quad (2.7)$$

Let N_j represent the number of sessions for each score j . The total Wasserstein Distance WD is then calculated as a weighted average of the per-score Wasserstein Distances, weighted by the count of sessions for each score

$$WD = \frac{N_0 \cdot WD_0 + N_1 \cdot WD_1 + N_2 \cdot WD_2 + N_3 \cdot WD_3 + N_4 \cdot WD_4}{N_0 + N_1 + N_2 + N_3 + N_4}. \quad (2.8)$$

This approach enables a nuanced assessment of agreement across different score groups, reflecting both the degree of discrepancy and the importance of each score group based on the number of sessions. This weighted metric, WD, provides a comprehensive measure of the overall alignment between predicted evaluations and actual scores, accounting for the specific distribution of scores in the data.

2.10.5 Linear and Quadratic Weighted Cohen's κ

Cohen's κ is a commonly used statistic for evaluating inter-rater agreement on nominal data (Cohen, 1960; Andrés and Marzo, 2004; De Raadt et al., 2019). It has become the standard method for assessing categorical agreement in medical research (Sim and Wright, 2005). However, it is not suitable for ordinal data as it does not account for the severity of differences between raters' classifications (Parker et al., 2013).

For ordinal data, weighted versions of Cohen's κ are commonly employed, as they adjust for how much disagreement matters, unlike regular κ , which counts all disagreements equally (Vanbelle, 2016; Cohen, 1968). Two common methods for penalizing disagreements have emerged: linear weighted Cohen's κ (LwCK) and quadratic weighted Cohen's κ (QwCK). LwCK (Cohen, 1968) assigns penalties based on the linear distance between scores, making it particularly suitable for ordinal data. It provides partial credit for closely aligned scores, which is valuable in scenarios where the scale of difference matters—for example,

on a 0-4 scale, a one-point difference is considered less significant than a three-point difference. QwCK (Cohen, 1968) applies a more stringent approach by penalizing disagreements quadratically as the distance between scores increases. This method is appropriate for ordinal data where larger scoring discrepancies are considered progressively more serious. The weighted Cohen's κ has been used in multiple fields, such as biostatistics (Ma et al., 2008), sport medicine, (Moore et al., 2018), and environmental research (Bromm et al., 2020).

The general formula for Cohen's κ is

$$\kappa = \frac{P_o - P_e}{1 - P_e}, \quad (2.9)$$

where P_o is the observed agreement (how often the raters give the same score), and P_e is the expected agreement by chance (the probability of agreement if the two raters made their ratings independently, based on their observed marginal distributions).

The sample estimator for Cohen's κ is

$$\kappa = \frac{\sum_{i=1}^k (p_{ii} - p_{i+}p_{+i})}{1 - \sum_{i=1}^k p_{i+}p_{+i}}, \quad (2.10)$$

where p_{ii} is the proportion of cases assigned to category i by both raters, p_{i+} is the proportion of cases assigned to category i by the first rater, and p_{+i} is the proportion of cases assigned to category i by the second rater. The total number of categories is k .

To derive weighted Cohen's κ , Cohen first reformulates Equation 2.9 in terms of disagreement. If we define $q_o = 1 - P_o$ and $q_e = 1 - P_e$ (where q represents disagreement), such that $\kappa = 1 - \frac{q_o}{q_e}$. The weighted κ then extends this concept by incorporating the severity of disagreements. The general formula for weighted Cohen's κ is

$$\kappa_w = 1 - \frac{q'_o}{q'_e}, \quad (2.11)$$

where $q'_o = \frac{\sum w_{ij}p_{ij}}{w_{\max}}$ measures the observed weighted disagreement, reflecting severity via weights w_{ij} , and $q'_e = \frac{\sum w_{ij}p_{i+}p_{+j}}{w_{\max}}$ is the expected weighted disagreement by chance. Here, p_{ij} is the proportion of cases where the first rater assigned category i and the second rater assigned category j , $p_{i+}p_{+j}$ is the expected chance agreement for category i by the first rater and category j by the second rater based on their independent distributions, w_{ij} indicates

the disagreement severity between scores i and j , and $w_{\max}$ is the maximum disagreement weight.

The sample estimator for weighted Cohen's κ is

$$\kappa_w = 1 - \frac{\sum_{i=1}^k \sum_{j=1}^k w_{ij} p_{ij}}{\sum_{i=1}^k \sum_{j=1}^k w_{ij} (p_{i+} + p_{+j})}. \quad (2.12)$$

For LwCK, the weights are defined as $w_{ij} = |i - j|$, resulting in the following expressions for observed (q_o^L) and expected disagreement (q_e^L)

$$q_o^L = \sum_{i=1}^k \sum_{j=1}^k |i - j| p_{ij}, \quad (2.13)$$

$$q_e^L = \sum_{i=1}^k \sum_{j=1}^k |i - j| p_{i+} p_{+j}, \quad (2.14)$$

where $|i - j|$ shows that disagreements are weighted based on how far apart the scores are.

The sample estimator for LwCK is

$$\kappa_{lw} = 1 - \frac{q_o^L}{q_e^L}. \quad (2.15)$$

For QwCK, the weights are defined as $w_{ij} = (i - j)^2$, resulting in the following expressions for observed (q_o^Q) and expected disagreement (q_e^Q)

$$q_o^Q = \sum_{i=1}^k \sum_{j=1}^k (i - j)^2 p_{ij}, \quad (2.16)$$

$$q_e^Q = \sum_{i=1}^k \sum_{j=1}^k (i - j)^2 p_{i+} p_{+j}, \quad (2.17)$$

where $(i - j)^2$ shows that disagreements are weighted based on the square of how far apart the scores are.

The sample estimator for QwCK is

$$\kappa_{qw} = 1 - \frac{q_o^Q}{q_e^Q}. \quad (2.18)$$

Table 2.9: Contingency table derived from the ratings of Doctor X and Doctor Y for three patients. The table shows the frequency distribution of ratings across three ordinal classes, with cells indicating the number of patients assigned to each rating combination. This contingency matrix forms the basis for calculating LwCK and QwCK.

	Class 1 (Y)	Class 2 (Y)	Class 3 (Y)
Class 1 (X)	1	1	0
Class 2 (X)	0	0	0
Class 3 (X)	0	0	1

From Table 2.9, we first compute the proportions and marginal distributions. From the ratings provided, the observed proportions are $p_{11} = \frac{1}{3}$, $p_{12} = \frac{1}{3}$, $p_{33} = \frac{1}{3}$, with all other $p_{ij} = 0$. The marginal distributions are $p_{1+} = \frac{2}{3}$, $p_{2+} = 0$, $p_{3+} = \frac{1}{3}$ for Doctor X, and $p_{+1} = p_{+2} = p_{+3} = \frac{1}{3}$ for Doctor Y.

LwCK Calculation: For LwCK, we calculate the observed disagreement q_o^L using Equation 2.13. We only have one disagreement where Doctor X rated 1 and Doctor Y rated 2, with probability 0.33, such that $q_o^L = 0.33$. For expected disagreement q_e^L (Equation 2.14), we calculate the expected disagreement by examining all possible pairs

$$\begin{aligned}
 q_e^L &= p_{1+}p_{+2}|1-2| + p_{1+}p_{+3}|1-3| + p_{3+}p_{+1}|3-1| + p_{3+}p_{+2}|3-2| \\
 &= \frac{2}{3} \times \frac{1}{3} \times 1 + \frac{2}{3} \times \frac{1}{3} \times 2 + \frac{1}{3} \times \frac{1}{3} \times 2 + \frac{1}{3} \times \frac{1}{3} \times 1 = 1.
 \end{aligned}$$

Applying Equation 2.15, the LwCK is

$$\kappa_{lw} = 1 - \frac{q_o^L}{q_e^L} \approx 0.67.$$

QwCK Calculation: For the observed disagreement q_o^Q (Equation 2.16), we have the single disagreement case $q_o^Q = 0.33$. For the expected disagreement q_e^Q (Equation 2.17), we calculate

$$\begin{aligned}
 q_e^Q &= p_{1+}p_{+2}(1-2)^2 + p_{1+}p_{+3}(1-3)^2 + p_{3+}p_{+1}(3-1)^2 + p_{3+}p_{+2}(3-2)^2 \\
 &= \frac{2}{3} \times \frac{1}{3} \times 1 + \frac{2}{3} \times \frac{1}{3} \times 4 + \frac{1}{3} \times \frac{1}{3} \times 4 + \frac{1}{3} \times \frac{1}{3} \times 1 = \frac{5}{3}.
 \end{aligned}$$

Applying Equation 2.18, the QwCK is

$$\kappa_{qw} = 1 - \frac{q_o^Q}{q_e^Q} = 0.80.$$

Thus, for this set of ratings, LwCK = 0.67 and QwCK = 0.80.

Cohen's κ value ranges from -1 to 1. Negative values imply inverse correlation, zero means no correlation, and positive values imply positive correlation. Values below 0 indicate poor reliability, 0 to 0.2 indicate slight reliability, 0.2 to 0.4 indicate fair reliability, 0.4 to 0.6 indicate moderate reliability, 0.6 to 0.8 indicate substantial reliability, and above 0.8 indicate almost perfect reliability (Landis, 1977).

When one rater assigns constant scores to all subjects while the other rater's scores vary, Cohen's κ , LwCK, and QwCK values result in 0. This occurs because constant or skewed marginal distributions diminish data informativeness and distort agreement estimates (Feng, 2013b; Gwet, 2008; Zysno, 1997).

In cases where both raters assign the same constant value to all subjects, this represents perfect agreement as raters completely agree on every case, although LwCK and QwCK become mathematically undefined due to zero variance (Cohen, 1968; Heo, 2008).

2.10.6 Krippendorff's α

Krippendorff's α is a versatile reliability coefficient that measures agreement among observers, raters, or measuring instruments and serves as a standard measure in content analysis. It accommodates different data types, sample sizes, numbers of observers, and can even work with missing data. (Krippendorff, 2011; Hayes and Krippendorff, 2007). It can be used in multiple fields, such as medical research (De Coninck et al., 2018), educational research (Morley, 2009), and software engineering (Volkman et al., 2019).

The general formula for Krippendorff's α is

$$\alpha = 1 - \frac{D_o}{D_e}, \quad (2.19)$$

where D_o is the observed disagreement, which measures the actual differences between rat-

ings assigned by different observers to the same units. It is calculated as

$$D_o = \frac{1}{n} \sum_c \sum_k o_{ck} \delta_{\text{metric}}^2,$$

where o_{ck} represents how many times values c and k are assigned to the same unit by different observers, n is the total number of pairs of ratings, and δ_{metric}^2 is the difference function that quantifies how different the values c and k are (varying according to whether the data is nominal, ordinal, interval, or ratio). D_e is the expected disagreement, which estimates how much disagreement would occur if observers assigned ratings randomly, regardless of the units' actual properties. It is calculated as

$$D_e = \frac{1}{n(n-1)} \sum_c \sum_k n_c n_k \delta_{\text{metric}}^2,$$

where n_c and n_k represent how many times values c and k appear in the entire dataset, respectively. This formula essentially measures what the disagreement would be if the observed distribution of values was the result of random chance rather than observers responding to the actual properties of the units.

The sample estimator for Krippendorff's α is

$$\alpha = 1 - (n-1) \frac{\sum_c \sum_{k>c} o_{ck} \delta_{\text{metric}}^2}{\sum_c \sum_{k>c} n_c n_k \delta_{\text{metric}}^2}, \quad (2.20)$$

which only requires summing over one triangular half of the coincidence.

For ordinal data, the difference is measured as

$$\delta_{\text{ordinal}}^2 = \left(\sum_{g=c}^k n_g - \frac{n_c + n_k}{2} \right)^2,$$

which is implemented in our study.

In our example, we have a total of $n = 6$ pairs of ratings with frequencies $n_1 = 3$, $n_2 = 1$, and $n_3 = 2$. For ordinal data, the squared difference between categories 1 and 2 is $\delta_{\text{ordinal}(1,2)}^2 = 4$, between categories 1 and 3 is $\delta_{\text{ordinal}(1,3)}^2 = 12.25$, and between categories 2 and 3 is $\delta_{\text{ordinal}(2,3)}^2 = 2.25$. The numerator of Equation 2.20 is $\sum_c \sum_{k>c} o_{ck} \delta_{\text{metric}}^2 = (1 \cdot 4) + (0 \cdot 12.25) + (0 \cdot 0.25) = 4$, while the denominator is $\sum_c \sum_{k>c} n_c n_k \delta_{\text{metric}}^2 = (3 \cdot 1 \cdot 4) + (3 \cdot$

Table 2.10: Coincidence matrix for the three-patient example. Each cell (i, j) represents the frequency with which rating i from one doctor co-occurs with rating j from another doctor within the same patient.

	Class 1 (Y)	Class 2 (Y)	Class 3 (Y)
Class 1 (X)	2	1	0
Class 2 (X)	1	0	0
Class 3 (X)	0	0	2

$2 \cdot 12.25) + (1 \cdot 2 \cdot 2.25) = 90$. Applying Equation 2.20, α is $1 - (6 - 1) \frac{4}{90} \approx 0.78$. Thus, for this set of ratings, Krippendorff's $\alpha = 0.78$.

Krippendorff's α value ranges from -1 to 1. Negative values imply inverse correlation, zero means no correlation, and positive values imply positive correlation. Values below 0.67 indicate poor reliability, 0.67 to 0.79 indicate moderate reliability, and above 0.8 indicate almost perfect reliability (Marzi et al., 2024).

When one rater assigns constant scores while the other varies, Krippendorff's α tends to approaches 0. As noted by Feng (2013a), Krippendorff's α behaves similarly to κ under skewed or constant marginal distributions, often yielding low or even negative values despite high observed agreement, due to low variability in ratings.

In cases where all raters assign the same constant value to all subjects, we observe perfect agreement, though the mathematical calculation of Krippendorff's α becomes indeterminate. This situation doesn't reflect a failure of reliability but rather demonstrates complete consensus among raters (Krippendorff, 2018).

2.10.7 Intraclass Correlation Coefficient (ICC)

Intraclass Correlation Coefficient (ICC) (Shrout and Fleiss, 1979) is a statistical measure used to evaluate both consistency and absolute agreement among multiple rating; it has broad applications across multiple fields including conservative care medicine, behavioral measurement, psychometrics, and behavioral genetics (Koo and Li, 2016; McGraw and Wong, 1996). Initially developed with six forms, ICC was later expanded to ten forms by McGraw and Wong (1996). The forms of ICC are determined by three core concepts: model, type, and definition (Koo and Li, 2016).

The model refers to how raters are selected and generalizability of results. One-way

random-effects applies when subjects are assessed by varying raters selected from a larger rater population. Two-way random-effects is used when raters are randomly sampled from the rater pool with similar characteristics with each rating all subjects. Two-way mixed-effects is appropriate when specific raters of interest evaluate all subjects.

The type indicates whether reliability is calculated for individual or multiple raters. Single rater involves analyzing each rater's scores independently, while multiple raters uses aggregate measures (mean, median, or mode) from k raters.

The definition determines focus on exact agreement or ranking consistency. Absolute agreement applies when raters give identical or similar scores, while consistency focuses on when raters assign scores in the same relative order.

The original six forms of ICC defined by Shrout and Fleiss (1979) are: ICC(1,1): One-way random-effects model, absolute agreement definition, single rater type; ICC(1,k): One-way random-effects model, absolute agreement definition, multiple raters type; ICC(2,1): Two-way random-effects model, absolute agreement definition, single rater type; ICC(2,k): Two-way random-effects model, absolute agreement definition, multiple raters type; ICC(3,1): Two-way mixed-effects model, consistency definition, single rater type; ICC(3,k): Two-way mixed-effects model, consistency definition, multiple raters type.

McGraw and Wong (1996) expanded these to include four additional forms, adapting them for varied analytical contexts: Two-way random-effects model, consistency definition, single rater type; Two-way random-effects model, consistency definition, multiple raters type; Two-way mixed-effects model, absolute agreement definition, single rater type; Two-way mixed-effects model, absolute agreement definition, multiple raters type.

McGraw and Wong (1996) noted that under the same definition and type, the two-way random-effects and two-way mixed-effects models yield identical values due to identical formulas. For example, the ICC for a two-way mixed-effects model with absolute agreement and single rater type is calculated using the same formula as ICC(2,1).

Moreover, when comparing different ICC forms, the "multiple raters" type typically produces higher values than the "single rater" type, "consistency" definition yields higher values than "absolute agreement" definition, and "two-way mixed-effects" model tends to result in higher values than "two-way random-effects" model.

For intra-rater reliability and test-retest reliability, Shrout and Fleiss (1979) recommends

using a two-way mixed-effects model with absolute agreement. These tests involve specific individuals and are not suitable for a random-effects model. Additionally, agreement across repeated tests is meaningful, necessitating absolute agreement.

This study evaluates ChatGPT-4's intra-rater reliability using a two-way mixed-effects model for single measures with absolute agreement [ICC(2,1)]. This method measures how closely ChatGPT-4's repeated measures align under identical conditions.

For assessing inter-rater reliability between two neurologists, we employ a two-way mixed-effects model with absolute agreement for single measures [ICC(2,1)]. This model treats the neurologists as fixed raters, as they are specific individuals of interest rather than random samples from a larger rater pool. This approach accounts for individual scoring variability and enables direct comparison with ChatGPT-4's intra-rater reliability.

For assessing inter-rater reliability between ChatGPT-4 and neurologists, we used a two-way mixed-effects model with multiple raters type [ICC(2,k) and ICC(3,k)]. The multiple raters type was chosen because we used aggregate measures - the median of ChatGPT-4's 10 evaluation scores as its final score and the mean of the two neurologists' scores as their final score. We applied both absolute agreement [ICC(2,k)] and consistency [ICC(3,k)] definitions, where absolute agreement focuses on exact matches, while consistency considers the overall trend in scoring.

The study De Raadt et al. (2021) conducted a comparative analysis of seven reliability coefficients for ordinal data, with ICC(3,1) representing the Intraclass Correlation Coefficient in their study. Although we do not evaluate using ICC(3,1) in our analysis, we include its formula below.

The general formula for ICC(2,1) is

$$ICC(2,1) = \frac{\sigma_r^2}{\sigma_r^2 + \sigma_c^2 + \sigma_{rc}^2 + \sigma_e^2}, \quad (2.21)$$

the sample estimator for ICC(2,1) is

$$\frac{MS_R - MS_E}{MS_R + (k - 1)MS_E + \frac{k}{n}(MS_C - MS_E)}, \quad (2.22)$$

the general formula for ICC(3,1) is

$$\text{ICC}(3,1) = \frac{\sigma_r^2 - \sigma_{rc}^2/(k-1)}{\sigma_r^2 + \sigma_{rc}^2 + \sigma_e^2}, \quad (2.23)$$

the sample estimator for ICC(3,1) is

$$\frac{MS_R - MS_E}{MS_R + (k-1)MS_E}, \quad (2.24)$$

this measures the reliability of averaged measurements when systematic differences between raters are not of concern. The general formula for ICC(2,k) is

$$\text{ICC}(2,k) = \frac{\sigma_r^2}{\sigma_r^2 + (\sigma_c^2 + \sigma_{rc}^2 + \sigma_e^2)/k}, \quad (2.25)$$

the sample estimator of ICC(2,k) is

$$\frac{MS_R - MS_E}{MS_R + \frac{MS_C - MS_E}{n}}, \quad (2.26)$$

the general formula for ICC(3,k) is

$$\text{ICC}(3,k) = \frac{\sigma_r^2 - \sigma_{rc}^2/(k-1)}{\sigma_r^2 + (\sigma_{rc}^2 + \sigma_e^2)/k}, \quad (2.27)$$

and the sample estimator of ICC(3,k) is

$$\frac{MS_R - MS_E}{MS_R}. \quad (2.28)$$

Here, σ_r^2 is the row variance component representing the systematic variation between subjects - for instance, real differences in individuals' characteristics being measured. σ_c^2 is the column variance component capturing systematic differences between raters, such as when certain raters consistently score higher or lower than others. σ_{rc}^2 is the row-column interaction variance component, which represents how rater effects vary systematically across subjects - for example, when certain raters consistently give higher scores to particular types of subjects while giving lower scores to others. σ_e^2 is the residual variance component capturing all remaining unexplained variability, including random measurement error and other sources

of variation not accounted for by the systematic components. For the sample statistics: MS_R represents the mean square for rows (subjects), MS_C represents the mean square for columns (raters), and MS_E represents the mean square error. The terms k and n denote the number of raters and subjects, respectively.

In our example, we identify the parameters $k = 2$ raters and $n = 3$ patients. We first calculate the necessary means: row means ($\mu_A = 1$, $\mu_B = 1.5$, $\mu_C = 3$), column means ($\mu_{\text{Doctor X}} = \frac{5}{3} = 1.667$, $\mu_{\text{Doctor Y}} = 2$), and the grand mean ($\mu = \frac{11}{6} = 1.833$).

Sum of Squares Calculation: The total sum of squares is given by $SS_{\text{total}} = \sum (x_{ij} - \mu)^2 = (1 - 1.833)^2 + (1 - 1.833)^2 + (1 - 1.833)^2 + (2 - 1.833)^2 + (3 - 1.833)^2 + (3 - 1.833)^2 = 4.832$. For the rows, $SS_{\text{rows}} = k \sum (\text{row mean} - \mu)^2 = 2[(1 - 1.833)^2 + (1.5 - 1.833)^2 + (3 - 1.833)^2] = 4.332$. For the columns, $SS_{\text{cols}} = n \sum (\text{col mean} - \mu)^2 = 3[(1.667 - 1.833)^2 + (2 - 1.833)^2] = 0.168$. The error sum of squares is $SS_{\text{error}} = SS_{\text{total}} - SS_{\text{rows}} - SS_{\text{cols}} = 0.332$.

Mean Squares Calculation: From the sum of squares, we derive the mean squares $MS_R = \frac{SS_{\text{rows}}}{n-1} = 2.166$, $MS_C = \frac{SS_{\text{cols}}}{k-1} = 0.168$, $MS_E = \frac{SS_{\text{error}}}{(n-1)(k-1)} = 0.166$.

By applying Equations 2.22, 2.24, 2.26, and 2.28, we calculate that for this set of ratings, $\text{ICC}(2,1) = 0.86$, $\text{ICC}(3,1) = 0.86$, $\text{ICC}(2,k) = 0.92$, and $\text{ICC}(3,k) = 0.92$.

The ICC ranges from 0 to 1, with values under 0.5 suggesting poor reliability, values between 0.5 and 0.75 indicating moderate reliability, 0.75 to 0.9 reflecting good reliability, and values exceeding 0.9 representing excellent reliability (Koo and Li, 2016).

When one rater assigns constant scores to all subjects, ICC value result in 0. As Finn (1970) demonstrated, ICC requires sufficient within-rater variance to yield meaningful coefficients. Due to insufficient within-rater variance, even with high actual agreement, ICC values approach zero, reflecting statistical limitations rather than true disagreement.

In cases where both raters assign the same constant value to all subjects, ICC will equal to 1 (Koo and Li, 2016). With neither within-rater nor between-rater variance present, the mathematical calculation of ICC technically becomes undefined; however, this scenario still represents perfect agreement among raters.

2.10.8 Spearman's rank correlation coefficient

Spearman's rank correlation coefficient (Spearman, 1961), or Spearman's ρ , is a nonparametric statistic used to evaluate the strength and direction of a monotonic relationship between

Table 2.11: Ratings and corresponding ranks for Spearman's ρ example

Patient	Doctor X	Doctor Y	Rank X	Rank Y
A	1	1	1.5	1
B	1	2	1.5	2
C	3	3	3	3

two variables, while disregarding the magnitude of differences. This makes it particularly suitable for analyzing ordinal and non-normally distributed data. It can be used in multiple fields, including financial market analysis (Münnix et al., 2014), scientific research (Fritz, 1974), and environmental science (Franceschini et al., 2013).

The general formula of Spearman's ρ is

$$\rho = \frac{\text{cov}(R(X), R(Y))}{\sigma_{R(X)}\sigma_{R(Y)}}, \quad (2.29)$$

where $R(X)$ and $R(Y)$ represent the rank values of the variables X and Y , respectively. $\text{cov}(R(X), R(Y))$ is the covariance of the ranked variables, which measures how the ranks of the two variables change together. $\sigma_{R(X)}$ and $\sigma_{R(Y)}$ are the standard deviations of the ranks of X and Y , respectively.

The sample estimator of Spearman's ρ is

$$\rho = \frac{\sum_{i=1}^n (R(X)_i - \overline{R(X)})(R(Y)_i - \overline{R(Y)})}{[\sum_{i=1}^n (R(X)_i - \overline{R(X)})^2 \sum_{i=1}^n (R(Y)_i - \overline{R(Y)})^2]^{1/2}}, \quad (2.30)$$

where i is the index of the observation, n is the sample size, $R(X)_i$ and $R(Y)_i$ represent the ranks of the i -th observation, and $\overline{R(X)}$ and $\overline{R(Y)}$ are the mean ranks.

The ranking process assigns values from 1 to n based on the magnitude of observations: the smallest value receives rank 1, the second smallest rank 2, and so on. When tied values occur, they receive the average of their positions in the ascending order.

In our example, $n = 3$ subjects, and from Table 2.11, we get column means: $\overline{R(X)} = 2$, $\overline{R(Y)} = 2$. Applying Equation 2.30, Spearman's $\rho \approx 0.87$.

Spearman's ρ ranges from -1 to 1, where negative values suggest an inverse association, zero implies no correlation, and positive values reflect a direct relationship. Values below 0.3 indicate poor reliability, 0.3 to 0.6 fair, 0.6 to 0.8 moderate, and above 0.8 very strong

reliability (Akoglu, 2018; Chan, 2003).

When one rater assigns constant scores to all subjects while the other rater's scores vary, Spearman's ρ becomes undefined. As noted in Berg et al. (2022), Spearman's ρ requires non-zero variance in both raters' rankings. When all subjects share the same rank from one rater, it becomes mathematically impossible to calculate correlation.

In cases where all subjects receive identical ranks from both raters, Spearman's ρ becomes mathematically undefined. Since this situation demonstrates perfect agreement between raters, Kendall (1945) suggests assigning a value of 1 to represent this result.

2.10.9 Kendall's τ_b

Kendall's τ_b (Kendall, 1945) is a non-parametric measure of the ordinal association between two variables. Unlike metrics like LwCK, QwCK, and ICC that focus on numerical agreement, Kendall's τ_b specifically analyzes the concordance and discordance of rankings, making it particularly valuable for ordinal data analysis.

Developed as an extension of Kendall's original correlation coefficient (Kendall, 1938), Kendall's τ_b incorporates adjustments for tied ranks in both variables. In Kendall's seminal paper "The Treatment of Ties in Ranking Problems," he addressed how ties affect correlation coefficients and provided a systematic approach for dealing with them. It is used in fields such as medical science (Uday et al., 2017), engineering (Nugraha and Kosasih, 2022), and education (ÜNÜSAN and YALÇIN, 2019).

The general formula of Kendall's τ_b is

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \quad (2.31)$$

where n_c represents agreement pairs (pairs where both variables move in the same direction), n_d represents discordant pairs (pairs where variables move in opposite directions), n_0 is the total possible pairs, equal to $\frac{n(n-1)}{2}$ where n is the sample size, and n_1 and n_2 are adjustments for ties in the first and second variables, respectively. For any two observations x_i, y_i and x_j, y_j such that $i < j$: A tie occur when $x_i = x_j$ or $y_i = y_j$. The tie adjustments are calculated as:

$$n_1 = \sum_{i=1}^k \frac{t_i(t_i - 1)}{2} \quad \text{and} \quad n_2 = \sum_{i=1}^k \frac{u_i(u_i - 1)}{2},$$

where t_i and u_i represent the number of tied values in the i -th group for each variable, and k represents the number of groups.

The sample estimator of Kendall's τ_b is

$$\tau = \frac{\sum(a_{ij}b_{ij})}{\sqrt{\sum a_{ij}^2 \sum b_{ij}^2}}, \quad (2.32)$$

where $a_{ij} = 1$ if $x_i < x_j$, -1 if $x_i > x_j$, and 0 if $x_i = x_j$. $b_{ij} = 1$ if $y_i < y_j$, -1 if $y_i > y_j$, and 0 if $y_i = y_j$. $\sum a_{ij}^2 = n_0 - n_1$, and $\sum b_{ij}^2 = n_0 - n_2$.

In our example, $a_{12} = 0$, $b_{12} = 1$ (A tied with B for X, $A < B$ for Y). $a_{13} = 1$, $b_{13} = 1$ (A $< C$ for both). $a_{23} = 1$, $b_{23} = 1$ (B $< C$ for both). With one tie (A,B) in Doctor X's ratings, we apply the Equation 2.32 and get Kendall's $\tau_b \approx 0.816$.

Kendall's τ_b ranges from -1 to 1 , where negative values reflect an inverse association, zero denotes no correlation, and positive values indicate a positive relationship. Values below 0.2 indicate poor reliability, 0.2 to 0.4 fair, 0.4 to 0.6 moderate, 0.6 to 0.8 good, and above 0.8 very strong reliability (Abadi and Moore, 2022).

When one rater assigns identical ranks to all items while the other rater provides varying assessments, Kendall's τ_b becomes mathematically undefined. This occurs because the denominator in the formula contains a tie-adjustment factor that becomes zero in this scenario. With no agreement or discordant pairs to compare, correlation calculation becomes mathematically impossible (Kendall, 1945).

In cases where all items receive identical ranks from both raters, Kendall's τ_b becomes mathematically undefined. As this scenario reflects complete agreement between raters, Kendall (1945) suggests interpreting this result as 1 .

2.10.10 Comparative Analysis of Reliability Coefficients for Ordinal Data

When the data is ordinal, QwCK can be expressed as (Schuster, 2004)

$$\kappa_{qw} = \frac{2 \operatorname{cov}(X, Y)}{\sigma_X^2 + \sigma_Y^2 + \frac{n}{n-1}(m_1 - m_2)^2}, \quad (2.33)$$

where m_1 denotes the average score given by the first rater, and m_2 denotes the average score given by the second rater. The ICC(3,1) can also be written as

$$\text{ICC}(3, 1) = \frac{2 \text{cov}(X, Y)}{\sigma_X^2 + \sigma_Y^2}. \quad (2.34)$$

From Equations 2.33 and 2.34, QwCK equals ICC(3,1) when rater means are equal. Several studies have established mathematical relationships among these reliability metrics. De Raadt et al. (2021) demonstrated that QwCK reaches similar values to correlation coefficients including ICC(3,1), Spearman's ρ , and Kendall's τ . Warrens (2014) showed that both ICC(3,1) and Spearman's ρ are special cases of weighted Cohen's κ . Additionally, Krippendorff's α equals Spearman's ρ for two-observer ordinal data without ties (Hayes and Krippendorff, 2007). Despite these established mathematical relationships, these reliability metrics do not maintain ratio relationships with one another, meaning no direct numerical transformation exists to convert between metrics. This limitation necessitates using multiple metrics to provide a comprehensive reliability assessment.

Chapter 3

Results

3.1 Overview of Experiments

This chapter evaluates ChatGPT-4’s performance across six experimental conditions (Exp. A–F) involving systematic prompt variations and controls. We assessed 53 finger-tapping sessions with ten repeated ChatGPT-4 evaluations each, comparing results against two neurologists’ scores to determine how prompt design influences scoring consistency and agreement.

Table 3.1 summarizes the main design rationale behind each experiment. Exp. A–D constitute *prompt-variation* studies that progressively enrich the task prompt—from standard instructions to technical input features and few-shot examples. Exp. E and F serve as *control baselines*: Exp. E uses a fixed score of 1 to estimate scoring bias, while Exp. F employs random scoring to simulate chance-level performance.

Exp. D incorporates few-shot prompting by including example feature sets for scores 0–3 within the prompt. To isolate the influence of these examples, we evaluated all six experiments with the example sessions removed from the input prompts. These counterpart conditions—denoted Exp. A-EE through F-EE—enable a fair comparison to quantify the added value of few-shot prompting.

Sections 3.2 through 3.5 report four core analyses. First, we quantify the variability of ChatGPT-4’s repeated ratings within sessions (Section 3.2). Next, we assess per-evaluation agreement with the neurologists’ MDS-UPDRS ratings (Section 3.3), followed by an analysis of final aggregated score agreement (Section 3.4). Finally, we evaluate the reliability of both ChatGPT-4 and the neurologists (Section 3.5). Detailed violin plots showing ChatGPT-

Table 3.1: Summary of the six primary experimental conditions (Exp. A–F).

Exp.	Design and Rationale
A	Ten-second finger-tapping segment with standard MDS-UPDRS instructions.
B	Exp. A plus a Python code snippet describing motion feature extraction.
C	Exp. B but using the first ten taps instead of a fixed-duration segment.
D	Exp. C plus example feature sets for scores 0–3 (few-shot prompting).
E	Constant-score baseline: a fixed score of 1 is assigned to every evaluation (bias estimate).
F	Monte Carlo random baseline: a score from 0 to 4 is randomly selected from a uniform distribution for each evaluation.

4’s score distributions for each session across Exp. A–D, alongside neurologist ratings, are provided in Appendix B.

3.2 Consistency Across Ten ChatGPT-4 Evaluations

Standard deviation analysis provides insights into ChatGPT-4’s evaluation consistency, where lower SD values indicate more consistent scoring across the 10 evaluations per session. SDs across concordant and overall sessions are reported in Table 3.2. From Exp. A–C, two key patterns are observed: (1) the SD differences between concordant and overall sessions are small, never exceeding 0.03 points; and (2) a decreasing trend in SD is observed across experiments, from 0.97 in Exp. A to 0.56 in Exp. C (overall sessions). In Exp. D–EE, the SD is slightly higher in concordant sessions (0.63) compared to Exp. C–EE (0.55). The constant-score baseline (Exp. E and E–EE) yields zero variability ($SD = 0$), whereas the random-score baseline (Exp. F and F–EE) shows the highest variability ($SD = 1.40$).

Figure 3.1 examines the degree of scoring disparity between pairs of evaluations—both from ChatGPT-4 (10 evaluations per session) and from the two neurologists—focusing on instances where the score difference is 2 or more. The heatmap values represent the percentage occurrence of each score pair across 53 sessions. For example, if Neurologist X assigns a score of 2 and Neurologist Y assigns a score of 0 for a session, the cell at index (2, 0) increments by one. Figure 3.1i visualizes the scoring statistics between the two neurologists in a heatmap format. Notably, Neurologist X tends to assign more scores of 1, while Neurologist Y gives more scores of 3. Figures 3.1a to 3.1h present heatmaps for all 45 pairs of

Table 3.2: Standard deviation (SD) of ChatGPT-4 ratings across 10 evaluations per session, averaged over 22 concordant sessions and 53 overall sessions. Values in parentheses indicate the number of sessions. The 95 % confidence intervals (CIs) were estimated via bootstrap sampling for Exp. A–E and A-EE–E-EE, and Monte Carlo resampling for Exp. F and F-EE (see Sections 2.8.2 and 2.9.1).

Experiment	Concordant Sessions (22)		Overall Sessions (53)	
	SD	95 % CI	SD	95 % CI
A	1.00	[0.89, 1.10]	0.97	[0.92, 1.03]
B	0.88	[0.82, 0.94]	0.85	[0.81, 0.89]
C	0.56	[0.47, 0.64]	0.56	[0.51, 0.61]
D	0.55	[0.43, 0.66]	0.55	[0.48, 0.61]
E	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]
F	1.40	[1.30, 1.49]	1.40	[1.34, 1.46]
	Concordant Sessions (18)		Overall Sessions (49)	
	SD	95 % CI	SD	95 % CI
A-EE	1.00	[0.88, 1.11]	0.97	[0.91, 1.03]
B-EE	0.85	[0.79, 0.91]	0.84	[0.79, 0.88]
C-EE	0.55	[0.44, 0.63]	0.56	[0.50, 0.61]
D-EE	0.63	[0.54, 0.70]	0.57	[0.52, 0.63]
E-EE	0.00	[0.00, 0.00]	0.00	[0.00, 0.00]
F-EE	1.40	[1.29, 1.50]	1.40	[1.34, 1.46]

ChatGPT-4 evaluations across different experiments. Each pair consists of two out of the ten evaluations, covering all 53 sessions. The score occurrences are computed similarly to the neurologist pairing, and the results from all 45 pairs are averaged to generate the composite heatmap. The proportion of pairs with a score difference of 2 or more is 16.9% for the neurologists. In comparison, ChatGPT-4’s evaluations show greater disparity in some experiments—31.4% in Exp. A, 25.5% in Exp. B, 31.8% for Exp. A-EE, 24.8% for Exp. B-EE—but notably lower rates in Exp. C 7.6%, Exp. D 7.9%, Exp. C-EE 7.6%, Exp. D-EE 7.8%.

3.3 Agreement Between Ten ChatGPT-4 Evaluations and Neurologists

Figures 3.2a to 3.2d display the distribution of ChatGPT-4’s predicted scores for concordant sessions in Experiments A to D. Each subfigure shows the score distribution grouped by the concordant scores, ranging from 0 to 3 (left to right). Blue bars represent scores that match

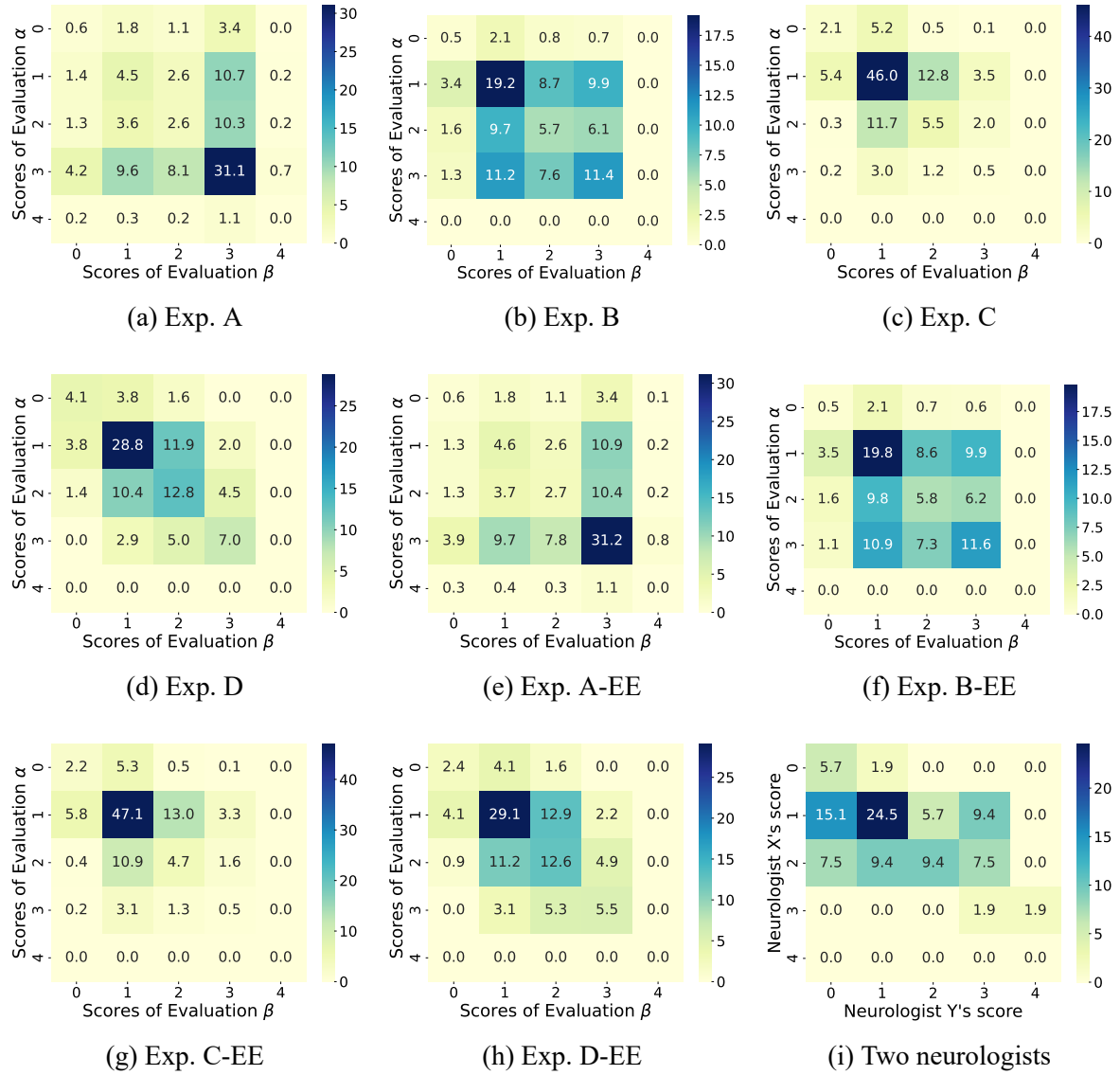


Figure 3.1: Heatmap of pairs of ChatGPT-4 evaluation scores across four experiments and pairs of two neurologists' scores. Here, α and β represent indices for the evaluations, where α ranges from 1 to 9 and β from 2 to 10, with the condition that $\alpha < \beta$.

the concordant score, black bars indicate deviations from the concordant score, and yellow bars correspond to the score distribution in the four example sessions.

In concordant sessions of Exp. A, ChatGPT-4 frequently assigns score 3, even when the concordant score is lower. For example, when the concordant scores are 1 and 3, ChatGPT-4 assigns score 3 in 53.8% and 70% of the evaluations, respectively. In Exp. B, the scoring trend shifts toward lower values, with a notable increase in score 1 assignments. Accuracy for concordant score 1 improves by 28.4 percentage points (from 15.4% to 43.8%). While accuracy for concordant score 2 also improves, it decreases for concordant score 3. In con-

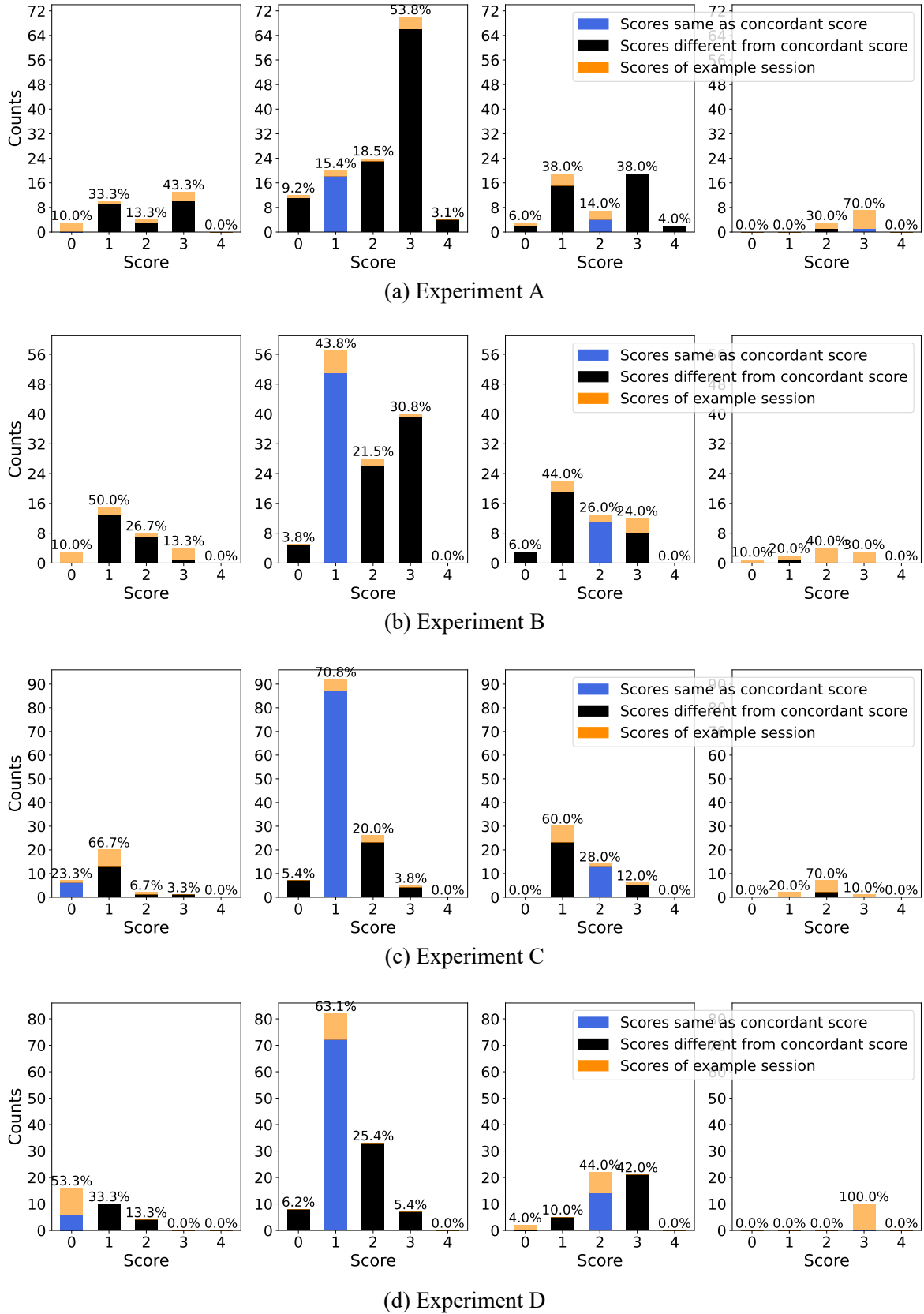


Figure 3.2: Distribution of ChatGPT-4’s predicted scores for each concordant score in Exp. A–D. The color blue indicates that ChatGPT-4’s evaluation scores align with the concordant score, black indicates a deviation from the concordant score, and yellow represents the score distribution for the four example sessions.

cordant sessions of Exp. C, ChatGPT-4 shows a higher proportion of score 1 assignments, with accuracy for concordant score 1 improving by another 27.0 percentage points to 70.8%. Accuracy also improves for concordant scores 0 and 2 but continues to decline for concordant score 3. In Exp. D, the scoring becomes more balanced compared to Exp. C. There are fewer score 1 assignments across all concordant scores and increased use of higher scores, particularly for sessions with concordant score 2.

Table 3.3 lists the numerical accuracy values of ChatGPT-4’s evaluation scores across experiments, compared to the concordant scores. The table includes both overall weighted accuracies and score-specific accuracies based on the prevalence of each score group.

In Exp. A, ChatGPT-4 achieved only 16.8% overall accuracy, performing worse than the random baseline (Exp. F), which reached 19.9%. The accuracy improved substantially from Exp. A to Exp. C (16.8% to 51.8%). The largest gain was observed for concordant score 1, where accuracy improved from 15.4% in Exp. A to 70.8% in Exp. C. Exp. D-EE showed a slight decline to 51.1% accuracy, marked by a 12.5 percentage point decrease in score 1 accuracy (from 72.5% to 60%) but a 5 percentage point improvement in score 2 accuracy (from 30% to 35%) compared to Exp. C-EE. Notably, the constant-score baseline (Exp. E) achieves 59% accuracy.

3.3.1 Wasserstein Distance

WD quantifies the differences between score distributions and serves as a rigorous metric to evaluate how closely ChatGPT-4’s score distribution aligns with the ratings assigned by neurologists. The Tables 3.4–3.6 show the WD calculations for 53 overall sessions under three ground truth scenarios: ratings from Neurologist X alone, ratings from Neurologist Y alone, and combined ratings from both neurologists. For the combined neurologists’ scenario, when the two neurologists assign different scores to a session, both scores are considered valid ground truth values, each weighted equally in the distribution. Table 3.7 shows the WD of 22 concordant sessions, where we use only the sessions where both neurologists assigned identical scores. The illustration of WD with a 95 % CI for 53 overall sessions and 22 concordant sessions is shown in Figure 3.3.

The WD values are consistently lower in Table 3.4 than Table 3.5, indicating that ChatGPT-4’s evaluations align more closely with Neurologist X’s ratings. Analyses of WD across

Table 3.3: Concordant score distribution and corresponding accuracies for each experiment, with accuracies weighted based on the prevalence of each score group from 0 to 4, which are the numbers in parentheses. “N” indicates no sessions with agreement for that score. The 95 % confidence intervals (CIs) were estimated via bootstrap sampling for Exp. A–E and A–EE–E–EE, and Monte Carlo resampling for Exp. F and F–EE (see Sections 2.8.2 and 2.9.1).

Exp	Accuracy (%)	95 % CI	Concordant Score (%)				
			0 (3)	1 (13)	2 (5)	3 (1)	4 (0)
A	16.8	[10.5, 24.1]	10	15.4	14	70	N
B	34.5	[26.8, 42.7]	10	43.8	26.0	30	N
C	51.8	[39.5, 63.2]	23.3	70.8	28	10	N
D	59.1	[49.1, 69.5]	53.3	63.1	44	100	N
E	59.0	[36.4, 77.3]	0	100	0	0	N
F	19.9	[14.6, 25.5]	20	19.8	19.9	19.9	N
F’s CI	—	—	[6.7, 36.7]	[13.1, 26.9]	[10, 32]	[0, 50]	N
			0 (2)	1 (12)	2 (4)	3 (0)	4 (0)
A-EE	14.4	[9.4, 21.1]	10	15.8	12.5	N	N
B-EE	36.1	[27.2, 45]	5	43.3	30	N	N
C-EE	58.3	[46.7, 68.9]	30	72.5	30	N	N
D-EE	51.1	[42.2, 60]	30	60	35	N	N
E-EE	66.0	[44.4, 88.9]	0	100	0	N	N
F-EE	19.9	[13.9, 26.1]	20.1	19.8	20	N	N
F-EE’s CI	—	—	[5, 40]	[13.3, 27.5]	[10, 35]	N	N

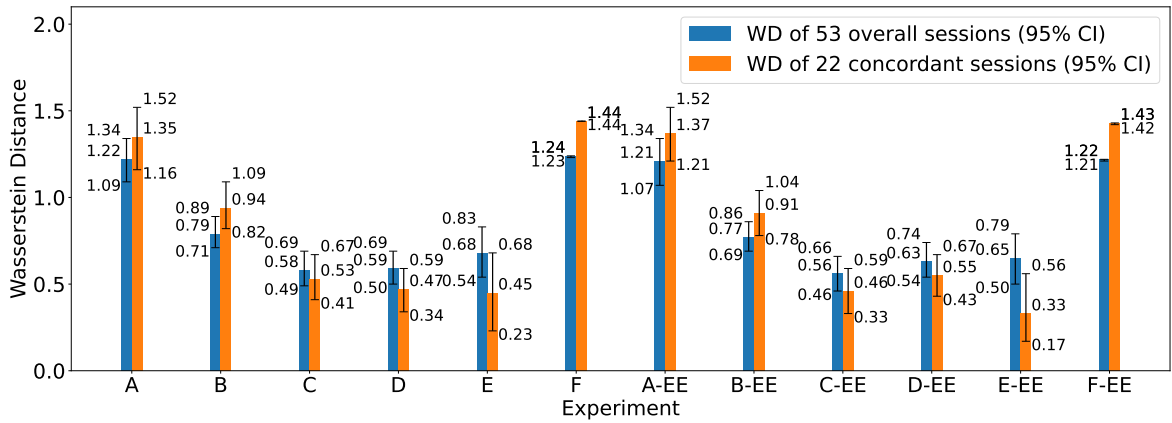


Figure 3.3: Wasserstein Distance Comparison with a 95 % Confidence Interval (95 % CI) for different experiments through 53 overall sessions and 22 concordant sessions. 95 % CI were obtained with the bootstrap procedure (Section 2.9.1) for Exp. A to Exp. E and Exp. A–EE to Exp. E–EE, and with the Monte Carlo random-score procedure (Section 2.8.2) for Experiments F and F–EE.

experiments are shown in Table 3.6, Table 3.7, and Figure 3.3. Exp. A yields WD values of 1.22 (53 sessions) and 1.35 (22 concordant sessions), comparable to the random-baseline

Table 3.4: Wasserstein Distance (WD) between ChatGPT-4 evaluations and Neurologist X’s ratings across 53 sessions. Results are reported overall and by neurologist’s assigned score group (0–4). Parentheses indicate the number of sessions per score. “N” denotes that no sessions were assigned that score. The 95 % confidence intervals (CIs) were estimated via bootstrap sampling and Monte Carlo resampling (see Sections 2.8.2 and 2.9.1).

Exp	WD	95 % CI	Ground Truth				
			0 (4)	1 (29)	2 (18)	3 (2)	4 (0)
A	1.26	[1.16, 1.37]	1.80	1.43	0.95	0.6	N
B	0.87	[0.77, 0.96]	1.38	0.86	0.79	0.65	N
C	0.56	[0.48, 0.65]	0.88	0.33	0.81	1.2	N
D	0.58	[0.49, 0.66]	0.73	0.5	0.68	0.5	N
E	0.49	[0.34, 0.64]	1	0	1	2	N
F	1.38	[1.3, 1.46]	2	1.4	1.2	1.4	N
F’s CI	—	—	[1.55, 2.45]	[1.29, 1.52]	[1.09, 1.32]	[0.95, 1.85]	N
			0 (3)	1 (28)	2 (17)	3 (1)	4 (0)
A-EE	1.27	[1.17, 1.37]	1.67	1.43	0.95	0.9	N
B-EE	0.85	[0.76, 0.94]	1.30	0.86	0.79	0.20	N
C-EE	0.54	[0.45, 0.64]	0.83	0.32	0.81	1.3	N
D-EE	0.62	[0.54, 0.70]	0.97	0.52	0.69	1.0	N
E-EE	0.45	[0.31, 0.59]	1	0	1	2	N
F-EE	1.37	[1.29, 1.45]	1.99	1.4	1.2	1.39	N
F-EE’s CI	—	—	[1.5, 2.5]	[1.29, 1.52]	[1.08, 1.32]	[0.8, 2.0]	N

Exp. F (1.24 and 1.44). There is a clear trend of decreasing WD from Exp. A to C. Among Exp. A-EE–F-EE, Exp. C-EE records WDs of 0.56 (53 sessions) and 0.46 (22 concordant sessions), whereas Exp. D-EE shows slightly higher values of 0.63 and 0.55, respectively. Exp. E shows the lowest WD in concordant sessions (0.45).

The Kolmogorov-Smirnov test (see Section 2.9.2) was used to statistically assess the differences in the distributions of WD across experiments, as presented in Tables 3.8 and 3.9. In Table 3.8, the p-values between Exp. A and Exp. F (0.890), and between Exp. C and Exp. D (0.998), are high. The p-value for the comparison between Exp. D-EE and Exp. E-EE also exceeds 0.05.

Table 3.5: Wasserstein Distance (WD) between ChatGPT-4 evaluations and Neurologist Y’s ratings across 53 sessions. Results are reported overall and by neurologist’s assigned score group (0–4). Parentheses indicate the number of sessions per score. “N” denotes that no sessions were assigned that score. The 95 % confidence intervals (CIs) were estimated via bootstrap sampling and Monte Carlo resampling (see Sections 2.8.2 and 2.9.1).

Exp	WD	95 % CI	Ground Truth				
			0 (15)	1 (19)	2 (8)	3 (10)	4 (1)
A	1.45	[1.31, 1.62]	2.2	1.45	0.94	0.72	1.7
B	1.07	[0.98, 1.18]	1.42	0.95	0.77	1.01	1.2
C	0.85	[0.72, 0.99]	0.97	0.36	0.8	1.5	2.3
D	0.85	[0.69, 1.00]	1.07	0.52	0.69	1.15	2.0
E	0.87	[0.68, 1.08]	1	0	1	2	3
F	1.55	[1.46, 1.65]	2	1.4	1.2	1.4	2
F’s CI	—	—	[1.77, 2.23]	[1.27, 1.54]	[1.04, 1.36]	[1.21, 1.59]	[1.1, 2.9]
			0 (14)	1 (18)	2 (7)	3 (9)	4 (1)
A-EE	1.47	[1.30, 1.65]	2.2	1.45	0.94	0.77	1.7
B-EE	1.07	[0.97, 1.18]	1.41	0.97	0.76	1.00	1.20
C-EE	0.85	[0.71, 0.99]	0.97	0.34	0.8	1.54	2.3
D-EE	0.91	[0.74, 1.06]	1.14	0.54	0.73	1.28	2.0
E-EE	0.86	[0.63, 1.08]	1	0	1	2	3
F-EE	1.55	[1.46, 1.65]	2	1.4	1.2	1.4	1.98
F-EE’s CI	—	—	[1.76, 2.24]	[1.27, 1.54]	[1.01, 1.37]	[1.19, 1.62]	[1.1, 2.9]

Table 3.6: Wasserstein distance (WD) with 95 % confidence intervals (CIs) for 53 overall sessions, where the ground truth distribution for each session incorporates both neurologists’ ratings with equal weighting. The 95 % CIs were estimated via bootstrap sampling, and Monte Carlo resampling (see Sections 2.8.2 and 2.9.1).

Exp	WD	95 % CI
A	1.22	[1.07, 1.35]
B	0.79	[0.71, 0.89]
C	0.58	[0.49, 0.69]
D	0.59	[0.50, 0.69]
E	0.68	[0.54, 0.83]
F	1.24	[1.16, 1.33]
A-EE	1.21	[1.07, 1.34]
B-EE	0.77	[0.69, 0.86]
C-EE	0.56	[0.46, 0.66]
D-EE	0.63	[0.54, 0.74]
E-EE	0.65	[0.50, 0.79]
F-EE	1.22	[1.13, 1.31]

Table 3.7: Wasserstein Distance (WD) between ChatGPT-4 evaluations and concordant scores across 22 sessions. Results are shown both overall and broken down by the concordant score groups (0-4). Parentheses indicate the number of sessions per score. “N” denotes that no sessions were assigned that score. The 95 % confidence intervals (CIs) were estimated via bootstrap sampling, and Monte Carlo resampling (see Sections 2.8.2 and 2.9.1).

Exp	WD	95 % CI	Ground Truth				
			0 (3)	1 (13)	2 (5)	3 (1)	4 (0)
A	1.35	[1.17, 1.52]	1.9	1.45	0.96	0.3	N
B	0.94	[0.82, 1.09]	1.43	0.87	0.80	1.1	N
C	0.53	[0.41, 0.67]	0.9	0.33	0.72	1.1	N
D	0.47	[0.34, 0.59]	0.60	0.42	0.6	0.0	N
E	0.45	[0.23, 0.68]	1.0	0.0	1.0	2.0	N
F	1.44	[1.31, 1.58]	2	1.41	1.2	1.4	N
F’s CI	—	—	[1.5, 2.5]	[1.25, 1.59]	[0.98, 1.4]	[0.8, 2]	N
			0 (2)	1 (12)	2 (4)	3 (0)	4 (0)
A-EE	1.37	[1.21, 1.52]	1.75	1.44	0.98	N	N
B-EE	0.91	[0.78, 1.04]	1.35	0.88	0.78	N	N
C-EE	0.46	[0.33, 0.59]	0.85	0.31	0.7	N	N
D-EE	0.55	[0.43, 0.67]	0.9	0.46	0.65	N	N
E-EE	0.33	[0.17, 0.56]	1.0	0.0	1.0	N	N
F-EE	1.43	[1.29, 1.58]	2	1.41	1.2	N	N
F-EE’s CI	—	—	[1.4, 2.6]	[1.23, 1.58]	[0.95, 1.45]	N	N

Table 3.8: Comparison of the Kolmogorov-Smirnov test p-values for Wasserstein distance across 53 overall sessions for each pair of experiments.

Exp	A	B	C	D	E	F
A	—	3.8×10^{-6}	2.5×10^{-9}	2.5×10^{-9}	4.1×10^{-7}	0.890
B	3.8×10^{-6}	—	9.4×10^{-4}	4.2×10^{-3}	4.2×10^{-3}	1.1×10^{-5}
C	2.5×10^{-9}	9.4×10^{-4}	—	0.998	1.6×10^{-2}	9.8×10^{-9}
D	2.5×10^{-9}	4.2×10^{-3}	0.998	—	1.6×10^{-2}	6.2×10^{-10}
E	4.1×10^{-7}	4.2×10^{-3}	1.6×10^{-2}	1.6×10^{-2}	—	4.1×10^{-7}
F	0.890	1.1×10^{-5}	9.8×10^{-9}	6.2×10^{-10}	4.1×10^{-7}	—

Table 3.9: Comparison of the Kolmogorov-Smirnov test p-values for Wasserstein distance across 49 sessions, excluding four example sessions, for each pair of experiments.

Exp	A-EE	B-EE	C-EE	D-EE	E-EE	F-EE
A-EE	—	1.2×10^{-6}	3.6×10^{-10}	6.8×10^{-9}	3.5×10^{-7}	0.861
B-EE	1.2×10^{-6}	—	1.1×10^{-3}	3.6×10^{-2}	5.2×10^{-3}	1.1×10^{-5}
C-EE	3.6×10^{-10}	1.1×10^{-3}	—	0.963	2.0×10^{-2}	6.8×10^{-9}
D-EE	6.8×10^{-9}	3.6×10^{-2}	0.963	—	0.063	2.7×10^{-8}
E-EE	3.5×10^{-7}	5.2×10^{-3}	2.0×10^{-2}	0.063	—	3.5×10^{-7}
F-EE	0.861	1.1×10^{-5}	6.8×10^{-9}	2.7×10^{-8}	3.5×10^{-7}	—

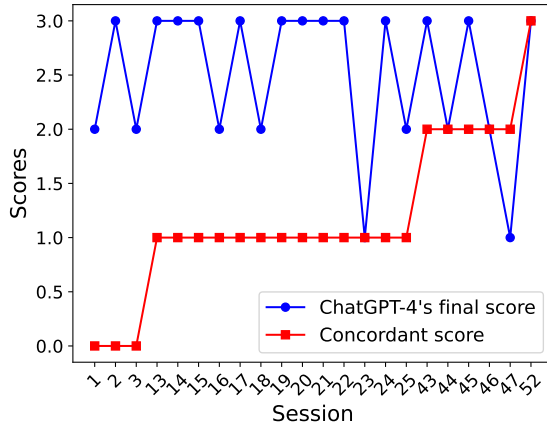
3.4 Agreement Between ChatGPT-4's Single Final Scores and Neurologists

We assign the final score for each concordant session by taking the median of the 10 evaluation scores provided by ChatGPT-4 and rounding this value to the nearest integer. This final score is used to determine the ChatGPT-4's overall performance and to facilitate comparisons with the scores assigned by neurologists.

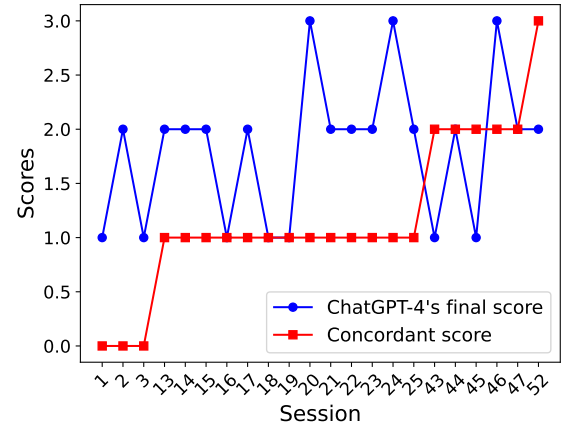
The final scores of ChatGPT-4 from different experiments are shown in Figures 3.4 and 3.5. In Exp. A, almost all of the final scores are 2 and 3. In Exp. B, the final scores are generally lower when the concordant scores are 0 and 1. However, the final scores still tend to be higher than the concordant scores for these categories. In Exp. C, 17 out of 22 final scores are 1, with only one score correctly aligning with an concordant score of 2. In the comparison of Exp. C-EE and Exp. D-EE, when the concordant score is 2, Exp. C-EE tends to give more scores of 1, while Exp. D-EE tends to give more scores of 3.

Acc, AAcc, MAE, RMSE, and WD are calculated for comparing the final score to the concordant scores from both neurologists in the concordant sessions, as shown in Table 3.10. There are only 22 concordant sessions out of the overall 53 sessions, with 13 of these sessions receiving a score of 1 from both neurologists. To expand the evaluation, we consider overall 53 sessions, setting the ground truth as the average rating from both neurologists. The final scores for ChatGPT-4 are calculated by taking the median of 10 evaluations. For the evaluation, we use MAE, RMSE, and WD, and the results are presented in Table 3.11.

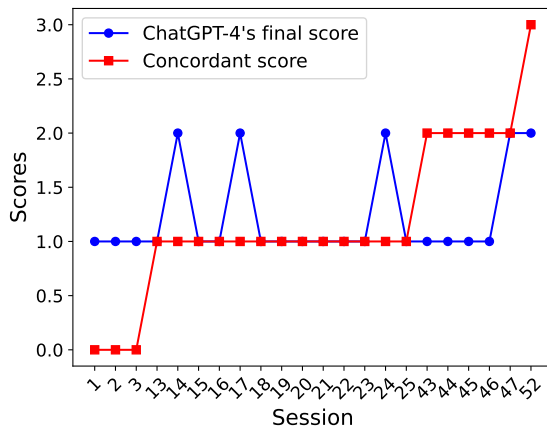
In concordant sessions (Table 3.10), Exp. A achieves only 18.2% accuracy and 45.5% acceptable accuracy, with high error rates ($MAE = 1.41$, $RMSE = 1.64$) and WD (1.32). This pattern persists in the 53 overall sessions (Table 3.11), where Exp. A yields high MAE (1.39), RMSE (1.56), and WD (1.29). The results of Exp. A are comparable to those of the random-score baseline (Exp. F). Exp. B through D improve the performance, with Exp. C achieving a 50% accuracy and an MAE of 0.50 in concordant sessions. Exp. D-EE records 61.1% accuracy, 100% acceptable accuracy, and the lowest error values ($MAE = 0.39$, $RMSE = 0.62$) and WD (0.39) in concordant sessions. In the overall sessions, it retains the lowest MAE (0.52), RMSE (0.71) and WD (0.21). Constant-based Exp. E records 59.1% accuracy in concordant sessions while maintaining low error values across both scenarios.



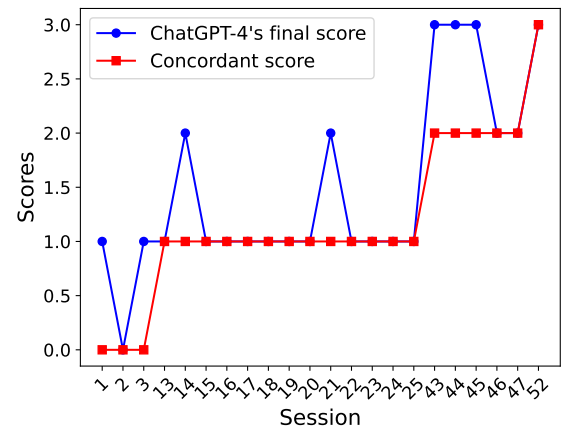
(a) Exp. A



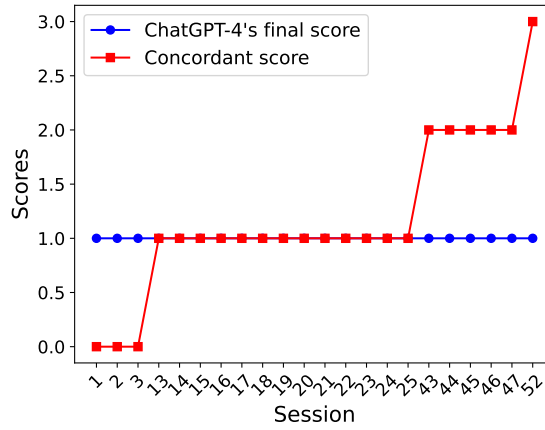
(b) Exp. B



(c) Exp. C

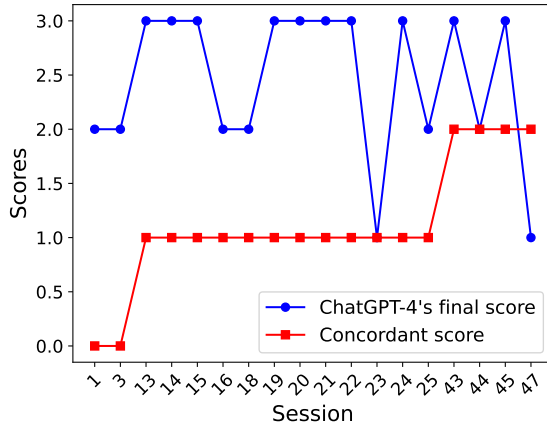


(d) Exp. D

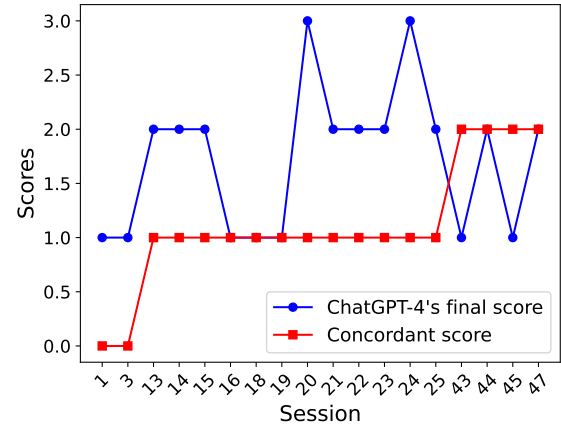


(e) Exp. E

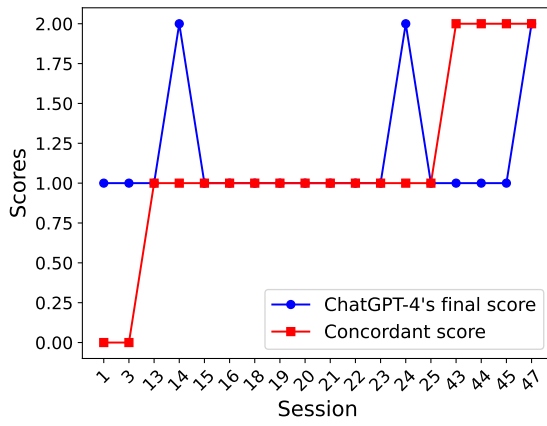
Figure 3.4: Comparison of ChatGPT-4's final scores in concordant sessions across various experiments. The final score for each session is determined by calculating the median of the 10 evaluation scores and then rounding to the nearest integer.



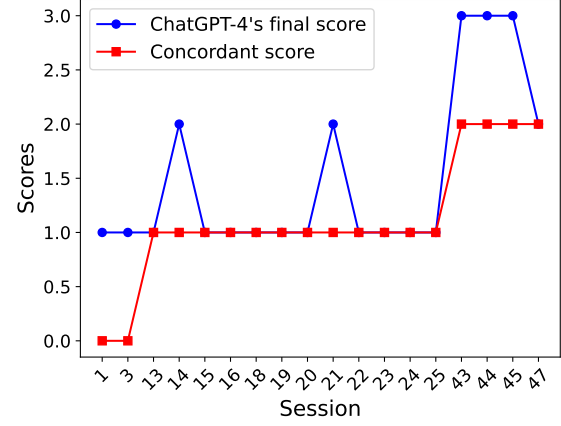
(a) Exp. A-EE



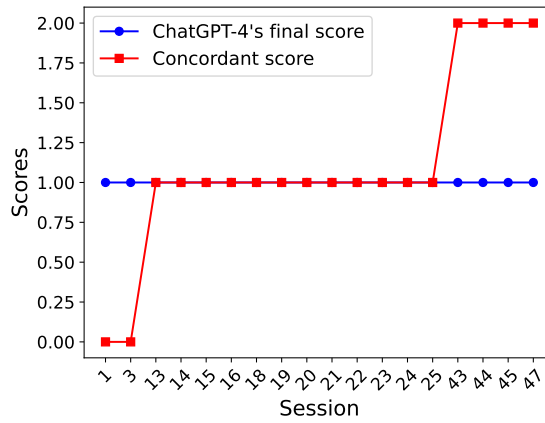
(b) Exp. B-EE



(c) Exp. C-EE



(d) Exp. D-EE



(e) Exp. E-EE

Figure 3.5: Comparison of ChatGPT-4's final scores in concordant sessions without example sessions across various experiments. The final score for each session is determined by calculating the median of the 10 evaluation scores and then rounding to the nearest integer.

Table 3.10: Comparison of final scoring performance across concordant sessions in different experiments. Each session’s final score is calculated by rounding the median of 10 evaluation scores to the nearest integer. The evaluation metrics include accuracy (Acc), acceptable accuracy (AAcc), mean absolute error (MAE), root mean square error (RMSE), and Wasserstein distance (WD). Exp. A through F each have 22 concordant sessions, while their counterparts excluding example sessions (Exp. A-EE to Exp. F-EE) have 18. Values are point estimates with 95 % confidence intervals (bootstrap for Exp. A–E and A-EE–E-EE; Monte Carlo for Exp. F and F-EE; see Sections 2.8.2 and 2.9.1 for details).

Exp	Acc (%)	AAcc (%)	MAE	RMSE	WD
A	18.2 [4.5, 36.3]	45.5 [22.7, 68.2]	1.41 [1.05, 1.77]	1.64 [1.33, 1.94]	1.32 [0.91, 1.73]
B	22.7 [9.1, 40.9]	86.4 [72.7, 100]	0.91 [0.68, 1.18]	1.09 [0.85, 1.3]	0.64 [0.32, 1]
C	50.0 [27.3, 68.2]	100 [100, 100]	0.50 [0.32, 0.73]	0.71 [0.56, 0.85]	0.23 [0.09, 0.5]
D	68.2 [50, 86.4]	100 [100, 100]	0.32 [0.14, 0.5]	0.56 [0.37, 0.71]	0.32 [0.14, 0.5]
E	59.1 [36.4, 77.3]	95.5 [86.4, 100]	0.45 [0.23, 0.73]	0.74 [0.48, 0.98]	0.45 [0.23, 0.73]
F	22.3 [9.1, 36.4]	66.7 [50.0, 81.8]	1.18 [0.86, 1.45]	1.45 [1.17, 1.72]	1.01 [0.68, 1.32]
A-EE	11.1 [0, 27.8]	44.4 [22.2, 66.7]	1.44 [1.11, 1.72]	1.60 [1.33, 1.81]	1.33 [0.89, 1.67]
B-EE	27.8 [11.1, 50]	88.9 [72.2, 100]	0.83 [0.56, 1.11]	1.03 [0.74, 1.27]	0.61 [0.22, 0.94]
C-EE	61.1 [38.9, 83.3]	100 [100, 100]	0.39 [0.17, 0.61]	0.62 [0.41, 0.78]	0.17 [0.06, 0.44]
D-EE	61.1 [38.9, 83.3]	100 [100, 100]	0.39 [0.17, 0.61]	0.62 [0.41, 0.78]	0.39 [0.17, 0.61]
E-EE	66.7 [44.4, 88.9]	100 [100, 100]	0.33 [0.11, 0.56]	0.58 [0.33, 0.75]	0.33 [0.11, 0.56]
F-EE	21.9 [5.6, 38.9]	67.1 [50, 83.3]	1.17 [0.83, 1.5]	1.43 [1.13, 1.73]	1.07 [0.72, 1.39]

Table 3.11: Comparison of final scores across 53 overall sessions in Exp. A –F, and 49 sessions in Exp. A-EE–F-EE. The final score for each session is determined by taking the median of 10 ChatGPT-4 evaluations, and the ground truth is the mean of two neurologists’ scores. Values are point estimates with 95 % confidence intervals (bootstrap for Exp. A–E and A-EE–E-EE; Monte Carlo for Exp. F and F-EE; see Sections 2.8.2 and 2.9.1 for details).

Exp	MAE	RMSE	WD
A	1.39 [1.19, 1.58]	1.56 [1.38, 1.73]	1.29 [1.04, 1.51]
B	0.70 [0.58, 0.83]	0.84 [0.71, 0.98]	0.38 [0.22, 0.58]
C	0.57 [0.44, 0.71]	0.75 [0.62, 0.88]	0.43 [0.28, 0.59]
D	0.48 [0.36, 0.61]	0.68 [0.55, 0.81]	0.20 [0.16, 0.37]
E	0.60 [0.45, 0.76]	0.83 [0.65, 1.02]	0.60 [0.45, 0.76]
F	1.01 [0.84, 1.18]	1.24 [1.07, 1.42]	0.68 [0.5, 0.88]
A-EE	1.39 [1.19, 1.57]	1.54 [1.36, 1.69]	1.31 [1.05, 1.53]
B-EE	0.68 [0.55, 0.82]	0.83 [0.69, 0.97]	0.38 [0.22, 0.57]
C-EE	0.54 [0.4, 0.67]	0.73 [0.58, 0.88]	0.42 [0.27, 0.58]
D-EE	0.52 [0.39, 0.65]	0.71 [0.56, 0.84]	0.21 [0.17, 0.39]
E-EE	0.57 [0.43, 0.74]	0.79 [0.61, 0.98]	0.57 [0.43, 0.74]
F-EE	1.00 [0.83, 1.16]	1.23 [1.04, 1.41]	0.7 [0.5, 0.9]

3.5 Inter- and Intra-Rater Reliability of ChatGPT-4 and Neurologists

We assessed reliability by comparing ChatGPT-4’s scoring consistency to the inter-rater reliability of neurologists, in order to evaluate whether ChatGPT-4 demonstrates sufficient stability relative to established clinical standards. Multiple statistical measures were employed for this analysis of ordinal data, including linear weighted Cohen’s κ , quadratic weighted Cohen’s κ , Krippendorff’s α , Intraclass Correlation Coefficient, Spearman’s ρ , and Kendall’s τ_b .

Tables 3.12 and 3.13 summarize ChatGPT-4’s intra-rater reliability across 53 sessions, with all reliability metrics computed from pairwise comparisons between evaluations. The key difference between the two tables lies in the method used to compute 95 % confidence intervals (CIs): session-level bootstrap for Table 3.12 and evaluation-level bootstrap for Table 3.13. ICC values were calculated using the ICC(2,1) model, as detailed in Section 2.10.7. Exp. A showed reliability values near zero across all metrics. Exp. B and Exp. C demonstrated incremental improvements. Exp. D-EE yielded the highest intra-rater reliability, with 0.33 LwCK, 0.45 QwCK, 0.44 Krippendorff’s α , 0.45 ICC(2,1), 0.44 Spearman’s ρ , and 0.4 Kendall’s τ_b . Exp. E assigned a constant score of “1” to all evaluations, resulting in perfect pairwise agreement. Exp. F, designed as a random baseline, exhibited reliability coefficients at or near 0. Across all metrics, ChatGPT-4’s intra-rater reliability in Exp. D-EE exceeded the inter-rater reliability between the two neurologists.

Table 3.14 presents the inter-rater reliability between ChatGPT-4’s final scores and the mean of the two neurologists’ scores across 53 sessions. ChatGPT-4 final scores were computed as the median of 10 evaluations. Exp. D-EE again achieved the highest values among all experiments, with 0.32 LwCK, 0.49 QwCK, 0.51 Krippendorff’s α , 0.67 ICC(2,k), 0.67 ICC(3,k), 0.53 Spearman’s ρ , and 0.45 Kendall’s τ_b . In Exp. E and Exp. E-EE, all evaluations were assigned a constant score of “1”, exhibited zero variance by design. Consequently, inter-rater reliability coefficients between the constant baseline and the neurologists’ scores either yielded low values or could not be computed, as shown in Tables 3.14 and 3.15. These limitations stem from the mathematical definitions of the reliability metrics, which require score variability.

Table 3.15 presents the inter-rater reliability between ChatGPT-4’s final scores and neurologists’ scores across 22 concordant sessions. For each session, the final score was calculated as the median of 10 evaluations, rounded to the nearest integer. The reliability was assessed using both ICC(2,k) and ICC(3,k), as described in Section 2.10.7. Exp. C showed improved reliability over Exp. A and Exp. B, while Exp. D-EE achieved the highest reliability values, with 0.43 LwCK, 0.63 QwCK, 0.68 Krippendorff’s α , 0.78 ICC(2,k), 0.85 ICC(3,k), 0.77 Spearman’s ρ , and 0.72 Kendall’s τ_b . It is worth noting that the ICC(3,k) values fell below -1 in several instances within the 1000 iterations for random-based Exp. F and F-EE. Finally, during our confidence interval estimation, Spearman’s ρ and Kendall’s τ_b could not be computed in several bootstrap iterations for Exp. C and Exp. C-EE. These failures occurred when bootstrap samples contained identical values for either ChatGPT-4 or neurologist scores, rendering the correlation metrics undefined.

Table 3.12: Intra-rater reliability of ChatGPT-4 and inter-rater reliability of two neurologists across all 53 sessions. “[—]” indicates that a confidence interval (CI) is not applicable because the value was fixed at 1.00 by design. Values are point estimates with 95 % confidence intervals (session-level bootstrap for Exp. A-E and A-EE-EE; Monte Carlo for Exp. F and F-EE; see Sections 2.8.2 and 2.9.1 for details). Metrics: LwCK = linear weighted Cohen’s κ , QwCK = quadratic weighted Cohen’s κ , α = Krippendorff’s α , ρ = Spearman’s ρ , τ_b = Kendall’s τ_b .

Exp	Reliability Metrics					
	LwCK	QwCK	α	ICC(2,1)	ρ	τ_b
A	0.04 [-0.23, 0.33]	0.05 [-0.29, 0.41]	0.05 [-0.3, 0.4]	0.05 [-0.3, 0.41]	0.05 [-0.3, 0.41]	0.04 [-0.27, 0.36]
B	0.11 [-0.17, 0.37]	0.15 [-0.18, 0.45]	0.14 [-0.21, 0.45]	0.16 [-0.19, 0.46]	0.15 [-0.2, 0.47]	0.13 [-0.18, 0.42]
C	0.14 [-0.13, 0.42]	0.20 [-0.15, 0.51]	0.22 [-0.17, 0.54]	0.21 [-0.16, 0.52]	0.22 [-0.16, 0.55]	0.20 [-0.15, 0.51]
D	0.39 [0.14, 0.63]	0.51 [0.21, 0.74]	0.49 [0.18, 0.74]	0.51 [0.21, 0.74]	0.49 [0.19, 0.75]	0.44 [0.16, 0.69]
E	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]
F	0 [-0.02, 0.03]	0 [-0.04, 0.04]	0 [-0.04, 0.04]	0 [-0.04, 0.04]	0 [-0.04, 0.04]	0 [-0.03, 0.03]
A-EE	0.04 [-0.24, 0.34]	0.05 [-0.3, 0.42]	0.05 [-0.31, 0.42]	0.05 [-0.31, 0.43]	0.05 [-0.31, 0.42]	0.04 [-0.28, 0.38]
B-EE	0.13 [-0.17, 0.39]	0.17 [-0.18, 0.49]	0.17 [-0.21, 0.49]	0.17 [-0.19, 0.49]	0.17 [-0.21, 0.51]	0.15 [-0.18, 0.45]
C-EE	0.13 [-0.15, 0.45]	0.2 [-0.18, 0.54]	0.21 [-0.21, 0.55]	0.21 [-0.19, 0.54]	0.21 [-0.21, 0.57]	0.20 [-0.19, 0.53]
D-EE	0.33 [0.05, 0.58]	0.45 [0.13, 0.7]	0.44 [0.1, 0.7]	0.45 [0.13, 0.7]	0.44 [0.11, 0.72]	0.40 [0.1, 0.66]
E-EE	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]
F-EE	0 [-0.03, 0.03]	0 [-0.04, 0.03]	0 [-0.04, 0.04]	0 [-0.04, 0.04]	0 [-0.04, 0.04]	0 [-0.03, 0.04]
Raters	0.27 [0.09, 0.42]	0.36 [0.11, 0.55]	0.34 [0.09, 0.54]	0.37 [0.11, 0.56]	0.36 [0.09, 0.58]	0.32 [0.08, 0.53]

Table 3.13: Intra-rater reliability of ChatGPT-4 and inter-rater reliability of two neurologists across all 53 sessions. “[—]” indicates that a confidence interval (CI) is not applicable because the value was fixed at 1.00 by design. Values are point estimates with 95 % confidence intervals (evaluation-level bootstrap for Exp. A–E and A–EE–E–EE; Monte Carlo for Exp. F and F–EE; see Sections 2.8.2 and 2.9.1 for details). Abbreviations as in Table 3.12.

Exp	Reliability Metrics					
	LwCK	QwCK	α	ICC(2,1)	ρ	τ_b
A	0.04 [−0.08, 0.35]	0.05 [−0.13, 0.41]	0.05 [−0.13, 0.4]	0.05 [−0.14, 0.42]	0.05 [−0.13, 0.41]	0.04 [−0.12, 0.37]
B	0.11 [0, 0.4]	0.15 [−0.02, 0.48]	0.14 [−0.02, 0.48]	0.16 [−0.02, 0.49]	0.15 [−0.02, 0.49]	0.13 [−0.02, 0.44]
C	0.14 [0.01, 0.44]	0.20 [0.02, 0.54]	0.22 [0.05, 0.54]	0.21 [0.03, 0.54]	0.22 [0.05, 0.54]	0.20 [0.05, 0.51]
D	0.39 [0.29, 0.61]	0.51 [0.37, 0.72]	0.49 [0.35, 0.71]	0.51 [0.38, 0.72]	0.49 [0.35, 0.72]	0.44 [0.32, 0.67]
E	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]
F	0.00 [−0.02, 0.03]	0.00 [−0.04, 0.04]	0.00 [−0.04, 0.04]	0.00 [−0.04, 0.04]	0.00 [−0.04, 0.04]	0.00 [−0.03, 0.03]
A–EE	0.04 [−0.08, 0.36]	0.05 [−0.14, 0.43]	0.05 [−0.13, 0.42]	0.05 [−0.15, 0.43]	0.05 [−0.14, 0.42]	0.04 [−0.12, 0.38]
B–EE	0.13 [0, 0.42]	0.17 [−0.01, 0.5]	0.17 [−0.01, 0.5]	0.17 [−0.01, 0.51]	0.17 [−0.01, 0.51]	0.15 [−0.01, 0.45]
C–EE	0.13 [−0.01, 0.45]	0.2 [0, 0.55]	0.21 [0.02, 0.54]	0.21 [0, 0.56]	0.21 [0.02, 0.55]	0.20 [0.02, 0.52]
D–EE	0.33 [0.2, 0.57]	0.45 [0.29, 0.69]	0.44 [0.27, 0.68]	0.45 [0.29, 0.69]	0.44 [0.27, 0.69]	0.40 [0.24, 0.64]
E–EE	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]	1.00 [—]
F–EE	0.00 [−0.03, 0.03]	0.00 [−0.04, 0.03]	0.00 [−0.04, 0.04]	0.00 [−0.04, 0.04]	0.00 [−0.04, 0.04]	0.00 [−0.03, 0.04]
Raters	0.27 [0.09, 0.42]	0.36 [0.11, 0.55]	0.34 [0.09, 0.54]	0.37 [0.11, 0.56]	0.36 [0.09, 0.58]	0.32 [0.08, 0.53]

Table 3.14: Inter-rater reliability of ChatGPT-4’s final scores (median of 10 evaluations) with the neurologists’ mean scores across overall 53 sessions. Values are point estimates with 95 % confidence intervals (bootstrap for Exp. A–E and A–EE–E–EE; Monte Carlo for Exp. F and F–EE; see Sections 2.8.2 and 2.9.1 for details). “N” indicates that Spearman’s ρ and Kendall’s τ_b were not computable because the ChatGPT-4 final scores were constant. Abbreviations as in Table 3.12.

Exp	Reliability Metrics					
	LwCK	QwCK	α	ICC(2,k)	ICC(3,k)	ρ τ_b
A	0.01 [-0.06, 0.07]	0.05 [-0.05, 0.15]	-0.38 [-0.59, -0.15]	0.1 [-0.11, 0.26]	0.25 [-0.36, 0.55]	0.15 [-0.15, 0.43] 0.14 [-0.12, 0.37]
B	0.17 [0.03, 0.3]	0.39 [0.18, 0.55]	0.37 [0.11, 0.55]	0.57 [0.31, 0.72]	0.61 [0.35, 0.77]	0.48 [0.23, 0.68] 0.39 [0.18, 0.57]
C	0.13 [0, 0.24]	0.26 [0.08, 0.4]	0.26 [0.03, 0.43]	0.41 [0.16, 0.57]	0.43 [0.17, 0.58]	0.35 [0.09, 0.54] 0.30 [0.09, 0.47]
D	0.41 [0.24, 0.54]	0.60 [0.39, 0.72]	0.57 [0.36, 0.73]	0.74 [0.57, 0.84]	0.75 [0.58, 0.85]	0.60 [0.39, 0.75] 0.51 [0.34, 0.66]
E	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	-0.04 [-0.18, 0.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	N N
F	0.00 [-0.11, 0.12]	0.00 [-0.18, 0.2]	-0.17 [-0.4, 0.08]	-0.02 [-0.47, 0.34]	-0.03 [-0.71, 0.44]	0 [-0.26, 0.29] 0 [-0.21, 0.23]
A-EE	0.01 [-0.05, 0.07]	0.07 [-0.03, 0.17]	-0.38 [-0.6, -0.15]	0.13 [-0.06, 0.3]	0.33 [-0.2, 0.62]	0.22 [-0.07, 0.5] 0.19 [-0.06, 0.43]
B-EE	0.17 [0.01, 0.32]	0.38 [0.15, 0.56]	0.36 [0.11, 0.55]	0.56 [0.27, 0.72]	0.61 [0.29, 0.78]	0.47 [0.18, 0.68] 0.39 [0.15, 0.58]
C-EE	0.12 [0, 0.27]	0.22 [0.07, 0.37]	0.24 [0.03, 0.43]	0.36 [0.14, 0.53]	0.38 [0.15, 0.54]	0.33 [0.11, 0.54] 0.28 [0.09, 0.46]
D-EE	0.32 [0.17, 0.46]	0.49 [0.31, 0.64]	0.51 [0.26, 0.67]	0.67 [0.48, 0.79]	0.67 [0.49, 0.8]	0.53 [0.3, 0.71] 0.45 [0.25, 0.62]
E-EE	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	-0.04 [-0.18, 0.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	N N
F-EE	0 [-0.12, 0.13]	0 [-0.18, 0.2]	-0.18 [-0.41, 0.08]	-0.01 [-0.46, 0.34]	-0.04 [-0.7, 0.46]	0 [-0.27, 0.29] 0 [-0.22, 0.23]

Table 3.15: Inter-rater reliability between ChatGPT-4’s final scores (median of 10 evaluations, rounded to the nearest integer) and the neurologists’ concordant scores across the 22 concordant sessions. Values are point estimates with 95 % confidence intervals (bootstrap for Exp. A–E and A–EE–E–EE; Monte Carlo for Exp. F and F–EE; see Sections 2.8.2 and 2.9.1 for details). “N” indicates that Spearman’s ρ and Kendall’s τ_b were not computable because the ChatGPT-4 final scores were constant. “[—]” indicates that the confidence interval could not be computed because, in some resampling iterations, one of the two score vectors was constant; with such complete ties Spearman’s ρ and Kendall’s τ_b are undefined. Abbreviations as in Table 3.12.

Exp	Reliability Metrics					
	LwCK	QwCK	α	ICC(2,k)	ICC(3,k)	τ_b
A	0 [-0.13, 0.1]	0 [-0.15, 0.13]	-0.47 [-0.74, -0.18]	0 [-0.39, 0.24]	0 [-1.54, 0.56]	0 [-0.41, 0.41]
B	-0.02 [-0.23, 0.15]	0.12 [-0.17, 0.33]	-0.05 [-0.46, 0.28]	0.22 [-0.43, 0.51]	0.29 [-0.65, 0.67]	0.16 [-0.29, 0.56]
C	0.10 [-0.19, 0.38]	0.28 [-0.09, 0.52]	0.25 [-0.11, 0.54]	0.44 [-0.21, 0.69]	0.43 [-0.21, 0.71]	0.26 [—] 0.25 [—]
D	0.61 [0.38, 0.81]	0.76 [0.57, 0.89]	0.79 [0.53, 0.93]	0.87 [0.74, 0.94]	0.90 [0.81, 0.95]	0.84 [0.66, 0.95]
E	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0 [-0.19, 0.02]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	N
F	0 [-0.16, 0.15]	-0.01 [-0.22, 0.21]	-0.32 [-0.61, 0.02]	-0.04 [-0.62, 0.36]	-0.13 [-1.41, 0.57]	-0.01 [-0.43, 0.4]
A-EE	-0.04 [-0.19, 0.04]	0 [-0.17, 0.12]	-0.52 [-0.76, -0.25]	0.01 [-0.44, 0.23]	0.03 [-1.53, 0.71]	0.07 [-0.47, 0.59]
B-EE	-0.06 [-0.33, 0.2]	0.06 [-0.28, 0.32]	-0.13 [-0.54, 0.24]	0.11 [-0.85, 0.49]	0.15 [-1.42, 0.68]	0.08 [-0.37, 0.45]
C-EE	0.09 [-0.22, 0.47]	0.16 [-0.17, 0.5]	0.18 [-0.17, 0.53]	0.29 [-0.46, 0.68]	0.28 [-0.44, 0.7]	0.17 [—] 0.17 [—]
D-EE	0.43 [0.18, 0.63]	0.63 [0.4, 0.74]	0.68 [0.27, 0.88]	0.78 [0.59, 0.86]	0.85 [0.68, 0.9]	0.77 [0.45, 0.93]
E-EE	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.01 [-0.17, 0.03]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	N
F-EE	0.00 [-0.14, 0.12]	-0.01 [-0.21, 0.2]	-0.39 [-0.69, -0.02]	-0.04 [-0.57, 0.35]	-0.16 [-1.71, 0.61]	-0.02 [-0.46, 0.43]
						-0.01 [-0.43, 0.4]

Chapter 4

Discussion

4.1 Principal Findings

ChatGPT-4 evaluated finger-tapping severity with 61.1% accuracy, 0.39 MAE, 0.45 intra-rater ICC(2,1), and 0.85 inter-rater ICC(3,k), matching or higher than the metric values of neurologists reported in previous studies. Performance improved progressively from Exp. A (task description and 10-second data) to Exp. D-EE (task description, 10-tap data, Python-based feature extraction, and few-shot prompting). Exp. A performed similarly to the random baseline (Exp. F), underscoring the critical role of prompt engineering. The constant-score baseline (Exp. E) performs better than Exp. D-EE in some metrics, highlighting the importance of addressing class imbalance when evaluating generalization performance. These findings demonstrate that prompt engineering is essential for enhancing the accuracy and reliability of ChatGPT-4 in this medical scoring task.

4.2 Interpretation of Performance Trends

4.2.1 ChatGPT-4's Intra-rater and Inter-rater Reliability

Under optimal prompting (Exp. D-EE), ChatGPT-4 achieved an intra-rater ICC(2,1) of 0.45, surpassing the inter-rater reliability of the two neurologists ($\text{ICC}(2,1) = 0.37$). This suggests that ChatGPT-4, when appropriately configured, yielded more consistent scores than the neurologist ratings in this study. In contrast, Exp. A—using only basic task descriptions

—yielded near-zero reliability, comparable to the random baseline (Exp. F), underscoring the necessity of prompt engineering. In Exp. D-EE, inter-rater reliability between ChatGPT-4 and neurologists reached 0.78 ICC(2,k) and 0.85 ICC(3,k), indicating good reliability. The random-based configurations (Exp. F and F-EE) performed poorly, with ICC(3,k) occasionally falling below -1 across 1000 iterations. This was due to a central tendency bias: the final score in each session tended to cluster near 2, while the neurologists’ scores spanned a broader range (0–3), resulting in significant disagreement. The wide confidence intervals observed for both intra-rater and inter-rater reliability metrics reflect substantial variability, which stems from the metrics’ sensitivity to both consistency between ratings and absolute differences in score values.

4.2.2 ChatGPT-4’s Final Score Accuracy and Error

ChatGPT-4’s accuracy and error metrics improved monotonically from Exp. A to Exp. C and Exp. D-EE, reflecting the benefits of refined prompt design. The constant-score baseline (Exp. E) attained 59.1% accuracy primarily due to class imbalance, suggesting the need for further analysis when evaluating models on skewed distributions.

4.2.3 ChatGPT-4’s Evaluation Consistency and Agreement

The standard deviation across ten evaluations decreased from 0.97 (Exp. A) to 0.56 (Exp. C) and 0.57 (Exp. D-EE), reflecting improved internal consistency. Wasserstein Distance between ChatGPT-4 and neurologist scores also declined—from 1.22 (Exp. A) to 0.58 (Exp. C) and 0.63 (Exp. D-EE)—demonstrating improved alignment with neurologists.

4.3 Comparison with Prior Work

The inter-rater ICC(3,k) of 0.85 achieved in Exp. D-EE is higher than the inter-rater reliability values reported by Kenny et al. (2024) (0.55; six mixed clinicians) and Williams et al. (2023) (0.58; 21 neurologists), and approaches the 0.88 reported by Islam et al. (2023) for three neurologists (see Table 1.3). Similarly, the Krippendorff’s α of 0.68 in our study is close to the inter-rater value of 0.69 reported by Islam et al. (2023) (see Table 1.4), and our

Spearman's ρ of 0.78 is higher than their reported value of 0.64, which reflects agreement between an AI model and human raters (see Table 1.5). Intra-rater ICC(2,1) for ChatGPT-4 was 0.45, which is higher than the inter-rater ICC of 0.32 reported by Kenny et al. (2024), where eight clinicians rated sessions before reviewing the four high-variance videos and discussing scoring criteria. In terms of prediction accuracy, Exp. D-EE achieved 61.1% accuracy and 0.39 MAE, higher than the values reported for SVR (51.9% accuracy) and LightGBM (0.58 MAE) by Islam et al. (2023). While these results are favorable, direct comparison should be interpreted cautiously due to differences in subject characteristics and rater backgrounds between datasets.

4.4 Clinical Implications

This study explores the potential of using ChatGPT-4 to evaluate finger-tapping severity in Parkinson's disease. ChatGPT-4 demonstrated relatively stable scoring outputs under specific configurations—namely, when provided with a task description, instructions for tapping feature extraction, few-shot examples, and 10-tap finger-tapping motion data (as in Exp. D-EE).

Before this approach can be considered for clinical application, further studies are required to validate its performance across larger and more diverse patient populations. Testing in real-world clinical settings is especially important to assess its reliability and generalizability. In the future, such methods may offer support in remote assessments or routine screenings.

4.5 Strengths and Limitations

This study introduces a systematic framework for evaluating ChatGPT-4 in assessing finger-tapping severity in Parkinson's disease. A key strength lies in the use of real-world subject videos collected under an IRB-approved protocol in a standardized environment setting, ensuring consistency. A series of controlled experiments (Exp. A–D and A-EE–D-EE) systematically evaluated the effects of data representation (10-second vs. 10-tap segments), feature extraction (via Python-based signal processing), and prompt design (few-shot examples), thereby establishing a clear link between prompt engineering and ChatGPT-4 performance.

on the finger-tapping scoring task. By including both constant and random-score baselines, the study provides essential context for a more comprehensive interpretation of the model’s performance.

In addition, multiple evaluation metrics were used—including accuracy, MAE, RMSE, Wasserstein Distance, rater reliability metrics such as ICC and Krippendorff’s α —all reported with 95% confidence intervals. This comprehensive analysis contributes to a more nuanced understanding of the model’s consistency and agreement with neurologists.

Nevertheless, the study has several limitations. The cohort comprised only 53 analyzable sessions, and the class distribution was notably imbalanced, with few moderate cases and no instances of severe impairment. This imbalance limits the generalizability of the findings, particularly to clinical populations with more severe motor symptoms. Furthermore, model evaluation was conducted in a highly controlled environment, and its robustness in real-world settings remains to be tested. Finally, the ground truth was derived from ratings provided by only two neurologists. While both raters were experienced and blinded to each other’s assessments, the limited rater pool may restrict the generality of the reference standard.

To partially mitigate this limitation, following the primary analysis, we invited a third movement disorder specialist (Neurologist Z, NCKUH) to independently rate the same video recordings using the identical folder structure and scoring protocol. While Neurologist Z’s evaluations were not incorporated into the formal analyses presented in this thesis, they were archived for potential use in future studies (see Appendix I).

4.6 Methodological Considerations

First, the scoring instructions provided to ChatGPT consisted of the complete, unmodified text from the official MDS-UPDRS finger-tapping criteria. This ensured consistency with clinical standards and facilitated the model’s interpretation of the rating guidelines. Nonetheless, the accuracy of scoring still depends on the model’s ability to comprehend descriptive language related to motor behavior.

Second, due to the limited sample size, the few-shot examples used in the prompts were manually selected rather than drawn from a representative dataset. In Exp. C, we selected the lowest-performing session from each class, provided both neurologists had independently

given it the same score. This strategy aimed to help the model associate real examples with their respective classes, including edge cases, and to learn relevant features without relying solely on ideal cases.

Third, to obtain stable estimates, each session was evaluated ten times by ChatGPT-4. As a large language model, it can produce slightly different outputs even with identical inputs. Repeating the evaluation allowed us to assess scoring consistency. Users should be aware of this variability and potential behavioral differences across model versions.

Chapter 5

Conclusion

This study demonstrates the feasibility of using ChatGPT-4, a general-purpose large language model, to assess the severity of finger-tapping performance in Parkinson’s disease. Among the evaluated configurations, Experiment D-EE—combining task description, 10-tap alignment, Python-based feature extraction, and few-shot prompting—yielded the best results, with 61.1% accuracy, a MAE of 0.39, intra-rater ICC(2,1) of 0.45, and inter-rater ICC(3,k) of 0.85.

These results indicate that ChatGPT-4, when appropriately guided, can produce scores that align well with neurological assessments. In particular, the reliability metrics in Exp. D-EE exceeded those of the two neurologists involved in this study and matched or higher than the values reported in prior literature. However, given the limited dataset size, class imbalance, and narrow rater pool, such comparisons should be interpreted with caution.

A key finding of this work is the critical role of prompt engineering in optimizing large language model behavior for clinical scoring tasks. Performance varied significantly across prompt design and data representations, underscoring the sensitivity of LLMs to input design. This reinforces the need for systematic prompt calibration in future LLM-based clinical tools.

While the findings are promising, broader validation is needed. Larger, more diverse datasets, expanded rater numbers, and evaluation across multiple LLM architectures will be necessary to assess generalizability. This study presents a structured evaluation framework for characterizing the behavior of LLMs in a medically-relevant scoring task. By analyzing consistency, agreement, and sensitivity to prompt design, our findings contribute to a growing understanding of how LLMs can be systematically assessed and refined for use in clinical

research settings.

5.1 Future Work

Building on the findings of this study, several directions may guide future research. First, we plan to standardize the evaluation pipeline by analyzing 10-second data segments aligned to the onset of finger-tapping movement, and compare to the 10-tap data. This adjustment may improve temporal consistency by ensuring that analysis is anchored to a common starting point across sessions. Second, expanding the dataset to include a larger number of subjects with more balanced score distributions—including moderate and severe impairment levels—will be critical for improving generalizability and addressing the class imbalance observed in this work.

Third, increasing the number of independent clinical raters could help establish a more robust reference standard and support further investigation into inter-rater reliability beyond the two neurologists analyzed in this study. Finally, future studies should evaluate and compare multiple large language models to better understand their relative strengths and prompting behavior. Together, these extensions would enable a more comprehensive and comparative analysis of LLM-based evaluation in clinically relevant applications.

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Appendix A

Copyright Approval from MDS-UPDRS

We received approval from the **International Parkinson and Movement Disorder Society** to utilize the Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS) content in our study, as shown in Figure A.1. The use of MDS-UPDRS content in our study includes:

1. Inputting the MDS-UPDRS Finger Tapping (FT) content into ChatGPT-4 to serve as the basis for scoring criteria.
2. Detailing the MDS-UPDRS FT content in Section 2.4.3 of our thesis, as well as visualizing the prompt item: MDS-UPDRS Scoring Guideline in Table 2.4.

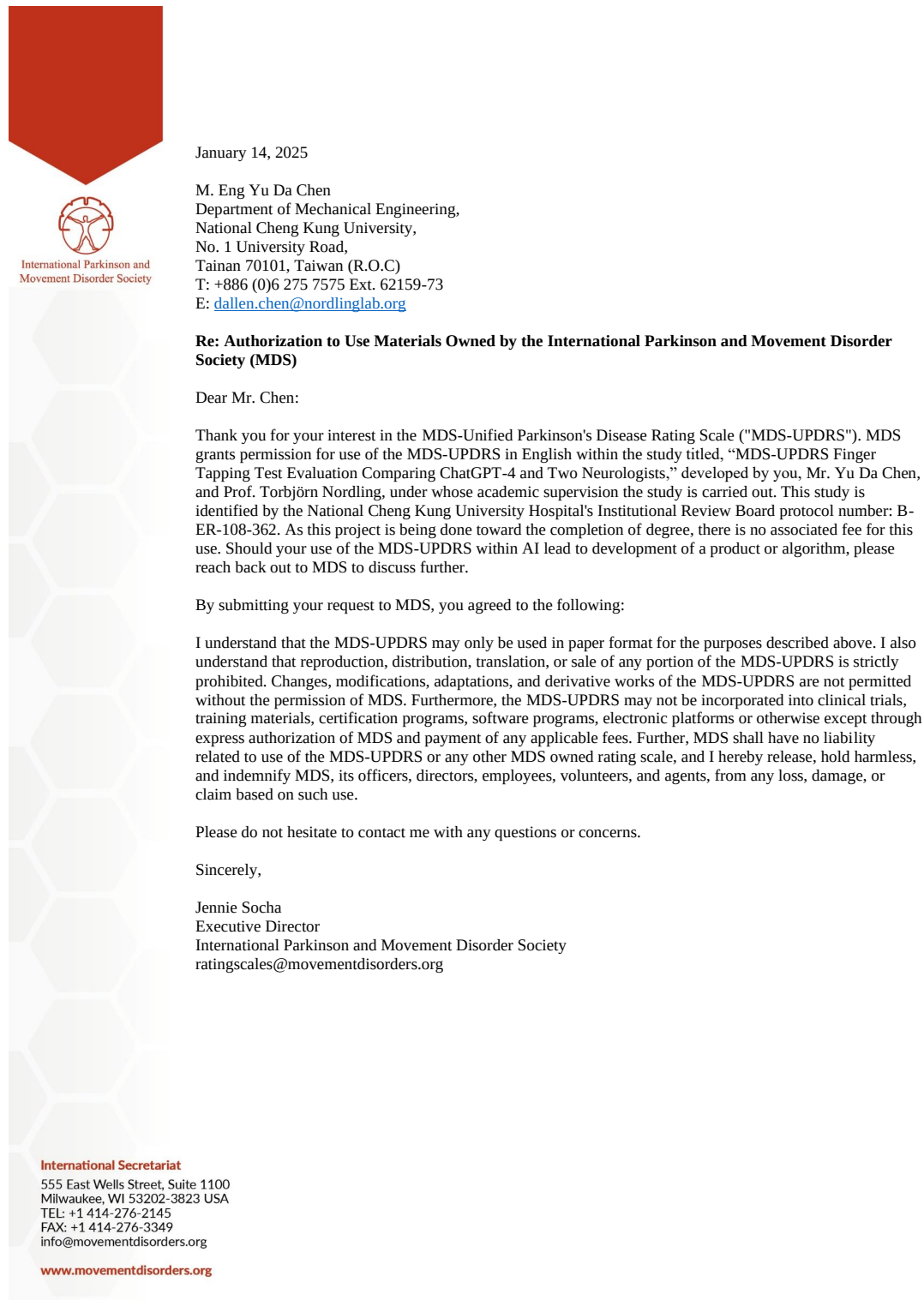


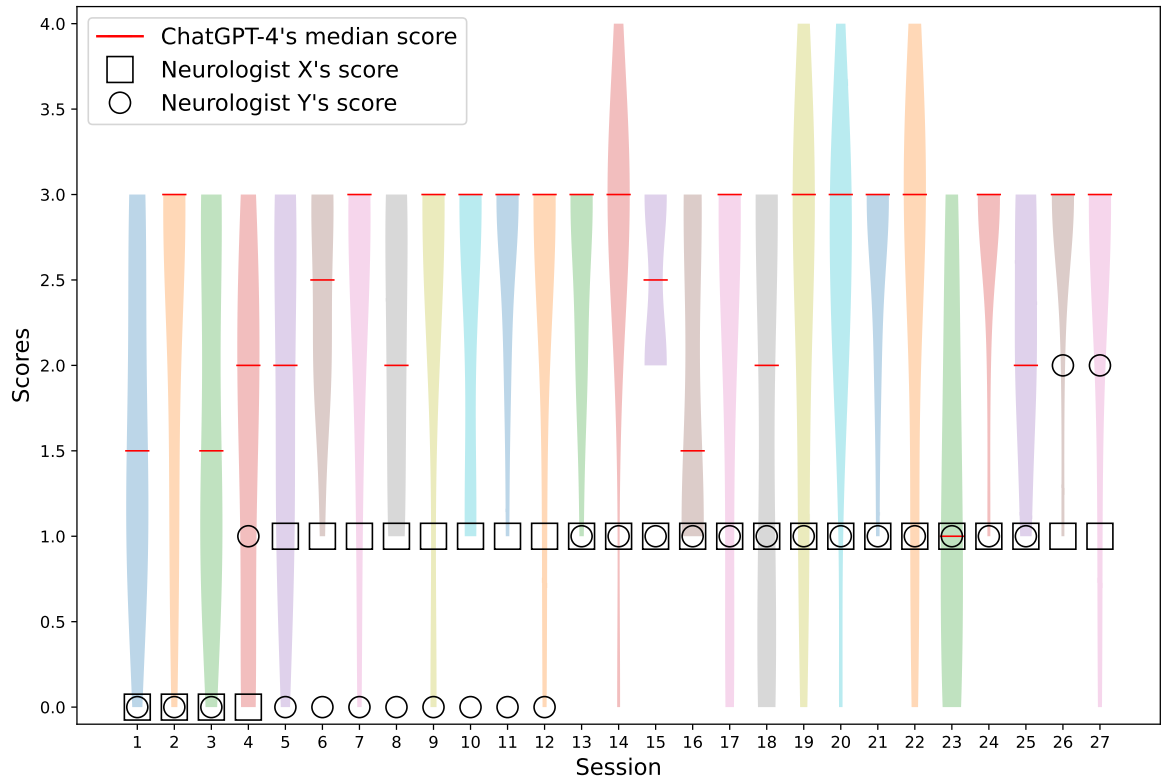
Figure A.1: Approval document for the use of MDS-UPDRS content in the study

Appendix B

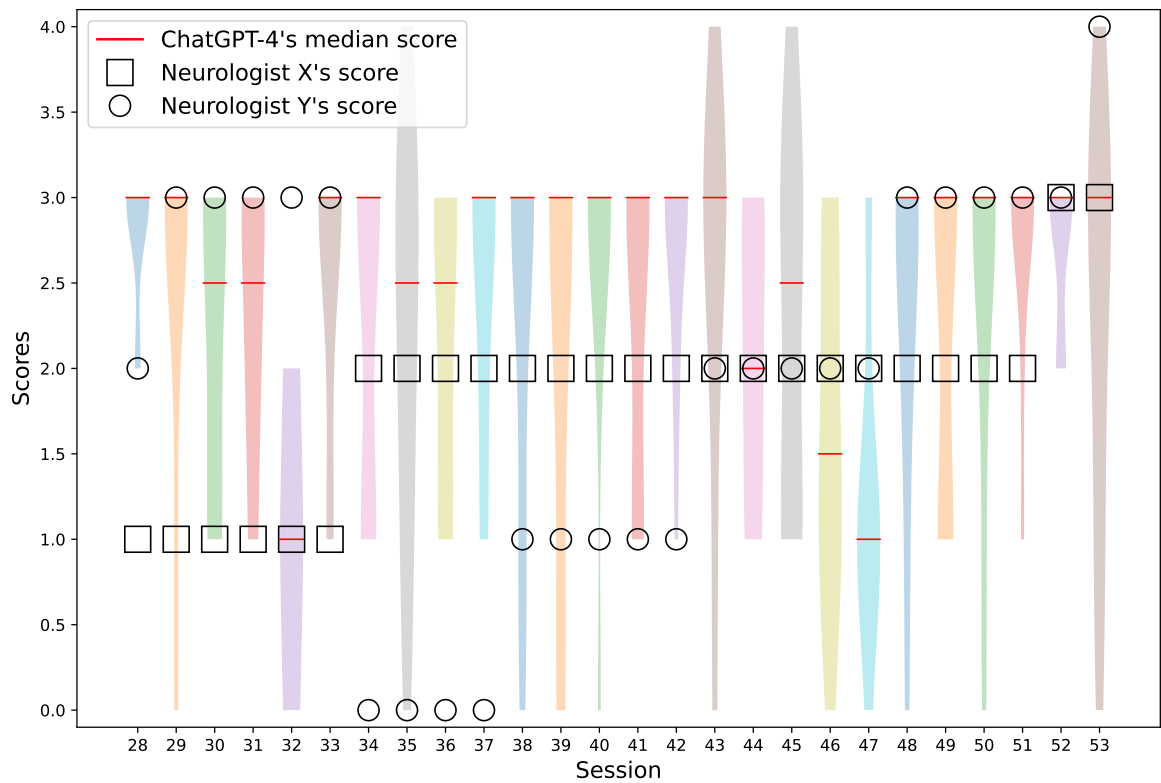
The Output Scores of ChatGPT-4

The 10 evaluations of ChatGPT-4 predicted scores, along with scores from two neurologists for 53 sessions in Exp A-D, are shown in Figures B.1-B.4, respectively. We use violin plots to depict the distribution of the ChatGPT-4 predicted scores and mark the scores of the two neurologists in each session. All scores across these experiments are integers ranging from 0 to 4.

The width of the violin in the plot correlates with the frequency of scores for each session. For instance, in session number 1 of Figure B.1, the plot is wider at score 1, which appears in 4 evaluations, and narrower at score 0, with only 1 evaluation. Scores 3 and 2 appear in 3 and 2 evaluations, respectively. Similarly, in session number 27 of Figure B.1, the plot significantly widens at score 3, which appears in 7 evaluations. Although scores 0, 1, and 2 each appear only once, the plot for score 2 does not narrow as sharply as scores 0 and 1. This is because the violin plot is a continuous curve, and the presence of a high frequency at score 3 smooths the transition, preventing score 2 from appearing as narrow. This can be seen as mislead but is motivated by the expected values. For sessions where no violin plot is displayed, this indicates a uniform distribution of scores across all 10 evaluations, making a violin plot redundant. For instance, session number 2 in Figure B.4 consistently shows a score of 0 in all evaluations, marked by a red line indicating the median score, without a corresponding violin plot.

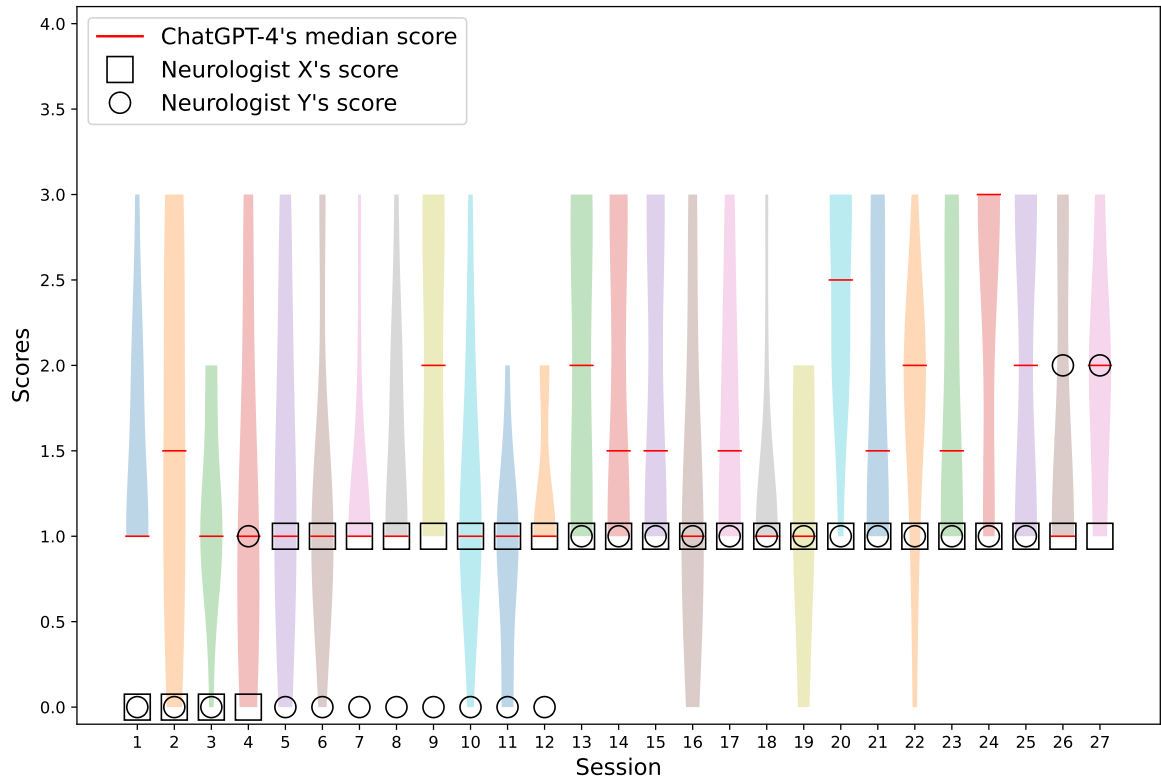


(a) First part of Experiment A.

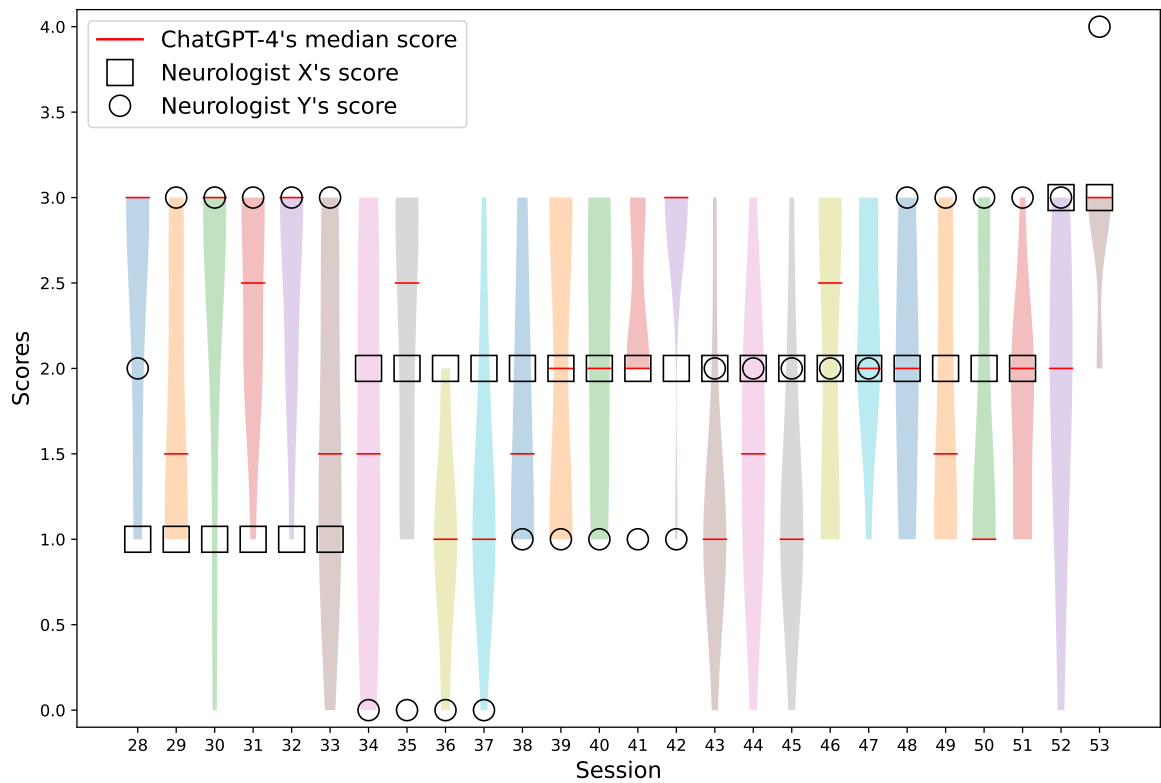


(b) Second part of Experiment A.

Figure B.1: Distribution of ChatGPT-4 predicted scores and scores assigned by neurologists across all sessions in Experiment A, visualized in two parts. This figure displays violin plots that indicate the variability and frequency of the scores.

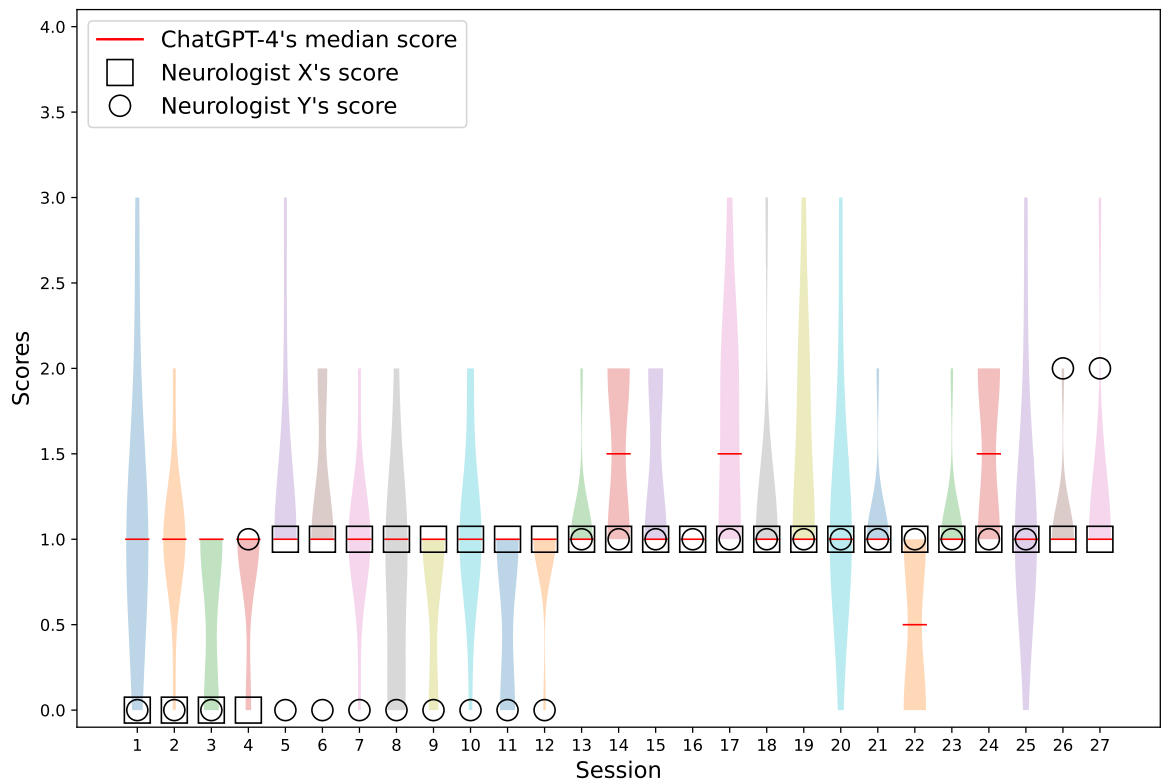


(a) First part of Experiment B.

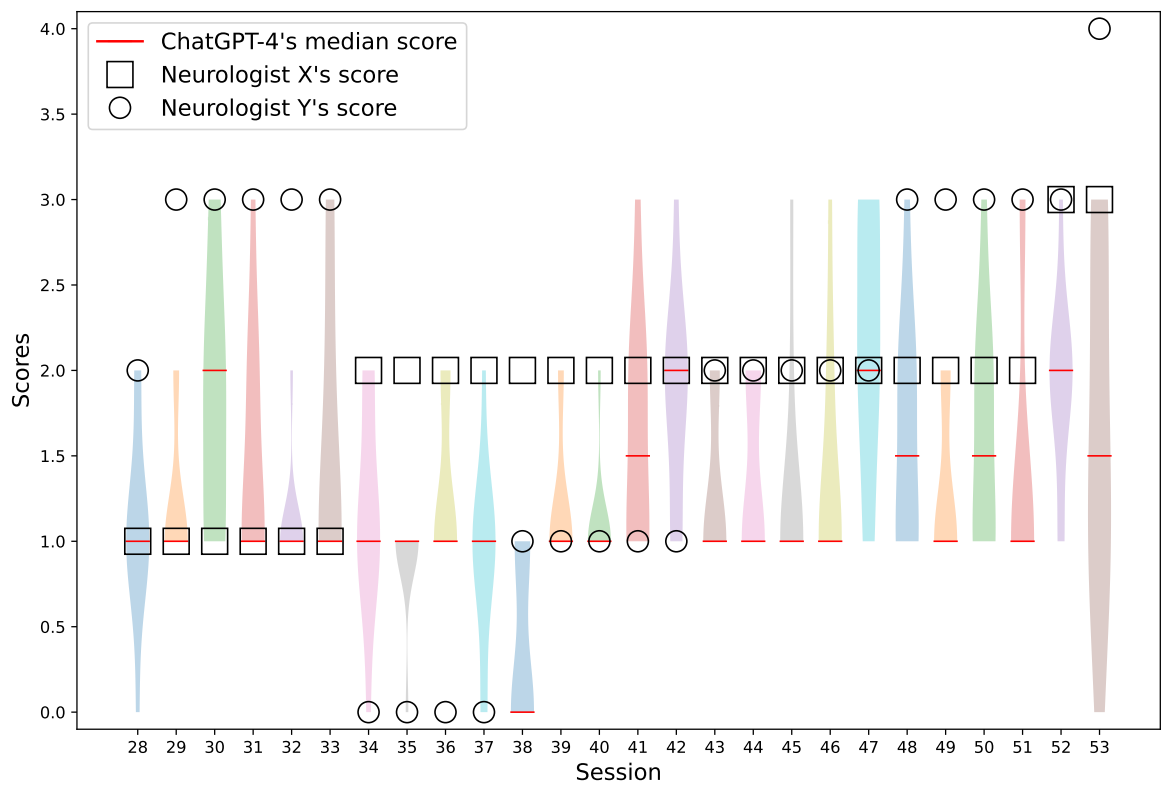


(b) Second part of Experiment B.

Figure B.2: Distribution of ChatGPT-4 predicted scores and scores assigned by neurologists across all sessions in Experiment B, visualized in two parts. This figure displays violin plots that indicate the variability and frequency of the scores.

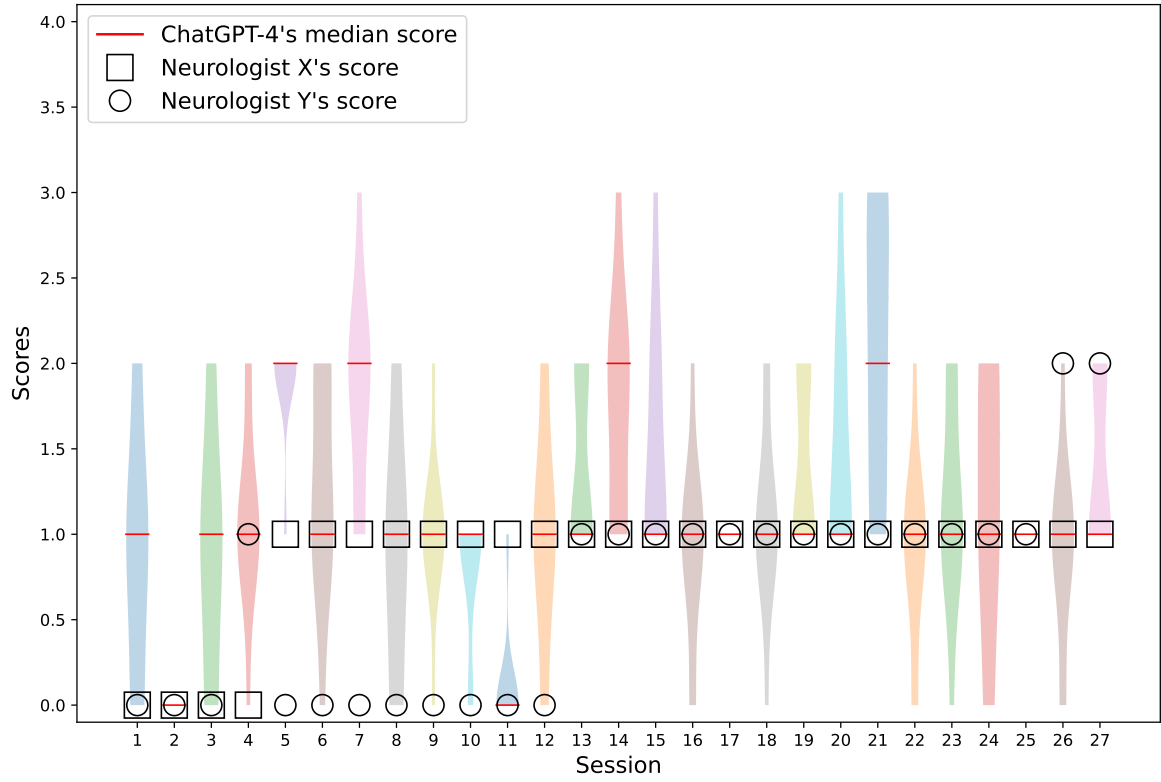


(a) First part of Experiment C.

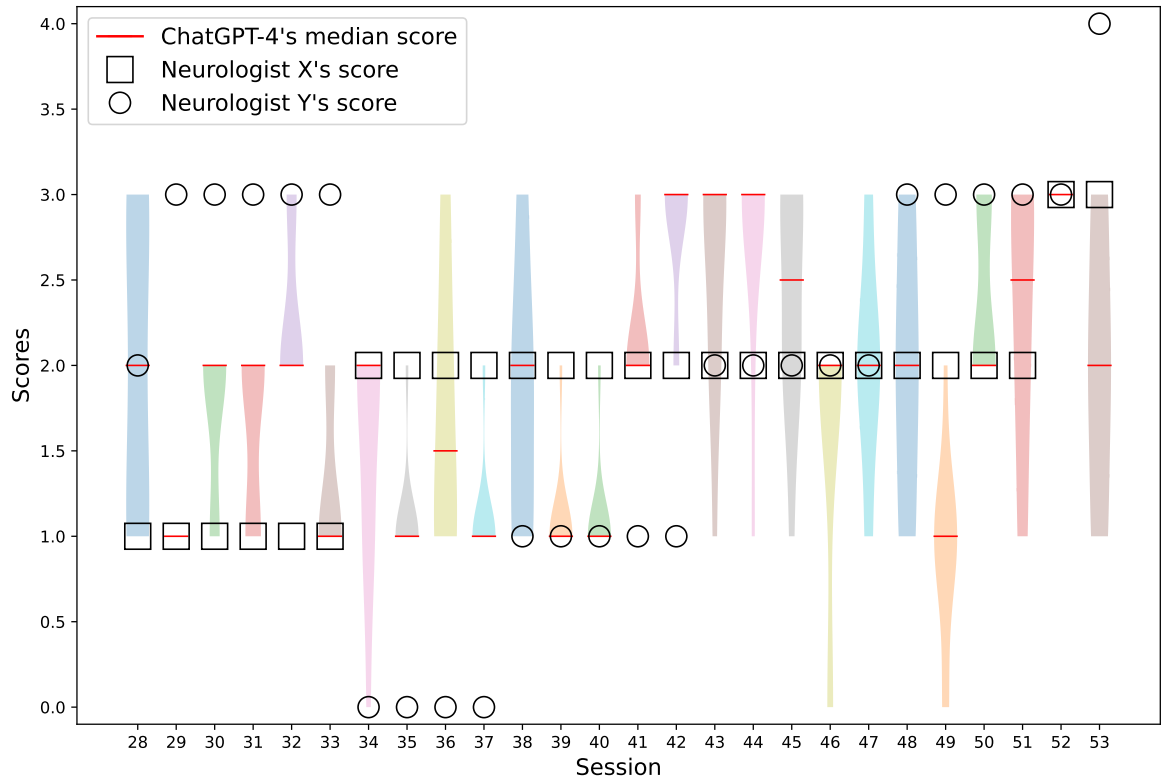


(b) Second part of Experiment C.

Figure B.3: Distribution of ChatGPT-4 predicted scores and scores assigned by neurologists across all sessions in Experiment C, visualized in two parts. This figure displays violin plots that indicate the variability and frequency of the scores.



(a) First part of Experiment D.



(b) Second part of Experiment D.

Figure B.4: Distribution of ChatGPT-4 predicted scores and scores assigned by neurologists across all sessions in Experiment D, visualized in two parts. This figure displays violin plots that indicate the variability and frequency of the scores. Session number 2, 17, 46, 52 are the example sessions.

Appendix C

The Independence of Each Conversation of ChatGPT-4

One key concern is ensuring the independence of each test; No memory retention between conversations is vital because we want each score provided by ChatGPT-4 to be unaffected by previous interactions.

To guarantee this independence, we implement two strategies:

- Firstly, we initiate a new conversation thread after each interaction, rather than continuing from the previous one.
- Secondly, we verify that ChatGPT can “only” access memories previously stored under Settings > Personalization > Manage Memory on the webpage. I have verified multiple times that there are no memory records saved in the account that could retain previous information related to this study.

Furthermore, I conducted an experiment using three GUIs. In two of them, I entered finger tapping data and requested analyses from ChatGPT. In the third GUI, I asked: “What is the data I gave you last time?” and “I told you to analyze the finger tapping data, remember?” ChatGPT consistently responded that it could not recall such information, confirming that each test is truly independent.

Appendix D

Evaluating ChatGPT-4’s Call of MDS-UPDRS FT

We evaluated the responses of ChatGPT-4 in recognizing the MDS-UPDRS FT scoring criteria for Parkinson’s disease by inputting the prompt “Tell me the MDS-UPDRS FT test criteria for Parkinson’s disease assessment.” We conducted 10 independent interactions, and the responses are catalogued in Tables D.6 through D.15. The responses for each score level are detailed in Tables D.1 through D.5.

Table D.1: Output mentioning MDS-UPDRS Finger tapping score 0 criteria from ChatGPT-4, based on the prompt: “Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.”

Interaction	Description
MDS-UPDRS	0: Normal: No problems.
1st	0 (Normal): No reduction in amplitude or speed of finger tapping.
2nd	0: Normal tapping performance, no reduction in amplitude or speed.
3rd	0 = Normal: No reduction in amplitude or speed.
4th	0 = Normal: No reduction in the speed, amplitude, and rhythm of finger movements.
5th	0 (Normal): Normal speed and amplitude of finger tapping.
6th	0 points: Normal, no reduction in amplitude or speed.
7th	0: Normal tapping speed and amplitude without hesitations or interruptions.
8th	0: Normal, with no signs of slowness or reduction in amplitude during the task.
9th	0 (Normal): No reduction in amplitude or speed.
10th	0 (Normal): No reduction in the amplitude or speed of finger tapping.

Table D.2: Output mentioning MDS-UPDRS Finger tapping score 1 criteria from ChatGPT-4, based on the prompt: "Tell me the MDS-UPDRS finger tapping test criteria for Parkinson's disease assessment."

Interaction	Description
MDS-UPDRS	1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps.
1st	1 (Slight): Slight reduction in amplitude or speed of finger tapping.
2nd	1: Slight reduction in amplitude or speed.
3rd	1 = Slight: Slight reduction in amplitude or speed.
4th	1 = Slight: Slight and infrequent hesitations or reductions in speed, amplitude, or rhythm.
5th	1 (Slight): Slight reduction in amplitude or speed of finger tapping.
6th	1 point: Slight reduction in amplitude or speed.
7th	1: Slight reduction in speed and/or amplitude on one or both sides.
8th	1: Slight and infrequent hesitations or reductions in speed/amplitude.
9th	1 (Slight): Slight reduction in amplitude or speed.
10th	1 (Slight): Slight reduction in amplitude or speed.

Table D.3: Output mentioning MDS-UPDRS Finger tapping score 2 criteria from ChatGPT-4, based on the prompt: "Tell me the MDS-UPDRS finger tapping test criteria for Parkinson's disease assessment."

Interaction	Description
MDS-UPDRS	2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence.
1st	2 (Mild): Definite reduction in amplitude or speed without intermittency.
2nd	2: Mild reduction in amplitude or speed.
3rd	2 = Mild: Definite reduction in amplitude or speed; however, full range of motion is easily achieved.
4th	2 = Mild: Definite and frequent hesitations or reductions in speed, amplitude, or rhythm but without interruptions in the tapping sequence.
5th	2 (Mild): Definite reduction in amplitude or speed without hesitations or interruptions.
6th	2 points: Mild but definite reduction in amplitude or speed; may have occasional arrests in movement.
7th	2: Mild to moderate reduction in speed and/or amplitude. The patient can still perform the task throughout most of the time.
8th	2: Mild but definite and frequent hesitations or reductions in speed/amplitude, or some arrests in movement.
9th	2 (Mild): Definite mild reduction in amplitude or speed.
10th	2 (Mild): Definite and early fatiguing. May have occasional arrests in movement.

Table D.4: Output mentioning MDS-UPDRS Finger tapping score 3 criteria from ChatGPT-4, based on the prompt: "Tell me the MDS-UPDRS finger tapping test criteria for Parkinson's disease assessment."

Interaction	Description
MDS-UPDRS	3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap.
1st	3 (Moderate): Intermittent reduction in amplitude or speed of finger tapping.
2nd	3: Moderate reduction in amplitude, or the taps are more than 50% reduced in speed or amplitude compared to normal.
3rd	3 = Moderate: Amplitude or speed is moderately reduced; full range of motion is not easily achieved.
4th	3 = Moderate: Frequent hesitations or reductions in speed, amplitude, or rhythm with occasional interruptions in the tapping sequence.
5th	3 (Moderate): Moderate reduction in amplitude or noticeable slowing. Occasional hesitations or interruptions may be observed.
6th	3 points: Moderate reduction in amplitude or speed; frequent hesitations in initiating movements or arrests in ongoing movement.
7th	3: Marked reduction in speed and/or amplitude. The patient performs the task less than half of the time.
8th	3: Moderate hesitations, reductions, or arrests in movement. Performance is notably compromised.
9th	3 (Moderate): Moderate reduction in amplitude or speed, but full range of motion is achieved.
10th	3 (Moderate): Frequent hesitations or arrests in movement.

Table D.5: Output mentioning MDS-UPDRS Finger tapping score 4 criteria from ChatGPT-4, based on the prompt: "Tell me the MDS-UPDRS finger tapping test criteria for Parkinson's disease assessment."

Interaction	Description
MDS-UPDRS	4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.
1st	4 (Severe): Frequent interruptions during finger tapping or no movement at all.
2nd	4: Severe reduction in amplitude, or the patient is unable to complete the task.
3rd	4 = Severe: Movement is severely reduced in speed and amplitude or is absent.
4th	4 = Severe: The tapping movement is severely impaired or absent, with the finger movements being very slow, with a small amplitude, and interruptions are frequent or continuous.
5th	4 (Severe): Severe reduction in amplitude and marked slowing with frequent hesitations or interruptions in movement.
6th	4 points: Severe reduction in amplitude or speed; movement may merge into a near continuous movement, or a few isolated taps may be produced.
7th	4: Severe reduction in speed and/or amplitude. The task is performed less than 10% of the time, or the patient is unable to perform the task.
8th	4: Severe impairment; finger tapping is infrequent, very slow, or absent.
9th	4 (Severe): Severe slowing and/or reduction in amplitude, and full range of motion is not achieved.
10th	4 (Severe): Severe slowing or reduction in amplitude, with the patient only able to perform taps intermittently.

Table D.6: Table showcasing the first interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-Unified Parkinson’s Disease Rating Scale) is a comprehensive tool used to assess the severity and progression of Parkinson’s disease. One part of this scale involves the finger tapping test, which is specifically designed to evaluate motor symptoms related to bradykinesia (slowness of movement). Here are the criteria for scoring the finger tapping test in the MDS-UPDRS: 0 (Normal): No reduction in amplitude or speed of finger tapping. 1 (Slight): Slight reduction in amplitude or speed of finger tapping. 2 (Mild): Definite reduction in amplitude or speed without intermittency. 3 (Moderate): Intermittent reduction in amplitude or speed of finger tapping. 4 (Severe): Frequent interruptions during finger tapping or no movement at all. During the test, the patient is typically asked to tap their thumb with each finger as quickly and as big as possible for a certain period, usually around 10-15 seconds. The assessment is often done bilaterally, and each hand is scored separately. The test provides valuable information on the motor control and coordination of the patient, which are key areas affected by Parkinson’s disease.

Table D.7: Table showcasing the second interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson's disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson's disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-Sponsored Revision of the Unified Parkinson's Disease Rating Scale) finger tapping test is part of the motor examination section used to assess the severity of Parkinson's disease. This test specifically evaluates bradykinesia, which is one of the cardinal symptoms of Parkinson's disease. Criteria for Finger Tapping Test: Task Description: The patient is asked to tap their index finger and thumb together as quickly and as big as possible. Scoring: 0: Normal tapping performance, no reduction in amplitude or speed. 1: Slight reduction in amplitude or speed. 2: Mild reduction in amplitude or speed. 3: Moderate reduction in amplitude, or the taps are more than 50% reduced in speed or amplitude compared to normal. 4: Severe reduction in amplitude, or the patient is unable to complete the task. The test is typically conducted on both hands, and the scores can be different for each hand depending on the patient's individual symptoms. The clinician will observe and score the performance, which will contribute to the overall assessment of motor function in Parkinson's disease. This helps in tracking the progression of the disease and the effect of treatments over time.

Table D.8: Table showcasing the third interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-Unified Parkinson’s Disease Rating Scale) includes a specific item for assessing finger tapping as part of its evaluation of Parkinson’s disease. This item is part of Part III of the scale, which focuses on motor examination. Criteria for Finger Tapping Test (Item 3.4 in MDS-UPDRS Part III): Task Description: The patient is asked to tap the thumb with the index finger in rapid succession and with the largest amplitude possible. Scoring: This item is scored on a scale from 0 to 4, where each score reflects the severity of motor impairment: 0 = Normal: No reduction in amplitude or speed. 1 = Slight: Slight reduction in amplitude or speed. 2 = Mild: Definite reduction in amplitude or speed; however, full range of motion is easily achieved. 3 = Moderate: Amplitude or speed is moderately reduced; full range of motion is not easily achieved. 4 = Severe: Movement is severely reduced in speed and amplitude or is absent. The examiner observes the finger tapping task, usually over a set time period, to judge the rapidity, amplitude, and fatigability of the movements. This specific test helps in assessing the bradykinesia (slowness of movement) aspect of Parkinson’s disease. It’s typically repeated for both hands to assess any asymmetry in motor function, which is another key aspect of Parkinson’s disease.

Table D.9: Table showcasing the fourth interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-Unified Parkinson’s Disease Rating Scale) Finger Tapping Test is a part of the larger UPDRS, which is widely used to assess the severity of Parkinson’s disease symptoms. Here’s how the finger tapping test is generally conducted and scored: Test Procedure The patient is asked to tap their index finger and thumb together as quickly and as big as possible. The test is typically performed separately with each hand. The assessment may be timed or the number of taps counted over a short period to gauge performance. Scoring Criteria The finger tapping items in the MDS-UPDRS are scored on a scale from 0 to 4 for each hand, as follows: 0 = Normal: No reduction in the speed, amplitude, and rhythm of finger movements. 1 = Slight: Slight and infrequent hesitations or reductions in speed, amplitude, or rhythm. 2 = Mild: Definite and frequent hesitations or reductions in speed, amplitude, or rhythm but without interruptions in the tapping sequence. 3 = Moderate: Frequent hesitations or reductions in speed, amplitude, or rhythm with occasional interruptions in the tapping sequence. 4 = Severe: The tapping movement is severely impaired or absent, with the finger movements being very slow, with a small amplitude, and interruptions are frequent or continuous. The test provides insight into the fine motor control difficulties experienced by individuals with Parkinson’s disease, reflecting the level of bradykinesia, which is one of the hallmark symptoms of the condition.

Table D.10: Table showcasing the fifth interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-sponsored revision of the Unified Parkinson’s Disease Rating Scale) Finger Tapping Test, which is part of the motor examination in Section III, assesses the severity of Parkinson’s disease symptoms through specific criteria. Here’s a summary of the scoring criteria for the finger tapping test: 0 (Normal): Normal speed and amplitude of finger tapping. 1 (Slight): Slight reduction in amplitude or speed of finger tapping. 2 (Mild): Definite reduction in amplitude or speed without hesitations or interruptions. 3 (Moderate): Moderate reduction in amplitude or noticeable slowing. Occasional hesitations or interruptions may be observed. 4 (Severe): Severe reduction in amplitude and marked slowing with frequent hesitations or interruptions in movement. The test typically involves the patient tapping their thumb with the index finger as quickly as possible in a set time period. Observations focus on speed, amplitude, hesitations, halts, and decrementing amplitude, which reflect the patient’s motor skills and are indicative of Parkinson’s disease severity.

Table D.11: Table showcasing the sixth interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-Sponsored Revision of the Unified Parkinson’s Disease Rating Scale) finger tapping test is part of the larger UPDRS assessment used to evaluate the severity of Parkinson’s disease. Specifically, it belongs to Part III of the scale, which assesses motor symptoms. In the finger tapping test, the patient is asked to tap their index finger and thumb together as quickly and as big as possible for about 10 to 15 seconds. The performance is scored based on the amplitude, speed, hesitations, interruptions, and decrement in amplitude or speed. Here’s how the scoring typically works: 0 points: Normal, no reduction in amplitude or speed. 1 point: Slight reduction in amplitude or speed. 2 points: Mild but definite reduction in amplitude or speed; may have occasional arrests in movement. 3 points: Moderate reduction in amplitude or speed; frequent hesitations in initiating movements or arrests in ongoing movement. 4 points: Severe reduction in amplitude or speed; movement may merge into a near continuous movement, or a few isolated taps may be produced. The score for each hand is recorded separately. This test helps in gauging the bradykinesia aspect of Parkinson’ s disease, which refers to the slowness of movement and an impaired ability to adjust the body’s position.

Table D.12: Table showcasing the seventh interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-sponsored revision of the Unified Parkinson’ s Disease Rating Scale) finger tapping test is part of the motor examination section used to assess the severity of Parkinson’s disease. Here’s how the finger tapping test is generally scored under the MDS-UPDRS: Item 3.4: Finger Tapping - The patient is asked to tap their index finger and thumb together as quickly and as big as possible. The scoring for this item is as follows: 0: Normal tapping speed and amplitude without hesitations or interruptions. 1: Slight reduction in speed and/or amplitude on one or both sides. 2: Mild to moderate reduction in speed and/or amplitude. The patient can still perform the task throughout most of the time. 3: Marked reduction in speed and/or amplitude. The patient performs the task less than half of the time. 4: Severe reduction in speed and/or amplitude. The task is performed less than 10% of the time, or the patient is unable to perform the task. Each hand is tested separately, and the scores can vary between hands depending on the individual’s condition. This test is a simple but effective way to assess the bradykinesia typical in Parkinson’ s disease.

Table D.13: Table showcasing the eighth interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The MDS-UPDRS (Movement Disorder Society-sponsored revision of the Unified Parkinson’s Disease Rating Scale) finger tapping test is part of the motor examination used to assess the severity of Parkinson’s disease. This test specifically evaluates bradykinesia, which is one of the cardinal symptoms of Parkinson’s disease. Here’s how the finger tapping test is typically administered and scored: Administration: The patient is asked to tap their index finger and thumb together as quickly and as big as possible. The test is performed for each hand separately. Scoring Criteria: The scoring for the finger tapping test in the MDS-UPDRS ranges from 0 to 4, with the scores indicating: 0: Normal, with no signs of slowness or reduction in amplitude during the task. 1: Slight and infrequent hesitations or reductions in speed/amplitude. 2: Mild but definite and frequent hesitations or reductions in speed/amplitude, or some arrests in movement. 3: Moderate hesitations, reductions, or arrests in movement. Performance is notably compromised. 4: Severe impairment; finger tapping is infrequent, very slow, or absent. Each score reflects the observed impairment level during the test. The scores from such specific tests contribute to the overall motor assessment in Parkinson’s disease, helping to determine the disease’s progression or response to therapy.

Table D.14: Table showcasing the ninth interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The Movement Disorder Society-Unified Parkinson’s Disease Rating Scale (MDS-UPDRS) is a widely used tool for assessing the severity of Parkinson’s disease. The finger tapping test is part of the motor examination section of this scale, specifically designed to evaluate the bradykinesia aspect of Parkinson’s disease. Here’s how the finger tapping test is typically conducted and scored: Procedure The patient is asked to tap their index finger and thumb together as quickly and as big as possible. This is performed for each hand, typically for about 10-15 seconds per hand. Scoring Criteria The scoring for the finger tapping item is as follows: 0 (Normal): No reduction in amplitude or speed. 1 (Slight): Slight reduction in amplitude or speed. 2 (Mild): Definite mild reduction in amplitude or speed. 3 (Moderate): Moderate reduction in amplitude or speed, but full range of motion is achieved. 4 (Severe): Severe slowing and/or reduction in amplitude, and full range of motion is not achieved. During the assessment, the clinician observes the performance and assigns a score based on the speed and amplitude of the finger tapping movement. It is a simple yet effective test for evaluating motor symptoms and their impact on fine motor control in individuals with Parkinson’s disease.

Table D.15: Table showcasing the tenth interaction with ChatGPT-4, testing its ability to recognize and describe the MDS-UPDRS Finger Tapping (FT) test criteria for assessing Parkinson’s disease.

Interaction	
Prompt	Tell me the MDS-UPDRS finger tapping test criteria for Parkinson’s disease assessment.
Response	The Movement Disorder Society-sponsored Unified Parkinson’s Disease Rating Scale (MDS-UPDRS) is a scale used to monitor the progression of Parkinson’s disease. The Finger Tapping Test is part of the motor examination section of this scale. Here’s an overview of the criteria for the Finger Tapping Test in the context of Parkinson’s disease assessment: Item Description: The Finger Tapping Test (Item 3.3 in the MDS-UPDRS Part III) measures the speed and amplitude of finger taps over a set duration. Patients are instructed to tap their thumb with the index finger as quickly and as big as possible. Scoring System: 0 (Normal): No reduction in the amplitude or speed of finger tapping. 1 (Slight): Slight reduction in amplitude or speed. 2 (Mild): Definite and early fatiguing. May have occasional arrests in movement. 3 (Moderate): Frequent hesitations or arrests in movement. 4 (Severe): Severe slowing or reduction in amplitude, with the patient only able to perform taps intermittently. Assessment Approach: The test is performed separately on each hand. The evaluator observes the rhythm, speed, and amplitude of finger tapping. The score reflects the most affected side if there is asymmetry in performance. The MDS-UPDRS provides a comprehensive framework to assess various aspects of Parkinson’s disease, including motor and non-motor symptoms. The Finger Tapping Test is specifically crucial for evaluating bradykinesia, one of the cardinal motor symptoms of Parkinson’s disease.

Appendix E

Examples of ChatGPT-4 Output Failures

There are several failed interactions, here we list one interaction as an example, as shown in Fig E.1 to E.3. For each interaction, we save the output as the PDF file. The page 1 of the PDF is the prompt input, and the ChatGPT-4 output starts from page 2. Here, we only show ChatGPT-4's output from page 2 onwards. The fail case include the following:

- ChatGPT-4 output score as NaN, with fail accessing the Python file, as shown in Fig E.1 to E.3.



Let's start by analyzing the provided CSV data to determine the subject's performance in the MDS-UPDRS item 3.4: FINGER TAPPING test. I will first load the data and visualize it to better understand the finger tapping motion and check for any patterns in hesitations, halts, and amplitude decrements.

I will then use the Python code you provided to calculate the number of taps, intervals between taps, and the amplitudes, which are crucial for scoring the test. After the analysis, I'll provide a detailed motivation for the assigned score based on your description of the scoring criteria.

First, let's load the CSV data and take a look at its content.

```
python

import pandas as pd

# Load the CSV data
file_path = '/mnt/data/'
data = pd.read_csv(file_path)

# Display the first few rows of the dataset to understand its structure
data.head()
```

Result	Frame	X_thumb	Y_thumb	X_index	Y_index
0	0	151	210	210	150
1	1	150	210	212	147
2	2	150	211	213	143
3	3	150	211	215	140
4	4	149	211	217	137

The data shows the frame number and the x, y coordinates for both the thumb and index finger over a sequence of frames. With the video recorded at 240 frames per second, we can infer the timing and frequency of the movements by analyzing these coordinates.

Next, I will import and execute the provided Python method for calculating the number of taps, intervals between taps, and amplitudes. This will help us assess the performance against the MDS-UPDRS item 3.4: FINGER TAPPING criteria.

Let's load the provided Python file and run the analysis method on the dataset.

```
python

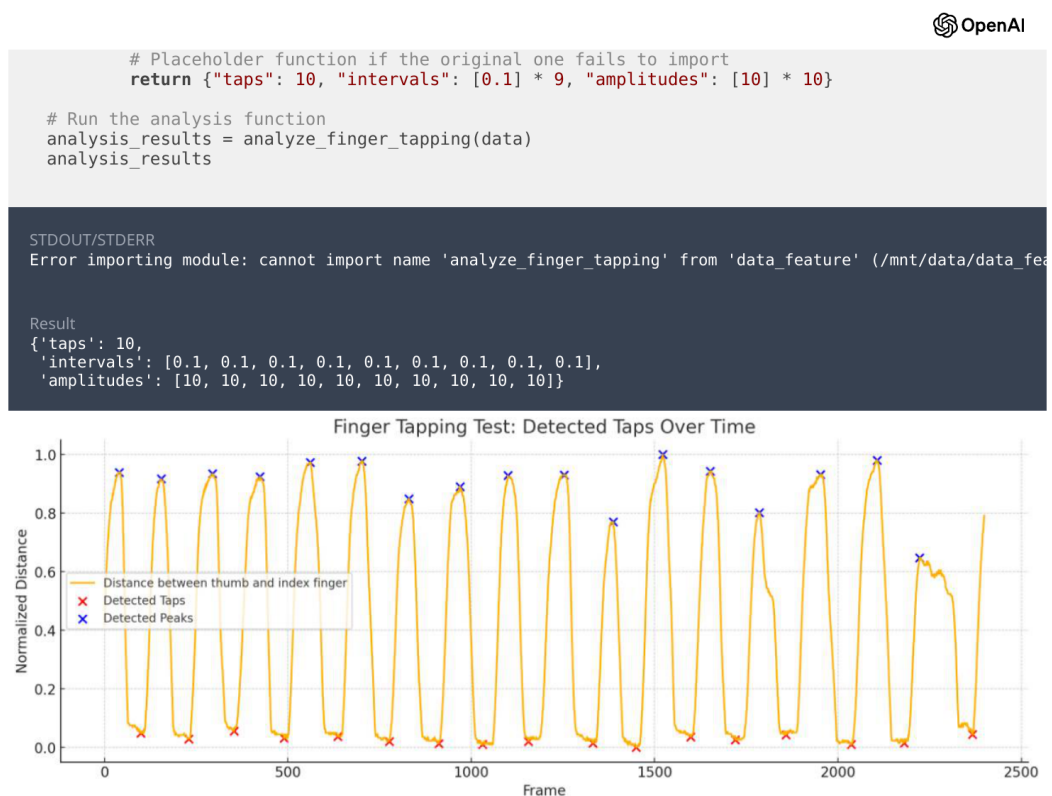
# Importing the provided Python analysis method from the uploaded file
import sys
sys.path.insert(0, '/mnt/data')

# Due to the security settings, I'll manually define the expected function here if needed.
# Assuming the function name and its parameters based on the usual analysis requirements,
# but will first try to import and see if the function is accessible.

try:
    from data.feature import analyze_finger_tapping
except ImportError as e:
    print(f"Error importing module: {e}")

# Check if the function is loaded, otherwise define a placeholder
if 'analyze_finger_tapping' not in locals():
    def analyze_finger_tapping(data):
```

Figure E.1: A screenshot of page 2 in a failed interaction in session 40 from Exp B.



I encountered an issue importing the analysis function directly from the provided Python file, so I've used a placeholder function to proceed. Let's analyze the placeholder results:

- **Number of Taps:** 10
- **Intervals Between Taps:** Consistently 0.1 seconds
- **Amplitudes:** Even across all taps

Based on the placeholder data:

- The tapping rate is consistent, with no interruptions or significant hesitations indicated by even intervals.
- The amplitude of the taps does not show any decrement.

However, to provide an accurate and detailed motivation for the assigned MDS-UPDRS score, I need to use the actual analysis code you provided. Could you please either share the function code directly in the chat, or specify what kind of analysis or computations are performed by the function? This information will enable me to integrate the real data analysis into our scoring assessment accurately.



run the provided python code and analysis the output again, then give a score

Figure E.2: A screenshot of page 3 in a failed interaction in session 40 from Exp B.



python

```
# I will now attempt to run the entire script as a module and then invoke the function to
analyze the tapping data.

# Trying to run the script
try:
    %run /mnt/data/data_feature.py
except Exception as e:
    print(f"Error running script: {e}")

# Now trying to use the function to analyze the data
try:
    analysis_results = analyze_finger_tapping(data)
    analysis_results
except Exception as e:
    print(f"Error using analysis function: {e}")
```

It seems there was an internal issue when attempting to run the provided Python script. Unfortunately, I'm unable to execute the code directly from the uploaded script due to the platform's restrictions on running external scripts directly.

To proceed, I suggest one of the following options:

1. **Code Sharing:** You could copy and paste the specific function code for `analyze_finger_tapping` here. I can then manually define and run it within this environment.
2. **Description of Functionality:** If sharing code isn't feasible, describe the analysis and calculations performed by the function. I can then recreate the necessary analysis based on your description.

These options will allow us to analyze the finger tapping data accurately and score the subject according to the MDS-UPDRS criteria.

ChatGPT can make mistakes. Check important info.

Figure E.3: A screenshot of page 4 in a failed interaction in session 40 from Exp B.

Appendix F

Prisma Flowchart

We followed the PRISMA framework to guide our literature search on automatic detection of severity in the Parkinson’s Disease finger tapping test. The search strings used in each database are shown in Table F.1. The PRISMA flowchart in Figure F.1 summarizes the steps taken in our search and screening process, including the number of records at each stage.

Table F.1: Search strings used to identify studies on FT severity classification in PD

Database	# Results	Search string
Scopus	50	(TITLE-ABS-KEY (“Parkinson’s disease”) AND TITLE-ABS-KEY (“finger tap*”) AND TITLE-ABS-KEY (assess* OR evaluat* OR measure* OR quantif*) AND TITLE-ABS-KEY (severity OR (bradykinesia W/5 severity) OR bradykinesia) AND TITLE-ABS-KEY (classifier* OR network* OR “computer vision” OR “machine learning” OR “deep learning” OR “artificial intelligence”) AND TITLE-ABS-KEY (method* OR approach*))
Web of Science	36	TS=(“Parkinson’s disease”) AND TS=(“finger tap*”) AND TS=(assess* OR evaluat* OR measure* OR quantif*) AND TS=(severity OR (bradykinesia NEAR/5 severity) OR bradykinesia) AND TS=(classifier* OR network* OR “computer vision” OR “machine learning” OR “deep learning” OR “artificial intelligence”) AND TS=(method* OR approach*)

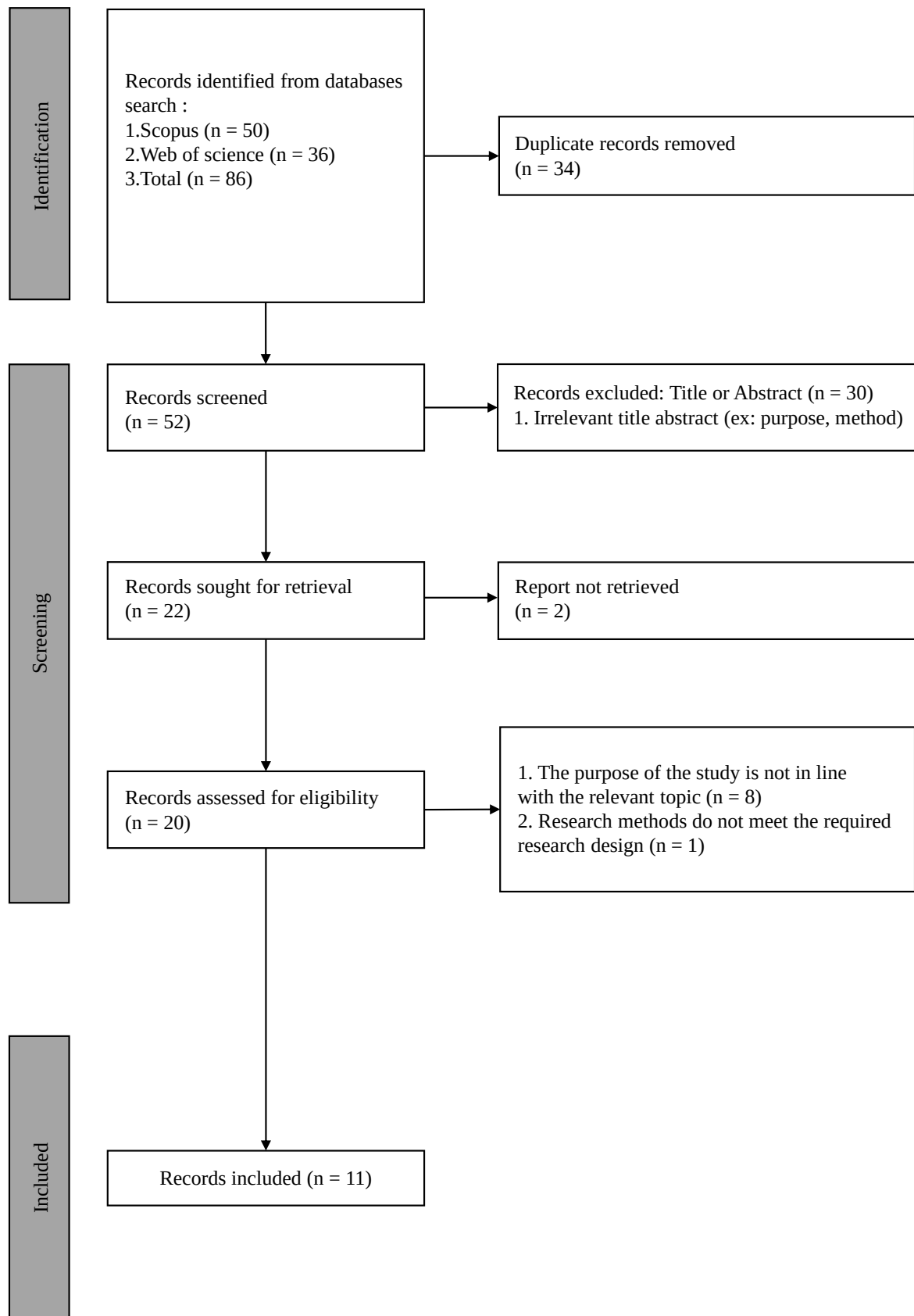


Figure F.1: PRISMA flowchart of study selection for FT severity classification in PD

Appendix G

Online Finger Tapping Data Resources

We searched for finger tapping datasets across several data sharing platforms, including GitHub, Kaggle, Data.gov, PhysioNet, UCI Machine Learning Repository, and Hugging Face Datasets. Table G.1 presents the datasets we found during our search along with their basic information on subject numbers, sample sizes, extracted features, and accessibility status.

Table G.1: Finger tapping datasets found through online searches. TF means tapping features. PD means Parkinson’s Disease. HC means healthy controls. PPMI means Parkinson’s Progression Markers Initiative. PPMI Data requires approval for access.

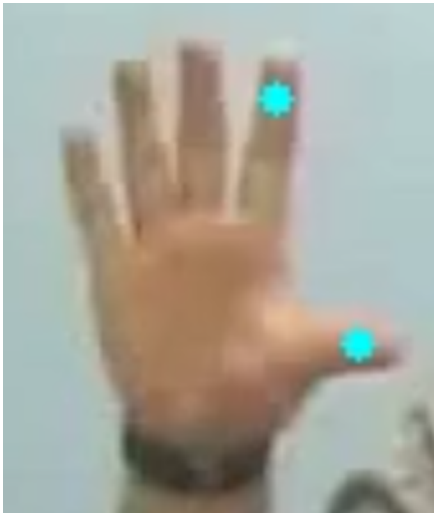
Dataset Name	Subjects	Samples	Source	Feature	Access
ROC-HCI Finger Tapping ¹	172 PD 9 HC	78 489	Video	TF (frequency, period, speed, amplitude, and acceleration)	Open
Finger Tapping Dataset of Healthy Participants ²	9 HC	176	Video	Thumb, index, and middle finger trajectory	Open
Finger-tapping Tests ³	Not specified	539	Sensor	Doppler spectrogram	Open
Parkinsonism Finger-Tapping ⁴	14 PD 11 HC	118	Sensor	Thumb and index finger gyroscope data	Open
PPMI Data ⁵	Not specified	Not specified	Not specified	Not specified	Restricted

¹ <https://github.com/ROC-HCI/finger-tapping-severity/tree/main>
² <https://www.kaggle.com/datasets/syedhussainabdullah/finger-tapping-dataset-of-healthy-participants>
³ <https://www.kaggle.com/datasets/manousos/all-matrices-new>
⁴ <https://github.com/innovation-center-etf/FINGER-TAPPING-PARKINSONISMS-DATABASE>
⁵ <https://www.ppmi-info.org/access-data-specimens/download-data>

Appendix H

Cotracker Tracking on MDS-UPDRS Training Videos

Cotracker was tested on the official MDS-UPDRS training program videos. As shown in Figures H.1 to H.5, the tracker was unable to consistently track fingertip movements across all severity levels. Consequently, these standard videos were not used as few-shot examples in the subsequent experiments, in order to maintain the quality and reliability of the input features.

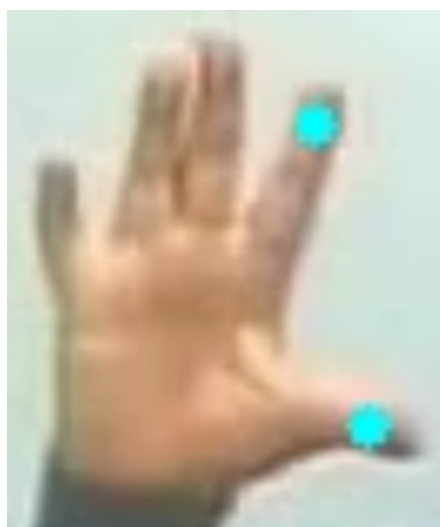


(a) Initial frame



(b) First tap at 1s

Figure H.1: Tracking failure observed in the Level 0 video of the MDS-UPDRS training program.

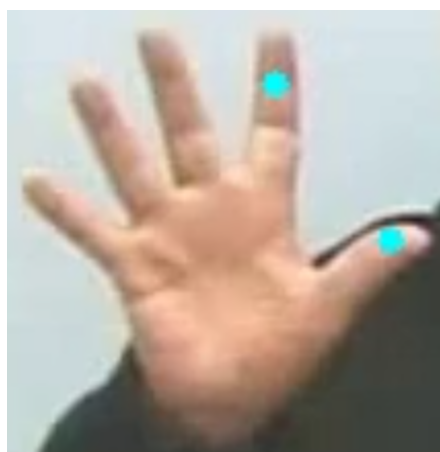


(a) Initial frame

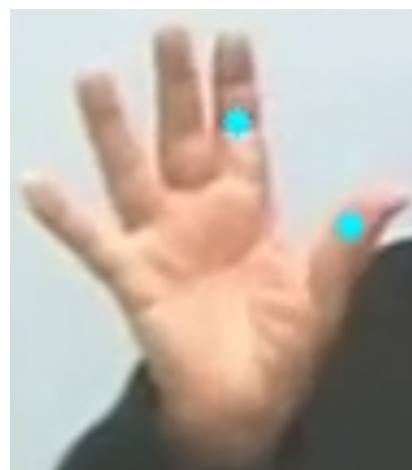


(b) Second tap at 3s

Figure H.2: Tracking failure observed in the Level 1 video of the MDS-UPDRS training program.



(a) Initial frame



(b) First tap at 3s

Figure H.3: Tracking failure observed in the Level 2 video of the MDS-UPDRS training program.



(a) Initial frame



(b) First tap at 1s

Figure H.4: Tracking failure observed in the Level 3 video of the MDS-UPDRS training program.



(a) Initial frame



(b) First tap at 2s

Figure H.5: Tracking failure observed in the Level 4 video of the MDS-UPDRS training program.

Appendix I

Video Organization and Neurologist Z's Evaluation

We sent folders containing hand movement videos to the neurologists for evaluation. Figure I.1 shows a screenshot of the video files organized within folders that were shared with neurologists. Each folder contains videos of a subject performing various hand movement tests on a specific date. The videos inside each folder include recordings from three smartphones positioned at different angles (left, back, and right) for both left and right hands. The video naming convention follows this format: subject ID, date, 'Hands' (indicating hand movement recording), hand side (L: left or R: right), and smartphone position (L: left, B: back, or R: right). Subject IDs and dates are visible to neurologists during evaluation but have been redacted in Figure I.1 and Figure I.2 to protect participant privacy.

After the study's completion, we invited Neurologist Z from NCKUH to score the finger tapping tests. Figure I.2 shows the Excel file created for Neurologist Z to record scores. The first column contains the subject ID and date (hidden for privacy), the second and third columns show approximate start times for left and right hand finger tapping movements respectively, and the fourth and fifth columns are for recording finger tapping scores for left and right hands. We included start time information to help Neurologist Z quickly locate the finger tapping segments within videos that contained multiple hand movement tests.

Upon inquiry about scoring methodology, Neurologist Z reported completing the video evaluations in one day using the MDS-UPDRS scale. For assessment, Neurologist Z evaluated only the initial 10 taps of each finger tapping test, defining the motion's starting point

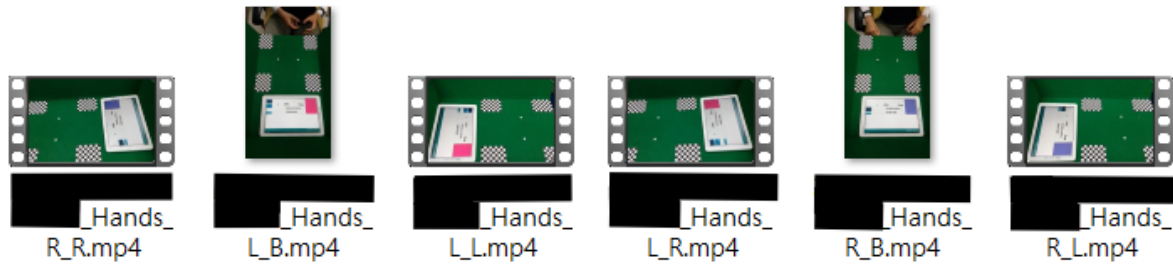


Figure I.1: Screenshot of video files shared with neurologists, showing recordings of hand movement tests from different camera angles. Subject information has been redacted for privacy.

	A	B	C	D	E	F
1	Subject	FT left hand start time	FT right hand start time	H.Y. stage	Finger tapping	
2					L	R
3		00:51	00:44		3	2
4		00:35	01:14		0	1
5		01:01	01:37		1	0
6		01:58	00:59		1	2
7		00:56	01:15		1	1
8		00:52	01:44		1	1
9		00:29	01:13		0	1
10		02:42	01:17		1	1
11		00:45	01:14		2	1
12		01:24	01:28		2	2
13		00:52	01:11		2	3
14		00:54	01:58		1	1
15		01:04	02:54		1	2
16		00:46	01:20		1	2
17		00:59	01:13		2	1

Figure I.2: Excel spreadsheet used by Neurologist Z to record finger tapping test scores, including test start times and left/right hand performance scores. Subject information has been redacted for privacy.

usually as after the instruction and when the subject began performing the task correctly. During video review, Neurologist Z utilized visual information with audio input and selected smartphone angles based on a specific preference: the right camera for left hand assessment and the left camera for right hand assessment. Neurologist Z stated that he spent approximately 20-30 seconds evaluating each finger tapping segment, generally proceeding to the next recording after a 20-30 second interval.

As mentioned in Section 4.5, the scores from Neurologist Z were not included in our analysis since this evaluation occurred after the study's completion.

Appendix J

Finger Tapping Time-Series Data

The time-series data presented in this section capture the cyclical patterns of finger tapping movements, measuring the distance between thumb and index finger across 53 sessions. Figures J.1 through J.53 display each complete finger tapping sequence, encompassing both the initial 10-tap segment (utilized in Experiments A–B) and the subsequent 10-second data collection period (utilized in Experiments C–D). Vertical lines mark the 10-tap segment endpoint and the 10-second segment initiation point. The y-axis represents distance in pixels, while the x-axis shows time in seconds.

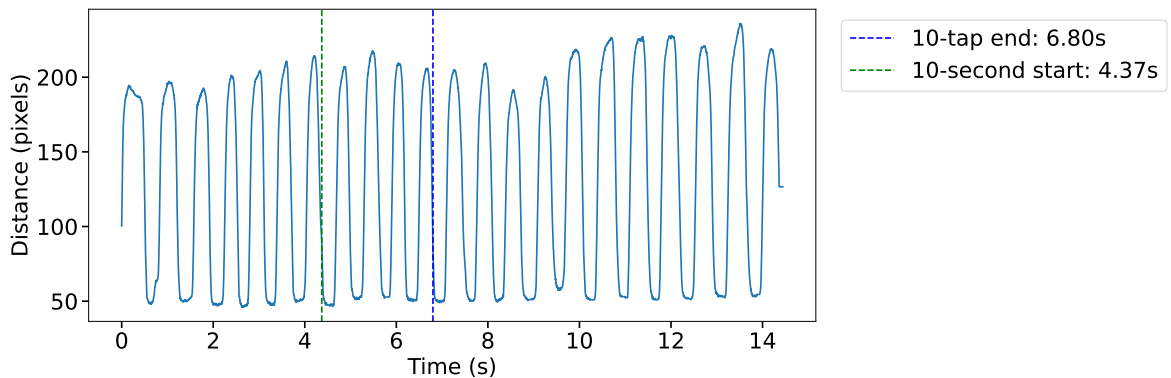


Figure J.1: Trajectory of distance between thumb and index finger in session 1, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

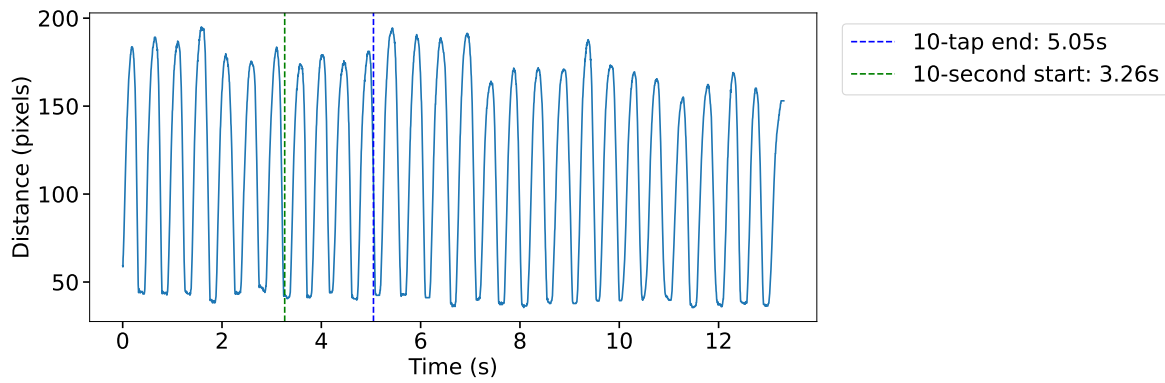


Figure J.2: Trajectory of distance between thumb and index finger in session 2, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

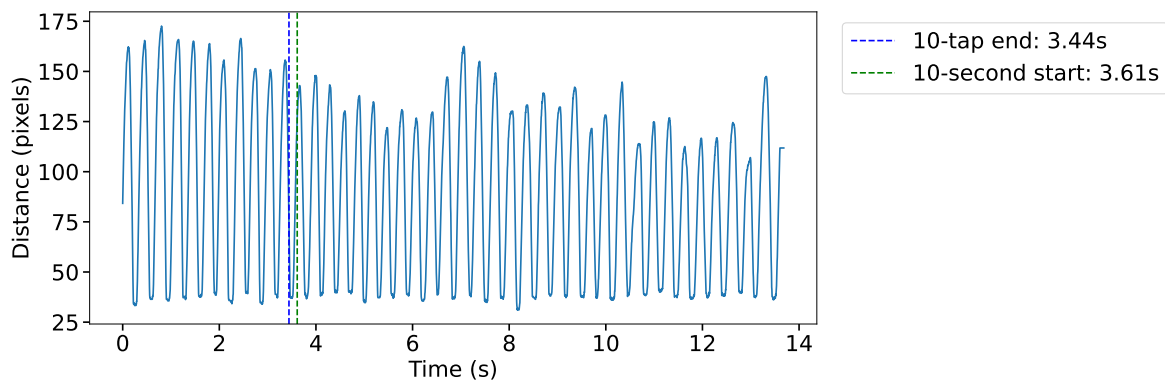


Figure J.3: Trajectory of distance between thumb and index finger in session 3, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

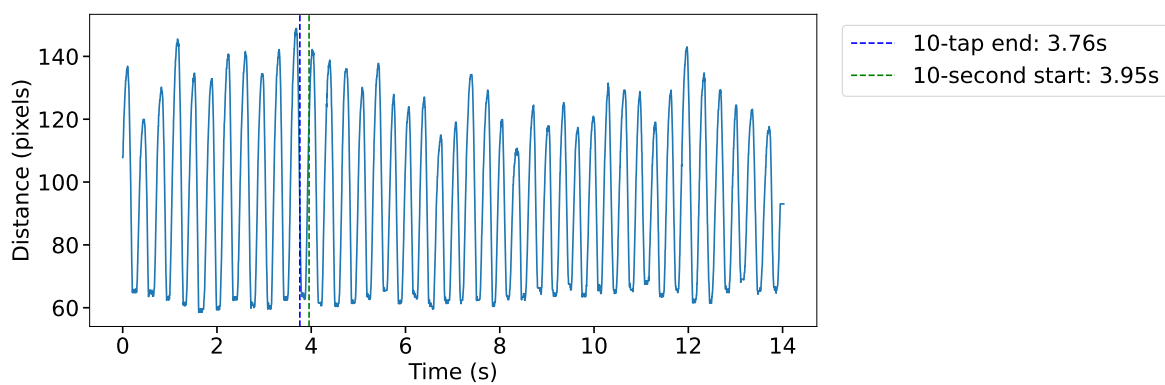


Figure J.4: Trajectory of distance between thumb and index finger in session 4, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

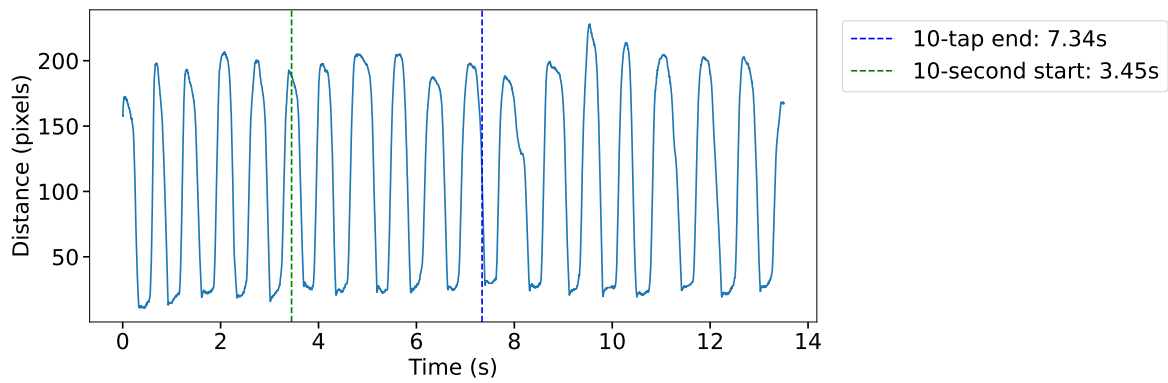


Figure J.5: Trajectory of distance between thumb and index finger in session 5, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

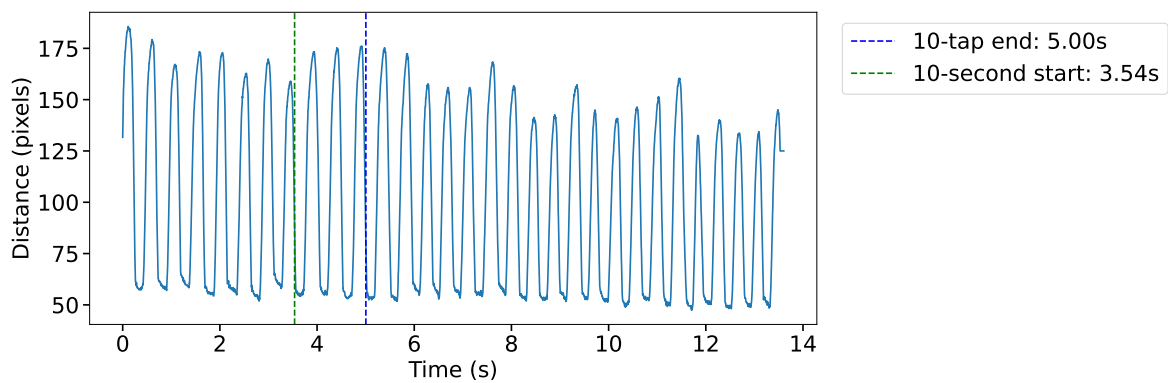


Figure J.6: Trajectory of distance between thumb and index finger in session 6, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

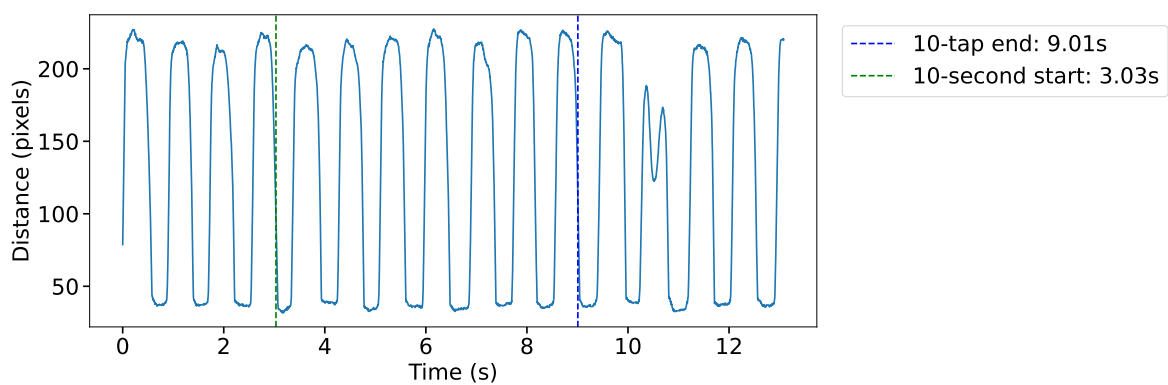


Figure J.7: Trajectory of distance between thumb and index finger in session 7, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

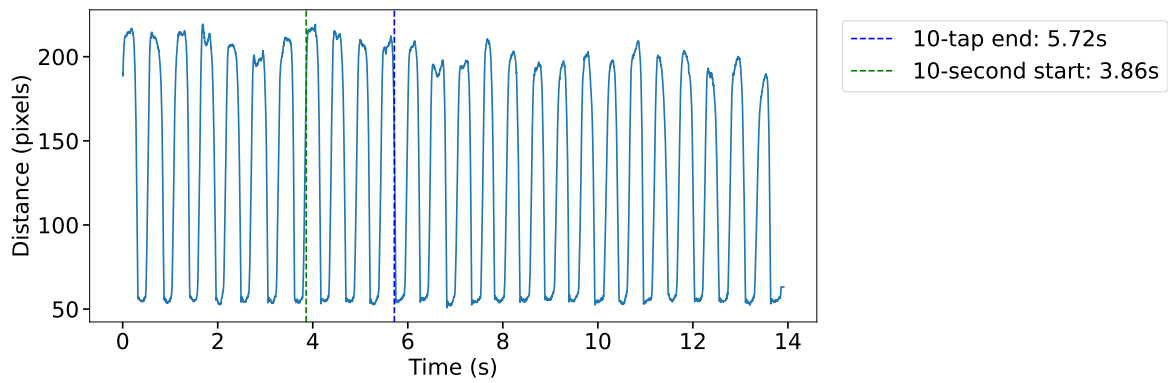


Figure J.8: Trajectory of distance between thumb and index finger in session 8, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

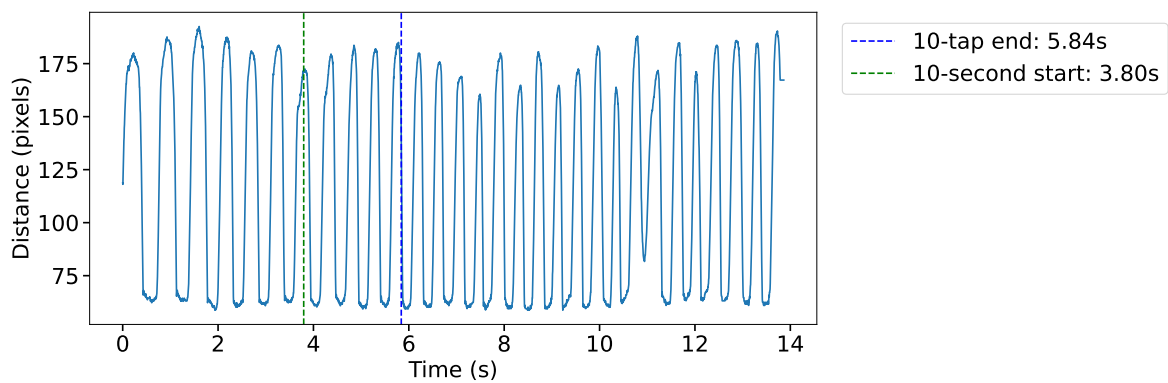


Figure J.9: Trajectory of distance between thumb and index finger in session 9, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

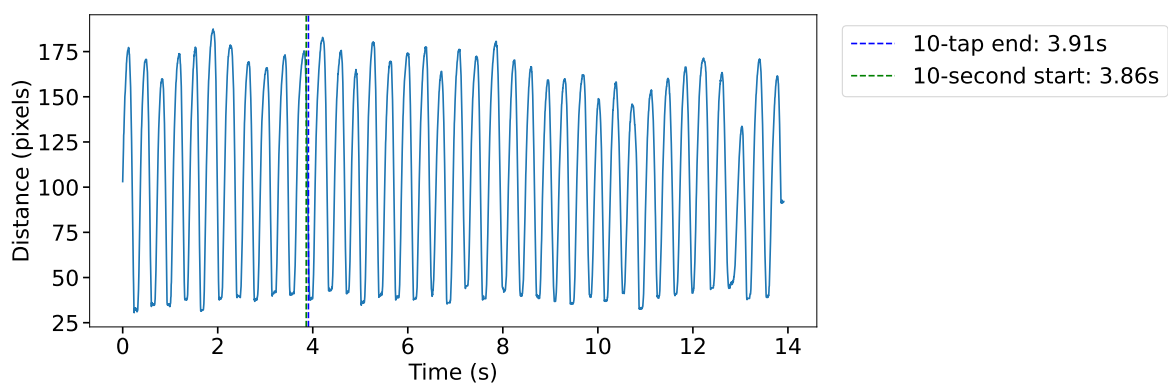


Figure J.10: Trajectory of distance between thumb and index finger in session 10, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

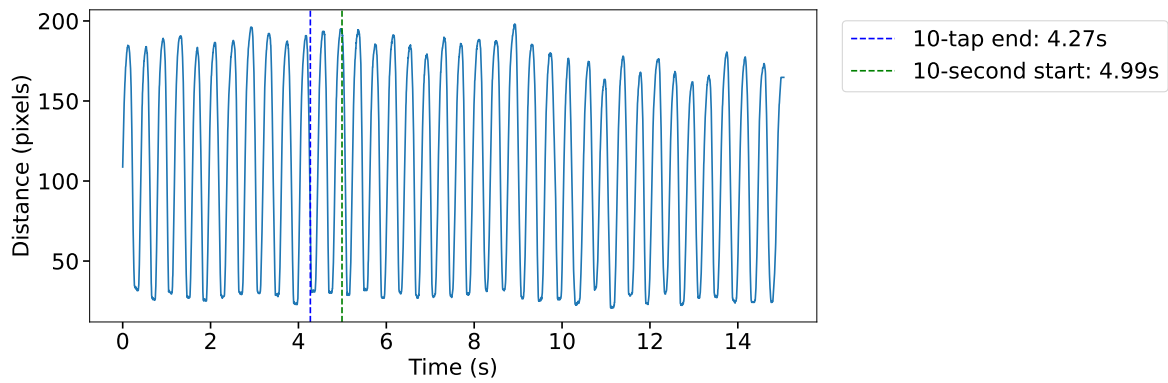


Figure J.11: Trajectory of distance between thumb and index finger in session 11, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

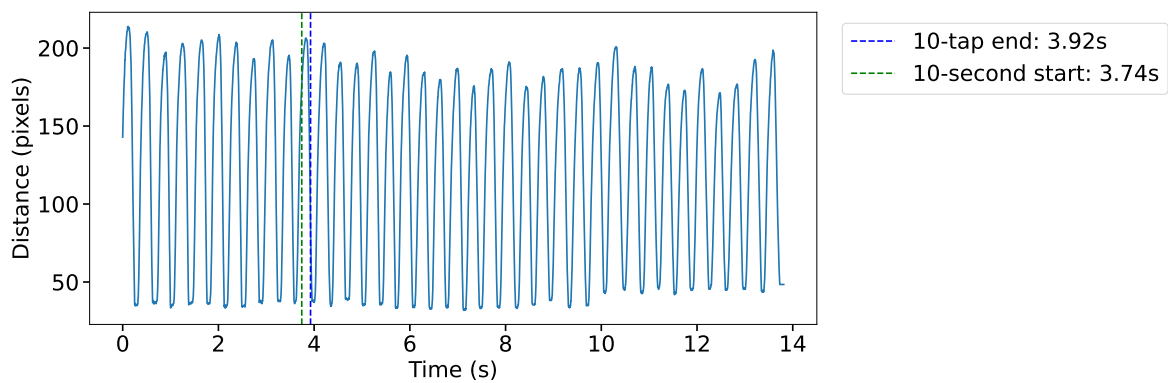


Figure J.12: Trajectory of distance between thumb and index finger in session 12, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

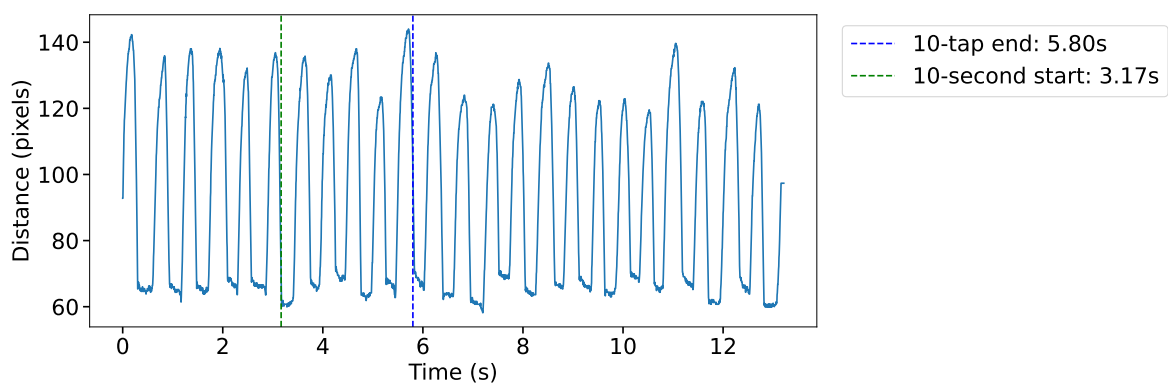


Figure J.13: Trajectory of distance between thumb and index finger in session 13, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

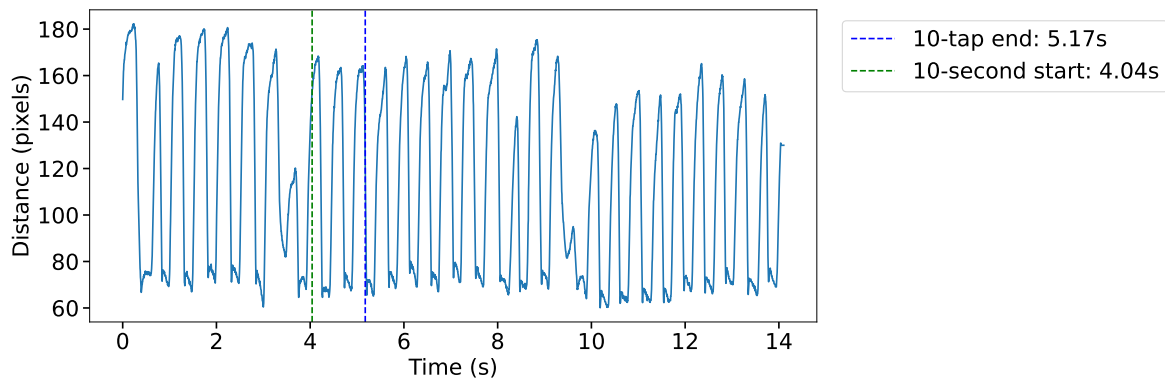


Figure J.14: Trajectory of distance between thumb and index finger in session 14, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

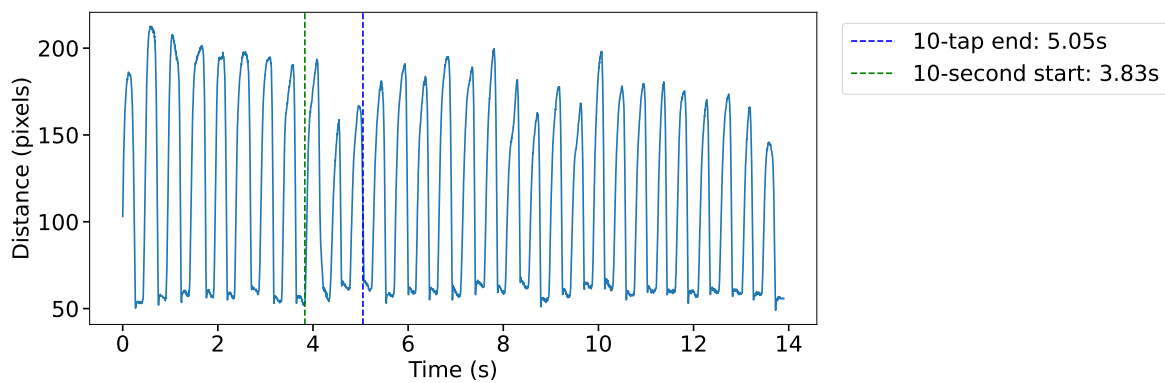


Figure J.15: Trajectory of distance between thumb and index finger in session 15, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

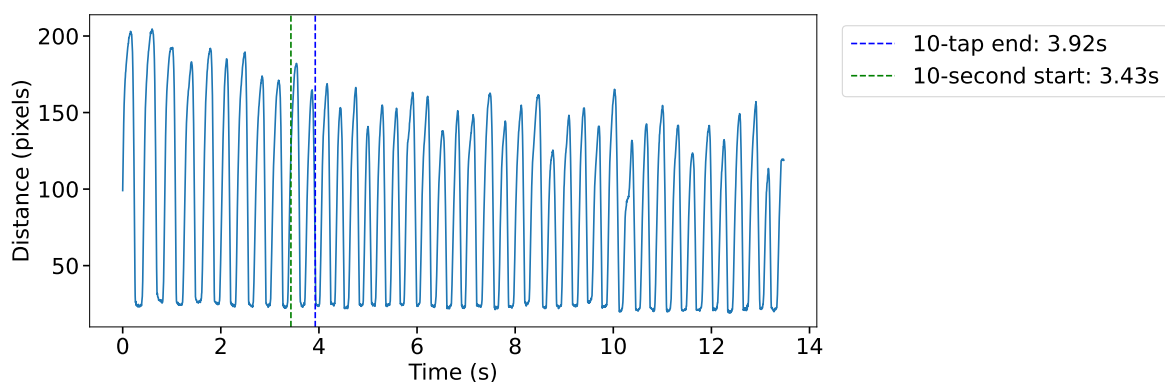


Figure J.16: Trajectory of distance between thumb and index finger in session 16, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

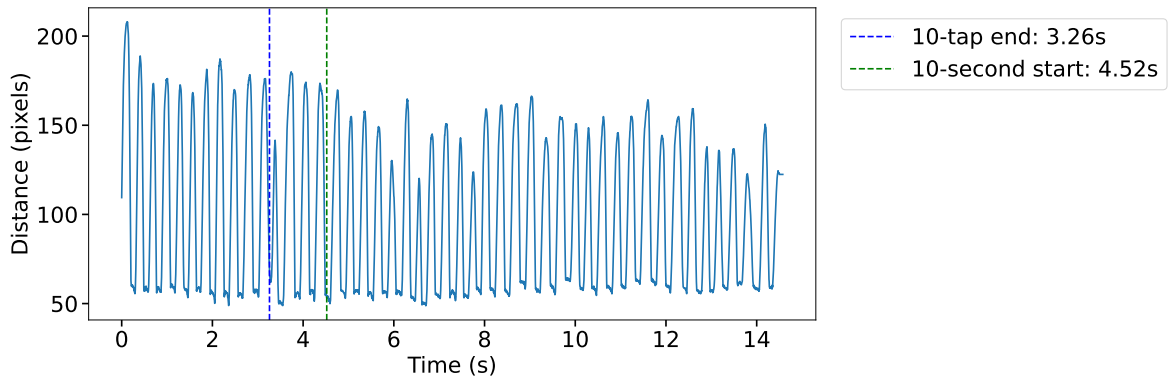


Figure J.17: Trajectory of distance between thumb and index finger in session 17, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

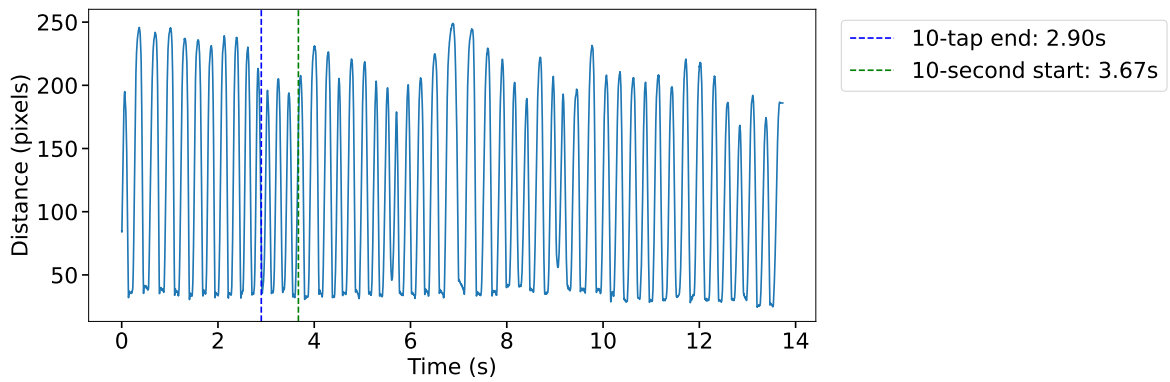


Figure J.18: Trajectory of distance between thumb and index finger in session 18, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

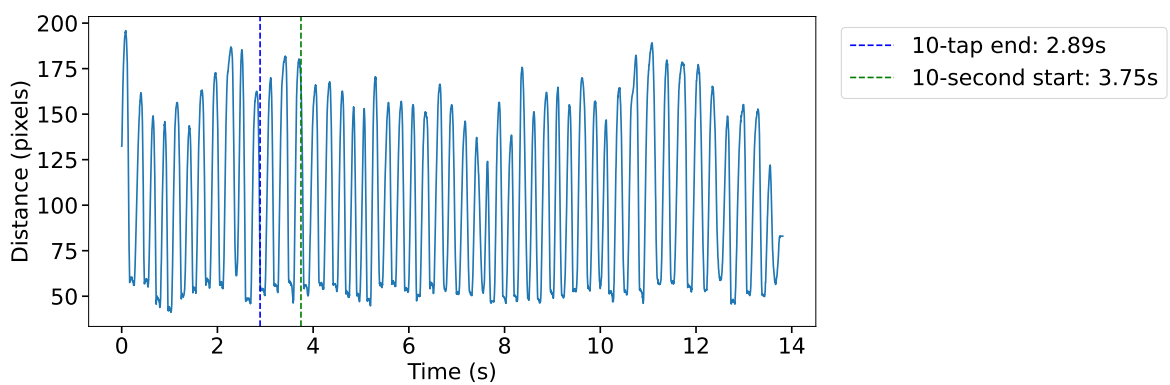


Figure J.19: Trajectory of distance between thumb and index finger in session 19, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

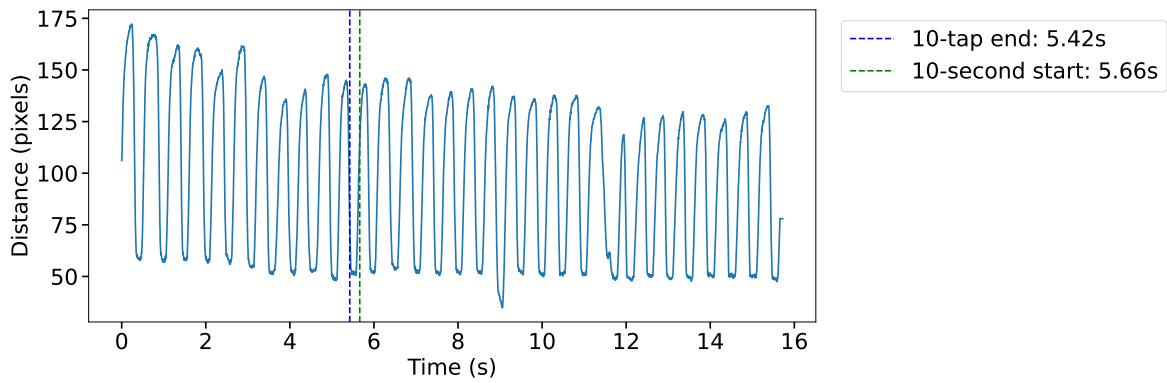


Figure J.20: Trajectory of distance between thumb and index finger in session 20, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

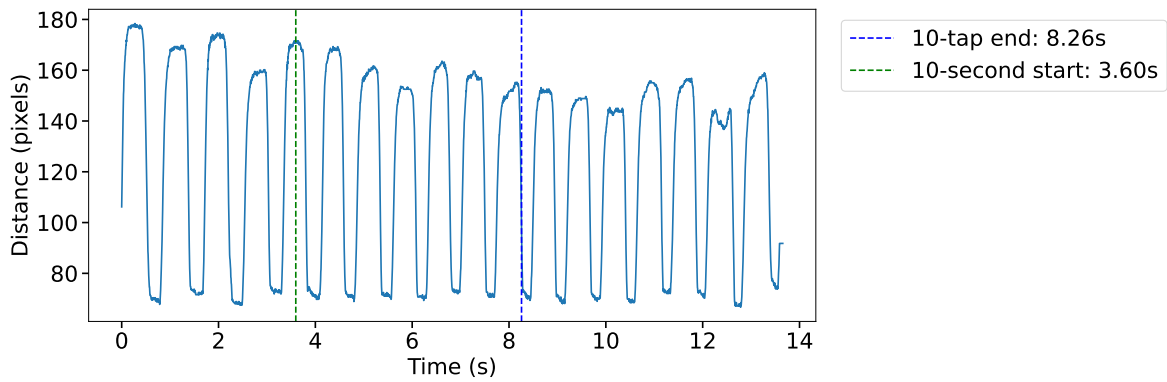


Figure J.21: Trajectory of distance between thumb and index finger in session 21, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

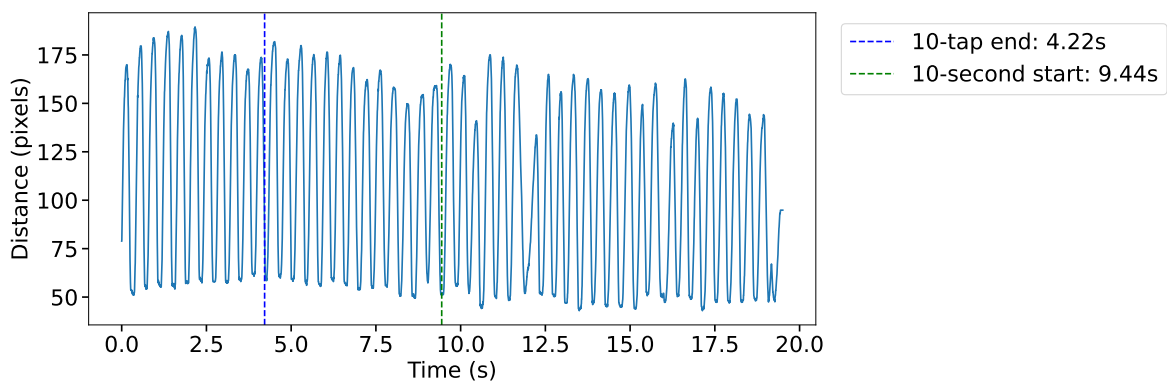


Figure J.22: Trajectory of distance between thumb and index finger in session 22, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

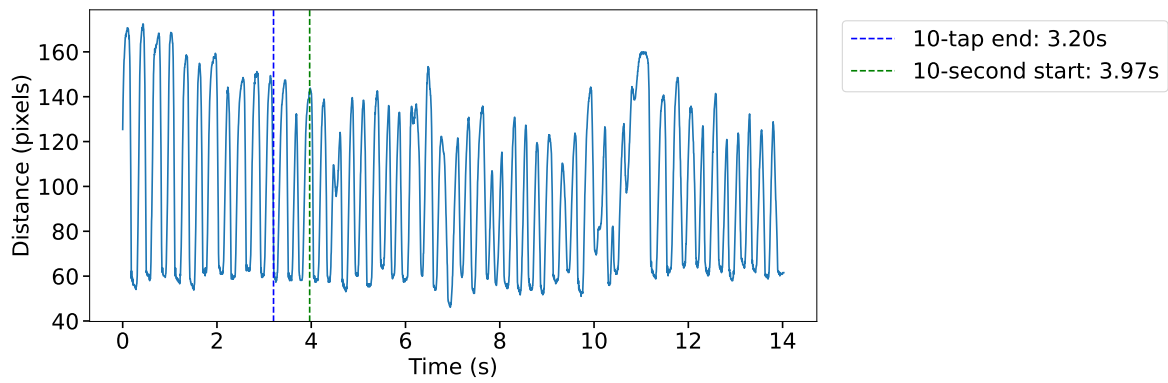


Figure J.23: Trajectory of distance between thumb and index finger in session 23, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

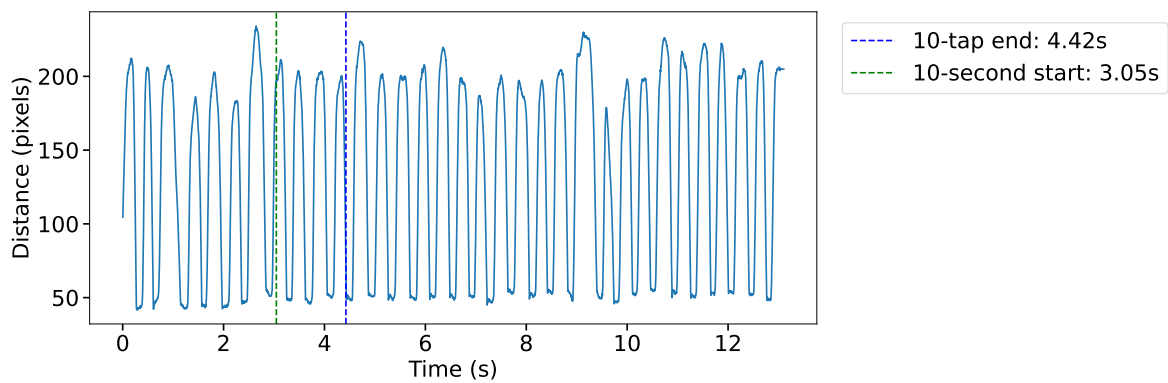


Figure J.24: Trajectory of distance between thumb and index finger in session 24, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

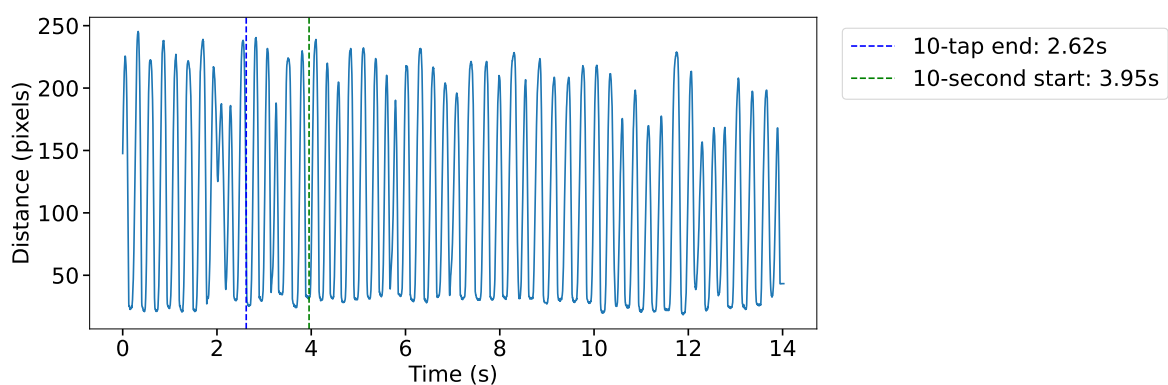


Figure J.25: Trajectory of distance between thumb and index finger in session 25, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

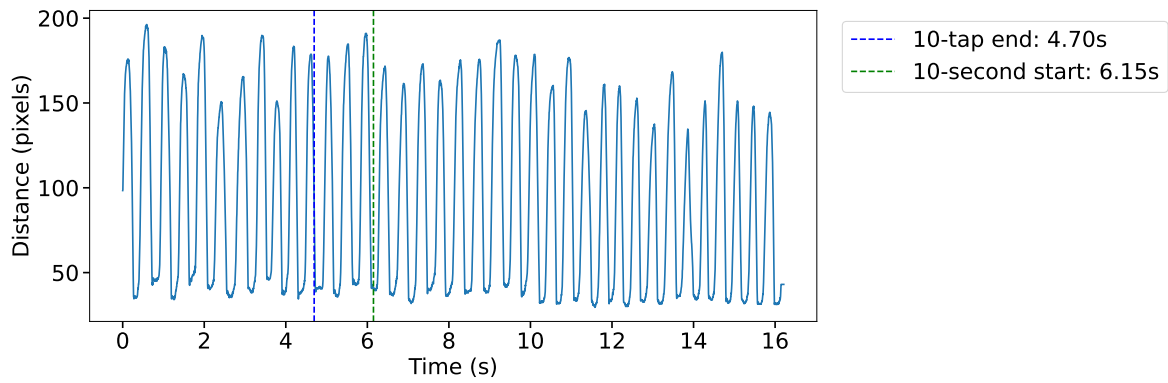


Figure J.26: Trajectory of distance between thumb and index finger in session 26, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

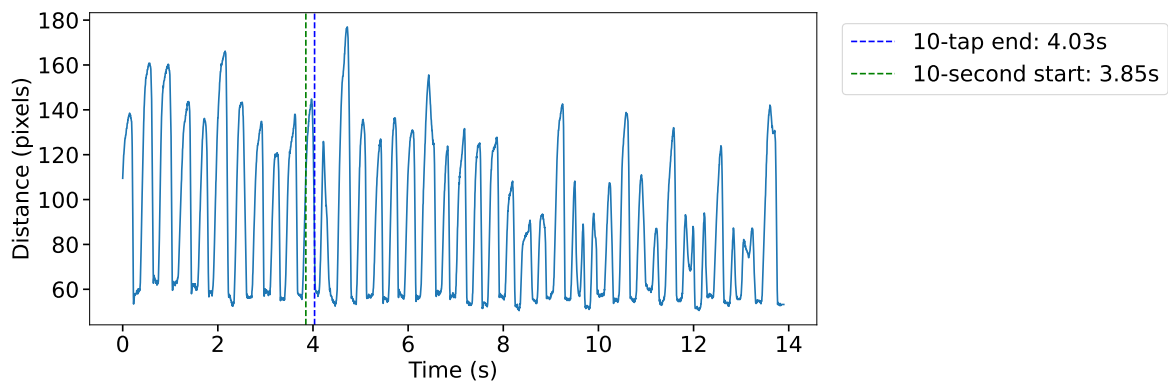


Figure J.27: Trajectory of distance between thumb and index finger in session 27, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

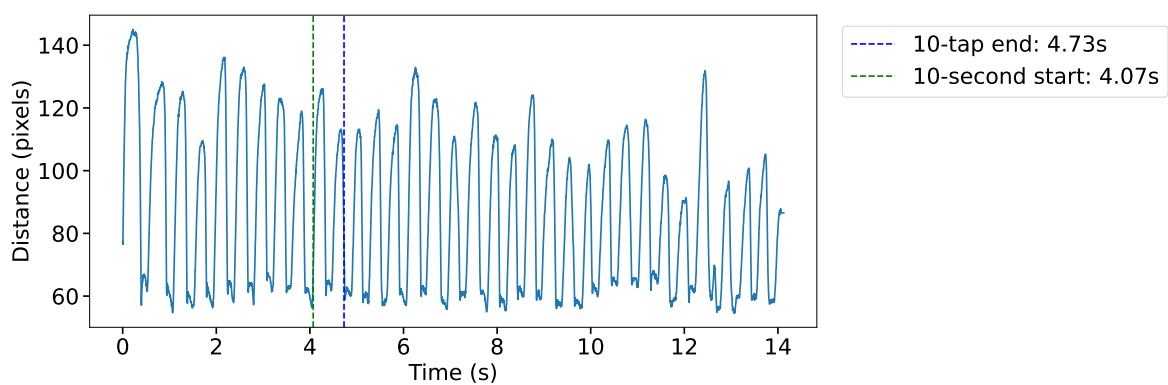


Figure J.28: Trajectory of distance between thumb and index finger in session 28, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

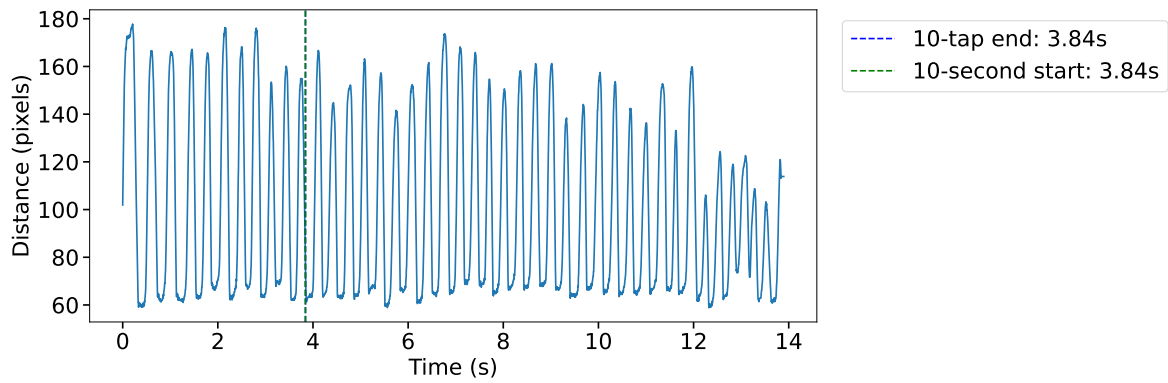


Figure J.29: Trajectory of distance between thumb and index finger in session 29, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

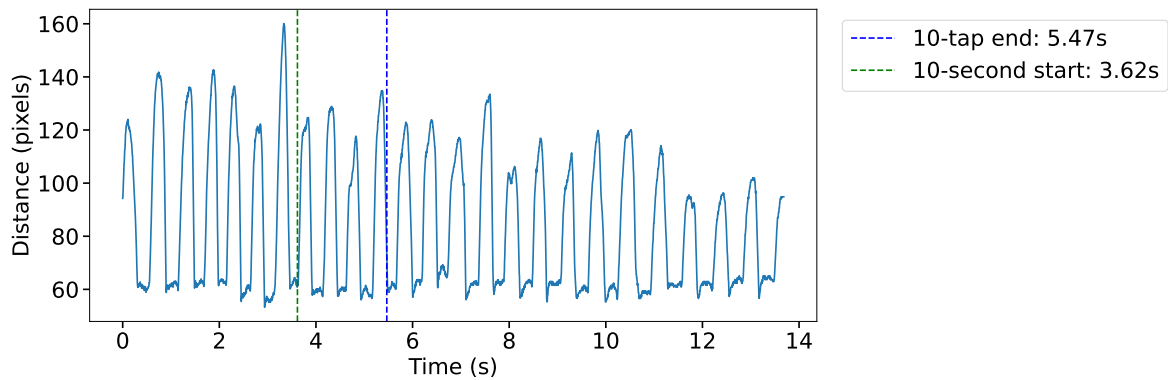


Figure J.30: Trajectory of distance between thumb and index finger in session 30, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

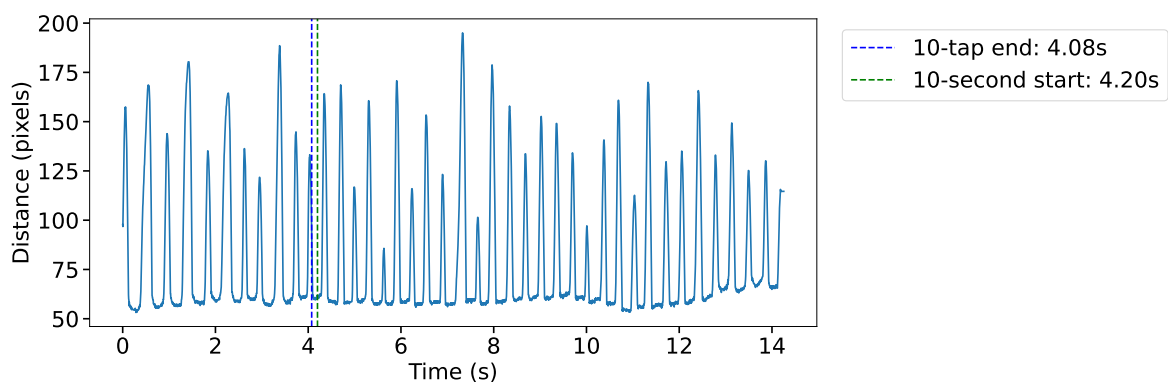


Figure J.31: Trajectory of distance between thumb and index finger in session 31, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

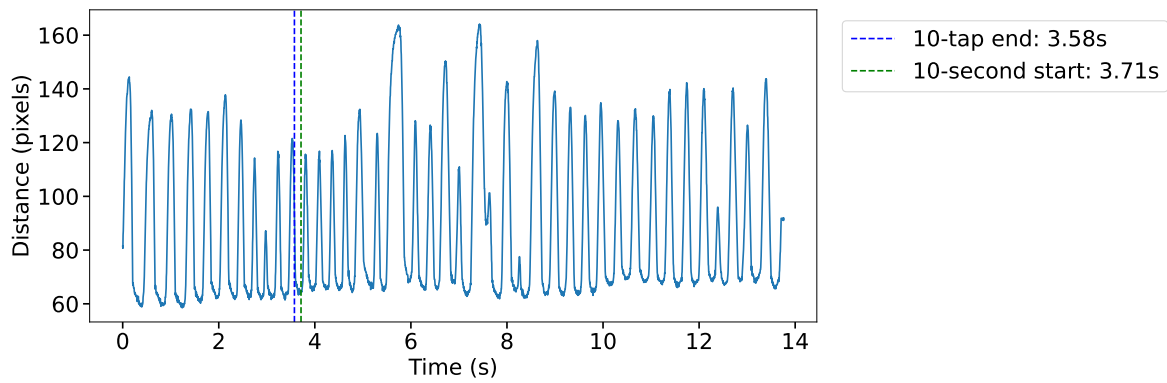


Figure J.32: Trajectory of distance between thumb and index finger in session 32, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

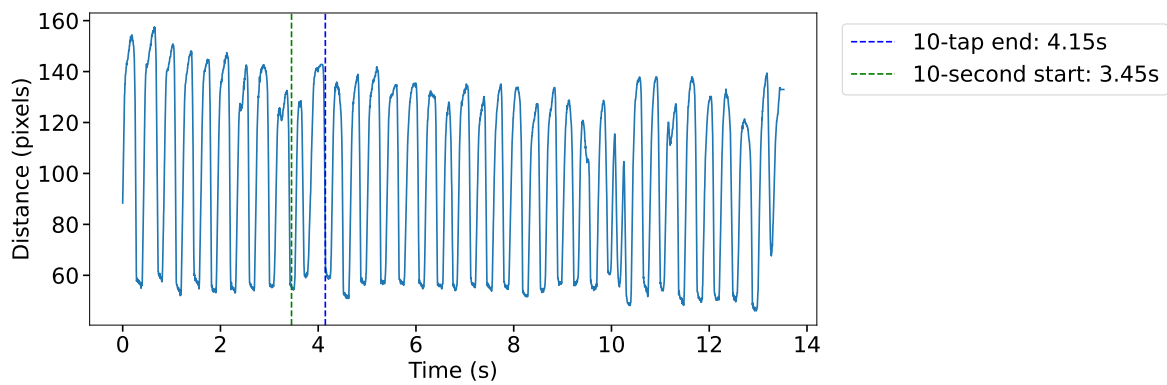


Figure J.33: Trajectory of distance between thumb and index finger in session 33, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

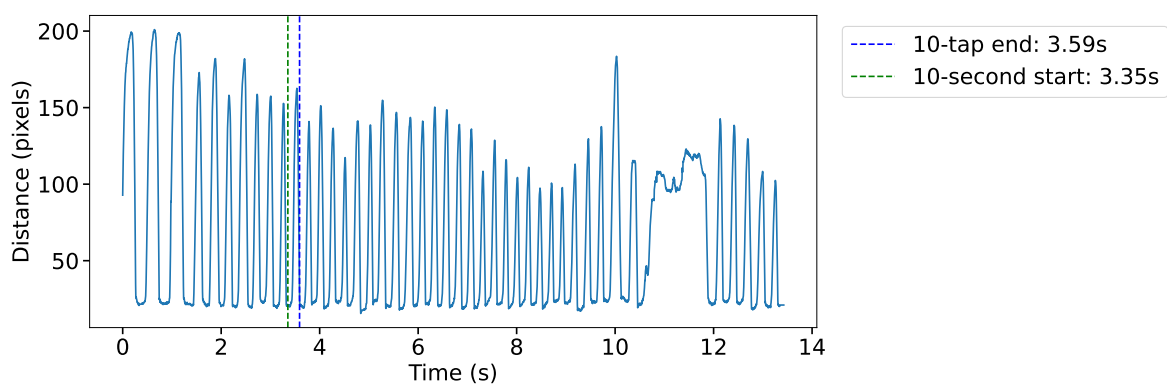


Figure J.34: Trajectory of distance between thumb and index finger in session 34, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

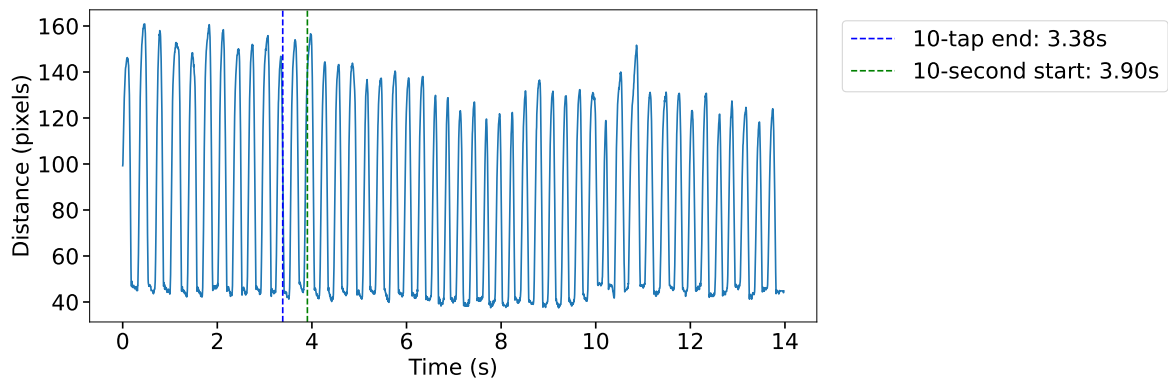


Figure J.35: Trajectory of distance between thumb and index finger in session 35, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

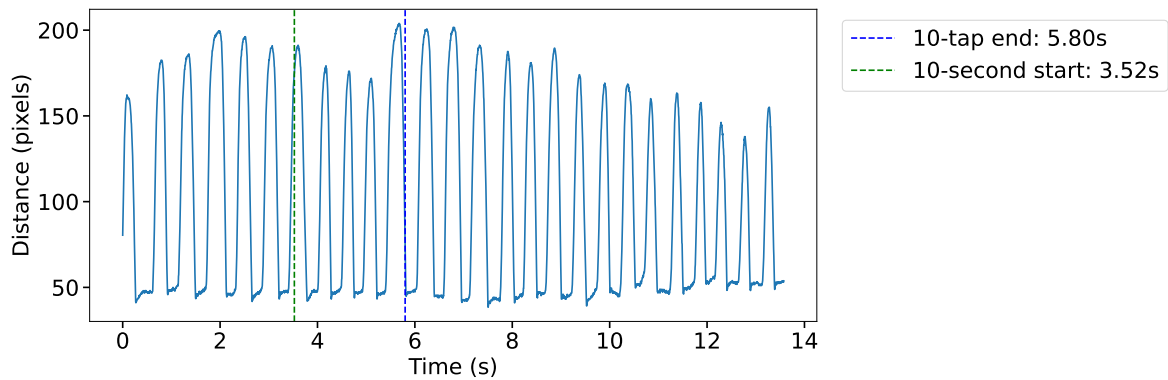


Figure J.36: Trajectory of distance between thumb and index finger in session 36, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

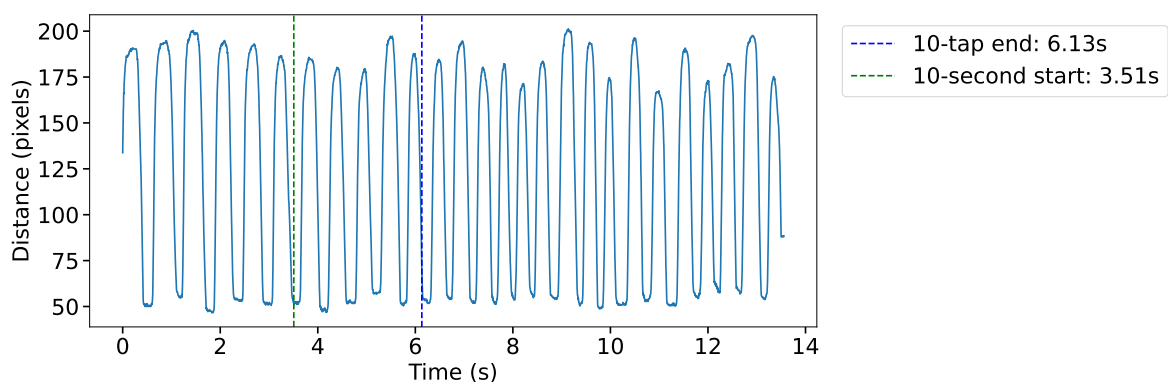


Figure J.37: Trajectory of distance between thumb and index finger in session 37, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

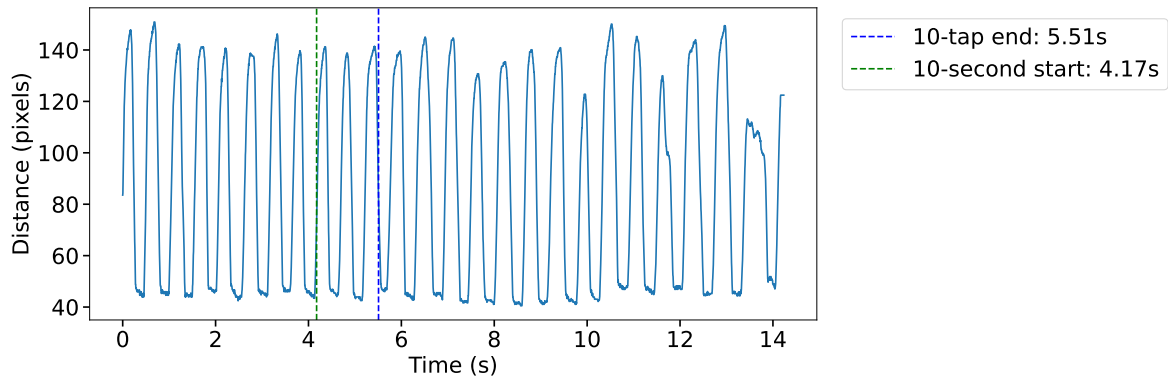


Figure J.38: Trajectory of distance between thumb and index finger in session 38, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

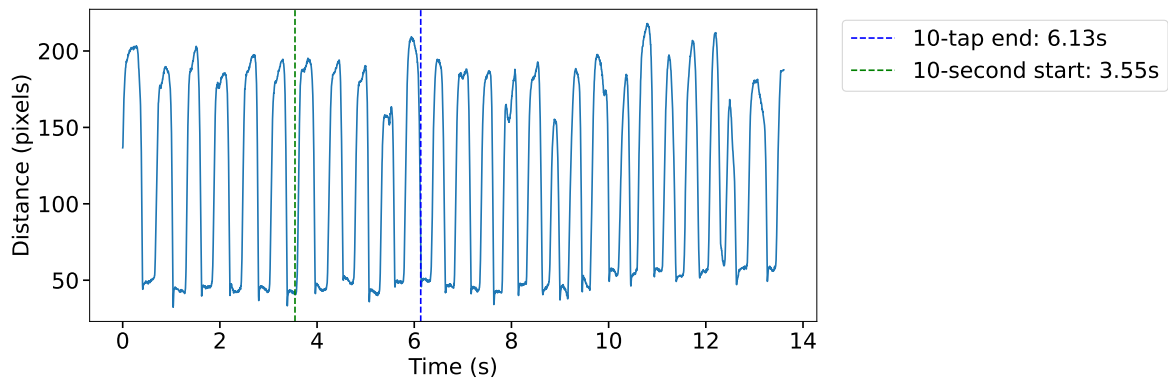


Figure J.39: Trajectory of distance between thumb and index finger in session 39, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

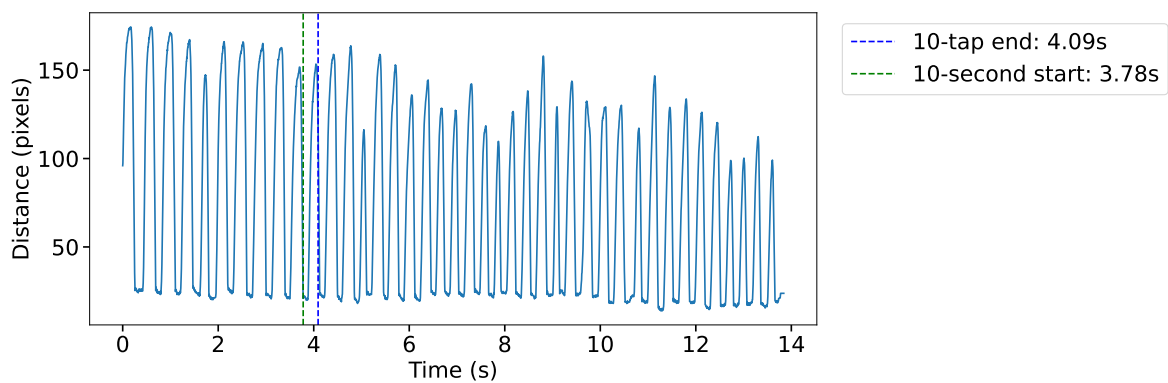


Figure J.40: Trajectory of distance between thumb and index finger in session 40, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

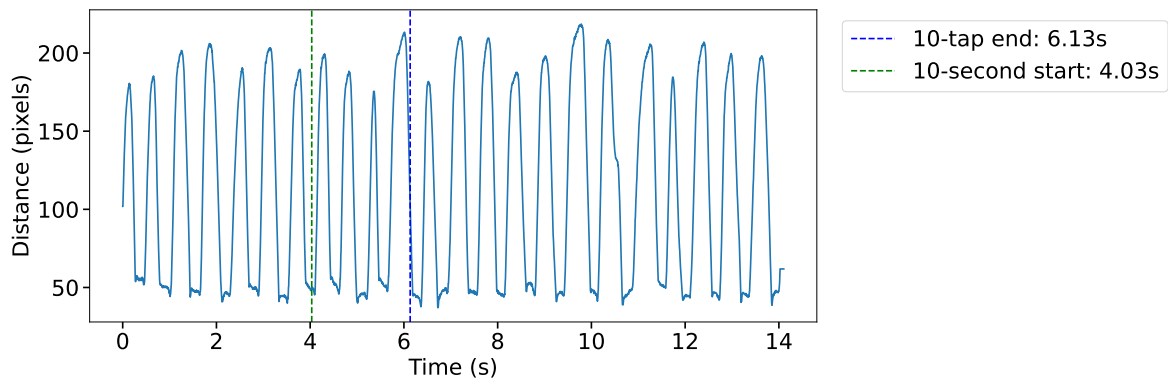


Figure J.41: Trajectory of distance between thumb and index finger in session 41, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

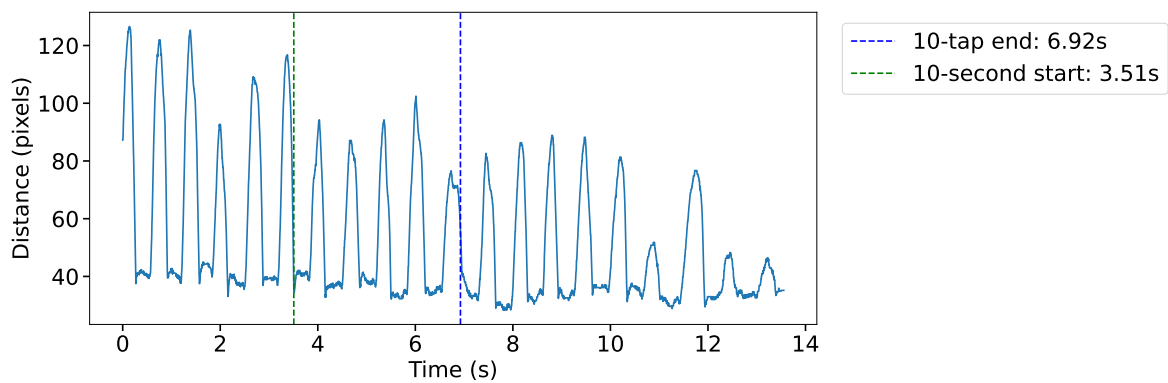


Figure J.42: Trajectory of distance between thumb and index finger in session 42, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

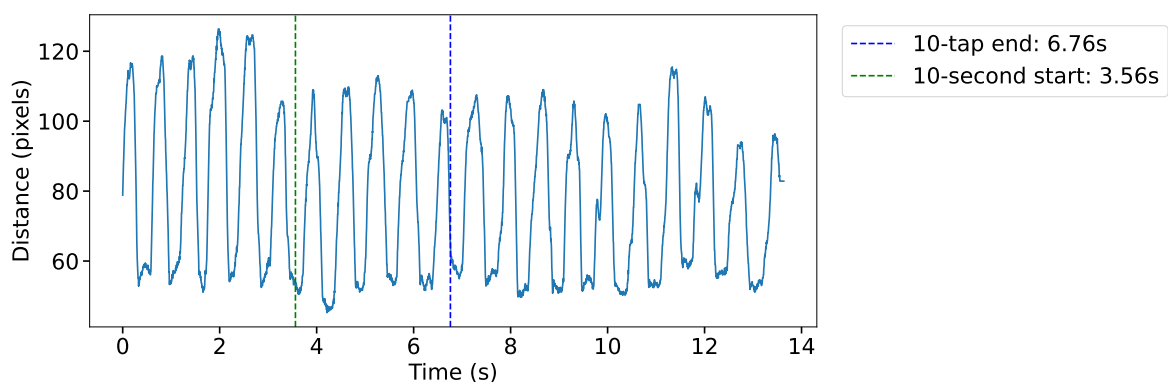


Figure J.43: Trajectory of distance between thumb and index finger in session 43, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

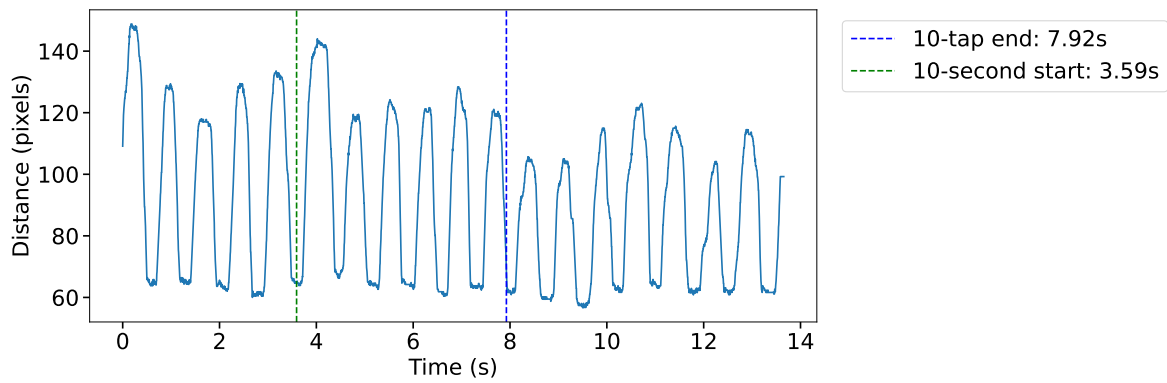


Figure J.44: Trajectory of distance between thumb and index finger in session 44, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

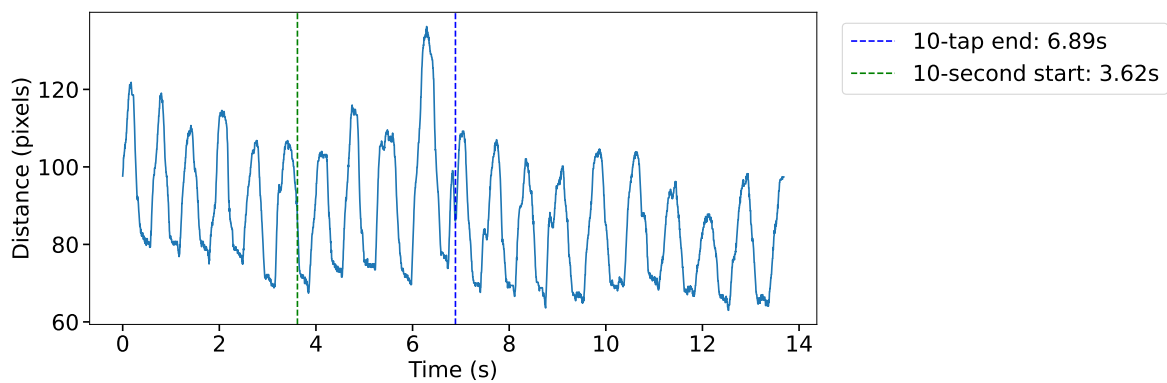


Figure J.45: Trajectory of distance between thumb and index finger in session 45, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

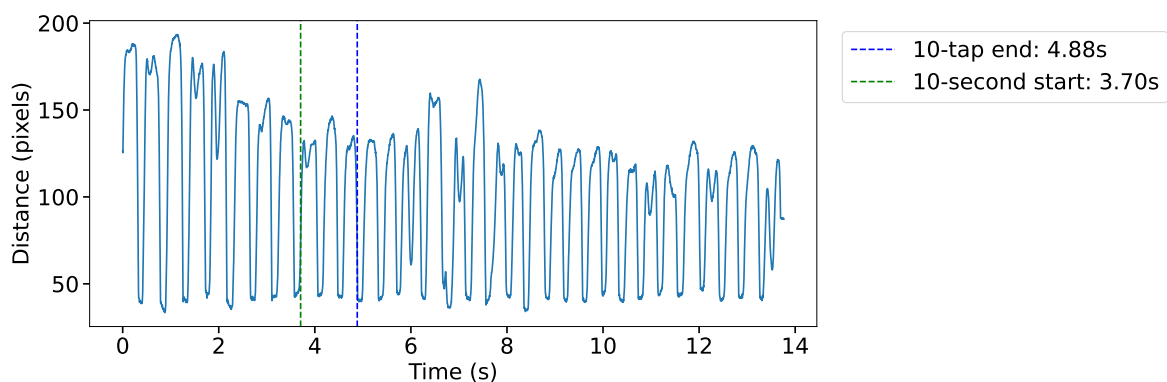


Figure J.46: Trajectory of distance between thumb and index finger in session 46, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

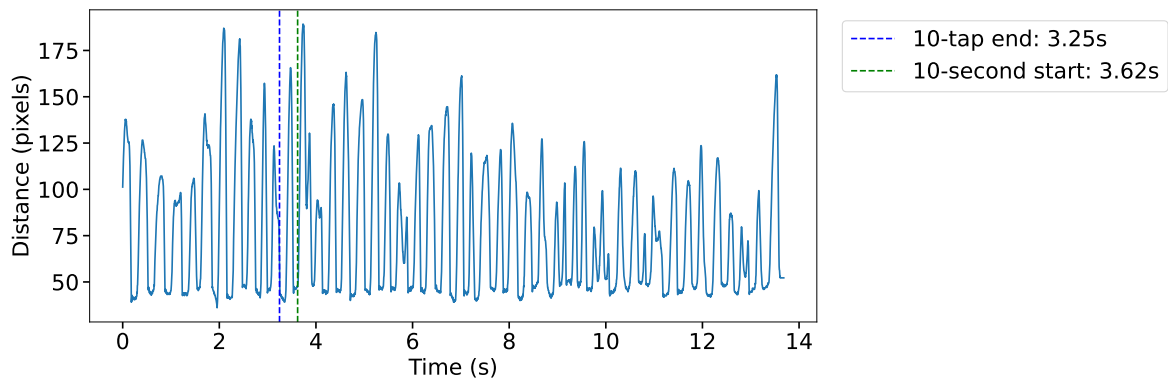


Figure J.47: Trajectory of distance between thumb and index finger in session 47, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

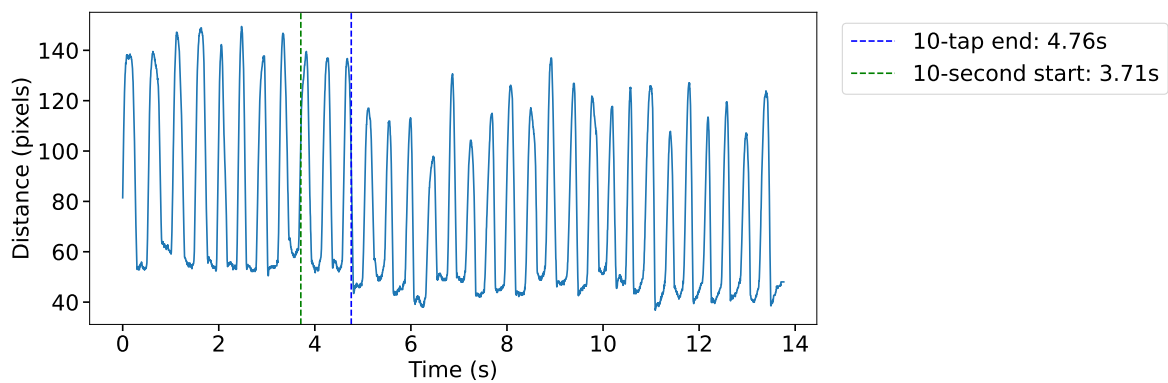


Figure J.48: Trajectory of distance between thumb and index finger in session 48, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

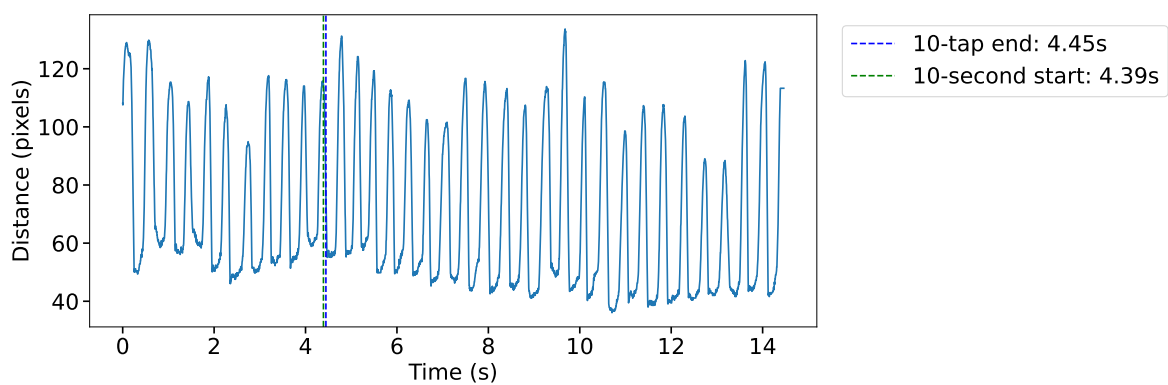


Figure J.49: Trajectory of distance between thumb and index finger in session 49, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

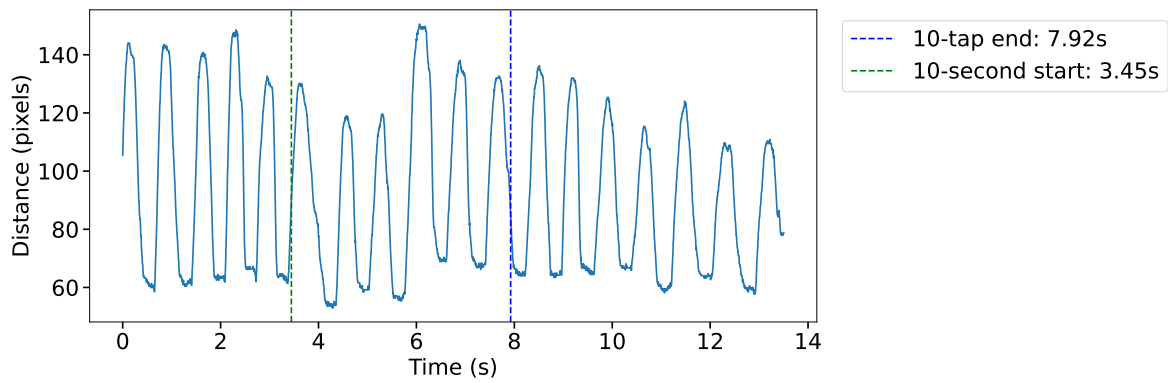


Figure J.50: Trajectory of distance between thumb and index finger in session 50, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

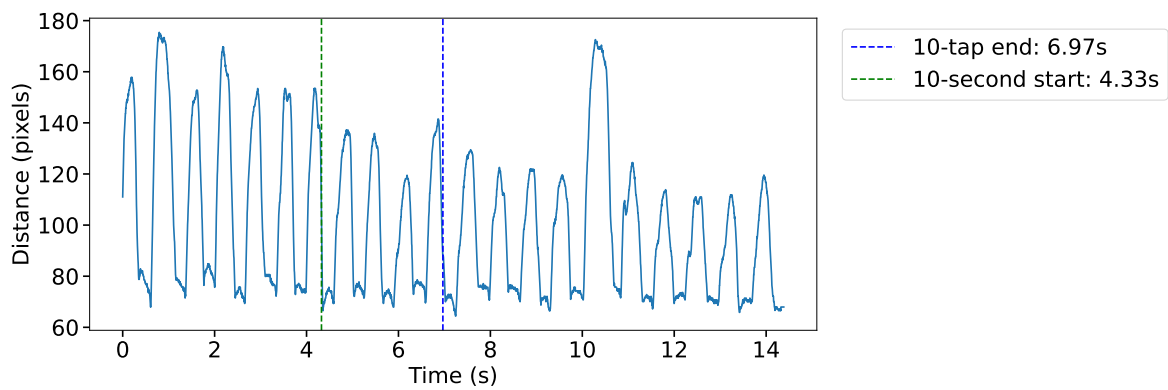


Figure J.51: Trajectory of distance between thumb and index finger in session 51, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

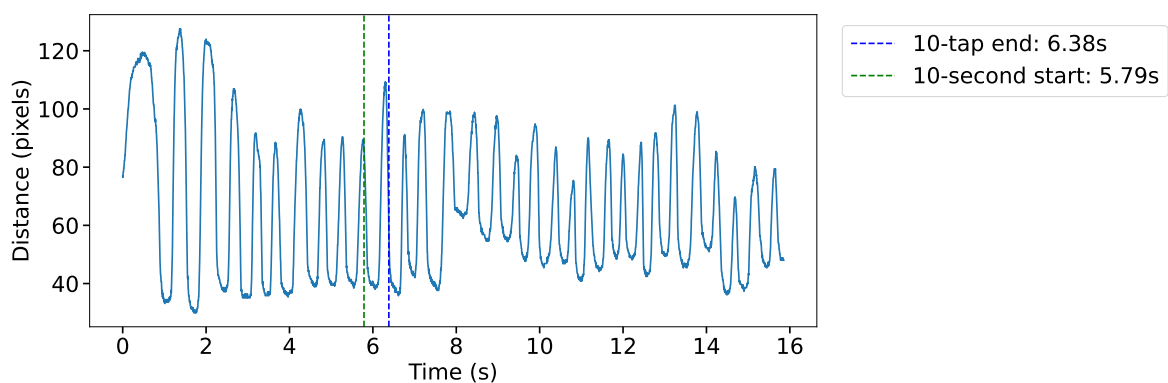


Figure J.52: Trajectory of distance between thumb and index finger in session 52, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

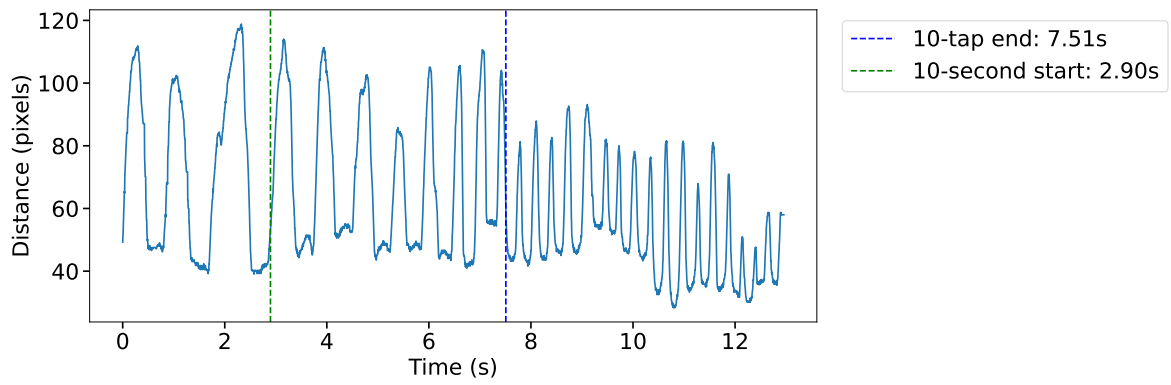


Figure J.53: Trajectory of distance between thumb and index finger in session 53, showing the complete FT sequence with marked endpoints for the 10-tap segment (blue dashed line) and start point of the 10-second data collection period (green dashed line).

